OpenSourceRail

Open Engineering Platform

Comprehensive illustrated edition | v0.6.1 published software | v0.7 development



Wheel-supported reference and candidate-bound research. Manufacturing, physical and operating acceptance remain open.

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I. Purpose and platform

The project, its three products, candidate identities and current capabilities.

1.1 Scope and reading guide

docs/BOOK.md

Reader edition: 13 November 2026. The published software release is v0.6.1. v0.7 is in development; its execution record separates completed software and numerical work from unresolved engineering and physical evidence.

OpenSourceRail develops an affordable, accessible urban railway and the open tools needed to plan, design, make, operate, maintain and renew it. Local design knowledge, manufacture and repair are central to the project. The platform connects city planning, railway engineering and manufacturing/lifecycle work; the three products must share explicit candidate and source identities.

How to read this edition

The PDF is a comprehensive illustrated ebook in six parts and 23 source-attributed chapters. It brings together the platform guide, published software and adoption scope, city and corridor work, vehicle and civil engineering, CAD/FEA methods, manufacturing and costing, business integration, and the v0.7 execution record and plan. The PDF outline navigates directly to each part and chapter. Chapter source paths and the build receipt let a reader return to the maintained documents and illustrations.

Part	Subject
I	Purpose, three products, candidate identity and current capability
II	Published v0.6.1 software, limitations and adoption
III	Samawah project continuity, native GIS and city design dossiers
IV	Coordinated railway, vehicle assembly, visual models, engineering controls and CAD/FEA
V	Model-based cost, delivery and business lifecycle integration
VI	v0.7 evidence, open interfaces, roadmap and agent delivery plan

The retained v0.6 engineering findings have their own clearly titled chapter. They remain research and open findings; the held integrated current dossier is not used as a passing release result. Other historical configurations remain available through their source-bound archives without silently changing the development reference.

What is available

Area	Supported scope	Boundary
Published software	v0.6.1 planning, research, Workbench and delivery tools; the clean-install Samawah journey is recorded.	Software publication does not approve hardware or train operation.
City comparison	Calculated outlines for 266 city networks and a candidate-bound Samawah station-edit comparison.	Route, demand, land, ground, access and prices can remain provisional.
Engineering research	Counted vehicle and civil concepts, SolidCraft/FieldCraft studies, independent controls and source-bound candidate packets.	A calculated or converged model is not a qualified installation.
Business and maintenance	ERPNext/Frappe HR and FUXA/gateway adapters, draft records and a bounded assembly custody contract.	Native full-module transactions, live control and purchasing authority remain separate.

Use the capability register for specific entry points and maturity. The v0.6.1 known limitations define the published release boundary. The stale integrated engineering dossier and portfolio selector are held; this reader does not draw conclusions from them.

Development reference and research alternatives

The development reference uses six 17 m cars with passive wheel bogies on conventional rails carried by concrete. Each car has 44 fixed seats and two wheelchair places, with no planned standing and two doors on one side. The passive aluminium secondary is a comparison interface. Installed propulsion, powered length, acceleration, braking and operating headway are selected per candidate. The former maglev configuration and the 120-second service case are retained research controls, not the selected physical railway.

Research candidates may change traction, running gear, formation, station spacing or civil form. Each change must carry its mass, power, heat, protection, service, manufacture, construction and whole-life cost effects. Mandatory safety, accessibility and passenger protection stay as requirements. The architecture explains how candidate, evaluation and selection identities prevent results from being mixed.

One corridor through v0.7

The v0.7 roadmap uses one Samawah corridor revision as the integration case. Two source station markers are about 79.977 m apart against a 120 m platform assumption. The reviewed midpoint edit changes raw markers from 26 to 25; the published comparison already groups them into 25 operational stops, so its fleet stays at 81. The project envelope identifies which service, quantity, cost and dossier results change and which unaffected results are reused. These are planning calculations, not a surveyed alignment.

Current v0.7 evidence also includes a headless native GIS two-edit cycle for Samawah and Baghdad, bounded civil and force/braking controls, and a measured opt-in Rust delivery scheduler. The measured complete delivery-CLI median fell from 12.49 to 10.79 seconds across three paired Samawah trials with a prebuilt binary. That is a 13.64% reduction for this compiler path; it is not a whole-city or engineering-solver speedup. The agent delivery plan keeps further Rust work tied to measured bottlenecks and exact output parity.

The door module and the next physical interfaces

The candidate-bound one-opening handoff counts 74 part types and 694 pieces: 97 locally fabricated, 568 bought to standard and 29 bought to qualification. Its 597 purchased pieces span 35 types. The software custody contract checks receipt-backed, individually serialized purchased children with serial, batch, incoming QA and parent consumption. A sealed, reviewed synthetic one-child pack exercises that contract. The complete module has not passed a native ERPNext receipt, assembly, replacement and handback journey.

The opening layout still lacks controlled header/body, rail/carriage, belt/pulley, drive and sensor mating datums. Installed transforms, approved inspection limits, clearance and removal paths, complete installed mass and physical acceptance remain open. A 2 m display proxy does not establish fit for the 2.9 m guide stock. The handoff and evidence record identifies each unresolved interface; no missing quantity or limit is zero.

Civil work likewise needs a represented rail/support/load domain, bearings, foundations, erection stability and a crane chart at the actual lift radius. Propulsion and braking need installed force, current and repeated thermal-duty evidence. Independent physical tests and reviews remain necessary before any manufacturing, movement or operating release.

Use and reproduce the software

Start with the published installation instructions and their checksum-verified downloads. The Workbench shows supported journeys and visible holds. The repository exposes bounded research commands:

```
./osr design railway plan
./osr design railway run --max-bundles 2 --native-jobs 1
./osr design railway verify
./osr book --check
```

Native CAD, FEM and GIS jobs need compatible tools, declared resource budgets and exact input identities. Keep a host disk reserve and retain failed or interrupted run evidence. Current v0.7 work uses SolidCraft, FieldCraft and RustGIS in their demonstrated domains; optional independent solvers remain useful for verification. A tool substitution creates a new evaluation, not a new physical design by itself.

Source, review and history

The project README remains the public entry point. This PDF includes 23 maintained source-attributed chapters, the illustrated platform guide, release scope and limitations, current development evidence, and the roadmap. The v0.7 execution record is a full chapter and remains source-bound to this edition. The platform guide retains deeper v0.6 reference and research background under its stated evidence limits. The documentation index lists further maintained guides. The v0.7 execution ledger records task status, candidate boundaries and evidence. The current book PDF has its own source receipt; its summaries do not supersede the sealed engineering packets.

The published v0.6.1 release and historical v0.6.0, v0.5 and v0.4 records keep their original identities. Earlier maglev and source-bound research remain retrievable through the historical manifest. Their results apply only to their declared candidates and methods. Preserve useful failures, open assumptions and original hashes when revising the design.

Software is licensed under Apache 2.0, open physical designs under CERN-OHL-S v2, and documentation under CC-BY-SA 4.0, subject to path-specific notices in the license guide.

1.2 Platform and railway concept

docs/platform-guide.md

Scope of this guide: detailed v0.6 reference and research background. For the published software scope, read v0.6.1; for current development and open engineering interfaces, read the v0.7 execution record and the current book. The integrated v0.6 dossier cited below has a source-binding hold and is not supported release evidence. Historical motor, service and civil examples remain tied to their original candidates and do not select installed propulsion or physical acceptance.

This guide preserves the detailed platform, engineering, delivery and research explanation. For the shorter introduction and practical starting points, see the project README. Current numerical results and evidence status belong to the linked source-bound publications.

OpenSourceRail develops affordable urban rail and the open tools needed to plan, design, procure, manufacture, construct, operate, maintain and renew it. The aim is to retain design authority, software, fabrication, integration and repair capability in-country, with competitive sourcing of specialist components.

v0.6 is the station-powered freewheeling reference railway: lightweight comfort cars run on passive steel-wheel bogies and conventional rails on precast concrete structures. Station-side propulsion is compared under candidate-specific designs; the vehicle retains passive reaction elements for LIM research alongside mechanical launch alternatives. Normal-service traction motors, traction batteries and levitation magnets are absent from the train; passenger services, essential power, protection and independent brakes remain.

The wider project encompasses city planning, CAD/BIM, manufacturing, business execution, equipment supervision and open controls. The current v0.6 front end exposes planning proposals, candidate engineering and delivery records; business and supervision functions require separate configuration, and complete city delivery remains open. The railway is a research and prototype design, without manufacturing or operating release.

The transport goal leads the search. The current vehicle, station propulsion, formation, guideway and service are revisable reference choices. Alternative propulsion, shorter or articulated vehicles, running gear, alignments, manufacturing methods and other transport comparators remain open when they address the same passenger needs. The open approach policy retains unfamiliar concepts as research proposals, calls for their own complete models and costs, and limits any “best” claim to the alternatives actually evaluated. Safety and accessibility remain mandatory.

Propulsion research: The station regenerative capture and mechanical-launch plan compares sprung contacts, hydraulic or flywheel storage, controlled forward release, wayside rollers and compact acceleration. Its executable Rust screen compares recovery energy, storage, grade and booster demands at variable speeds. This is an outline-stage research family. Speed, powered length, acceleration, headway and whole-life cost must follow passenger requirements; no installed propulsion system is selected.

Start here: how the whole system works · illustrated reader · project overview · current platform capabilities · current engineering · sourcing and cost evidence · city designs and maps · roadmap.



OpenSourceRail v0.6 proposed wheel-supported train at a compact elevated station

Architectural concept: a lightweight wheel-supported train, roof solar, conventional rails, repeatable precast beams and an accessible elevated station. Inspect the source-bound vehicle geometry and download the engineering models. The illustration communicates the design intent; geometry, installed equipment and physical qualification follow the candidate records.

Why OpenSourceRail

- Build useful capability locally. Develop concrete precasting, structural fabrication, GFRP panels, interiors, winding, wiring, assembly, inspection and maintenance alongside the railway. Qualify local suppliers rather than assuming every component can already be made locally.
- Keep ownership of the design and information. Open software, drawings, interfaces, component identities and maintenance records let an owner change suppliers and repair equipment without depending on one closed turnkey provider.
- Simplify the complete installed system. Compare vehicle mass, bogies, station power, beams, piers, foundations, equipment, factory tooling and erection together. A cheaper train that requires more expensive stations is not automatically a cheaper railway.
- Serve constrained urban corridors. Mostly elevated, prefabricated construction and compact stations aim to reduce surface occupation and site disruption. Land, utilities, buildings, ground conditions, drainage, access and evacuation still govern each route.
- Offer comfortable, frequent travel. The reference prioritises seated accommodation and wheelchair spaces; frequency is optimised against demand, with a two-minute comparison case retained. Capacity, dwell, safe separation, transfers and whole-system cost must be evaluated together.

Local content means origin-traced domestic value added, not the address of a distributor. Imported cells, semiconductors, bearings, sensors, materials and machinery retain their actual origin. No local-content percentage or cost saving is established until the installed BOM, processes and supplier evidence support it.

Three connected products

Product	Purpose	Present boundary
Design and delivery tools	City inputs, design alternatives, engineering evidence, quantities, procurement, manufacturing and project records	Current v0.6 Workbench, planning editor, candidate exports and integration components; complete city-specific delivery remains open
Reference railway and manufacturing system	Lightweight vehicles, stations, civil works, tooling, assembly and repair methods	v0.6 physical candidates and numerical studies; equipment, drawings, suppliers and physical qualification are incomplete
Open GoA4 control software	ATP, interlocking, station-led dispatch, train supervision and fault response	Rust research software, simulation and formal/method checks; no live railway or safety approval

The first adoption path describes the current planning and engineering interfaces, candidate delivery drafts and separately configured business and supervision integrations. Owners can use them to organise research and review before a train is qualified; they do not provide a live railway operating interface.

From a city brief to a working railway

The intended end-to-end loop is: define the transport need generate competing designs calculate and test them plan their manufacture and construction estimate complete costs revise and compare approve, build and operate. Every revision should flow through the same vehicle, station, civil, factory, service and financial models. A change to a panel, bogie, beam, winding or service assumption must automatically update all affected quantities, analyses, tasks and costs.

Stage	What the platform should do	Output and next decision
1. Define the remit	Record city geography, passengers, station locations, comfort, speed, climate, accessibility, safety, local manufacturing capability and target cost	One versioned requirements and site scenario; distinguish mandatory requirements from preferences
2. Generate alternatives	Explore vehicle materials/shapes, structures, bogies, propulsion, stations, beams, foundations and production methods—not just small parameter changes	Several fully identified competing physical candidates
3. Detail and check	Produce CAD/BIM, part-by-part installed BOM, weight/CG/inertia, joints, service routes and manufacturing interfaces; run structural, motor, braking, thermal and fault models	Explicit pass, fail or missing-evidence records for every candidate
4. Repair and iterate	On failure, revise geometry, material, component, assembly method or entire architecture; rerun dependent calculations without relaxing safety requirements	New candidate generations with ancestry and Pareto comparisons
5. Calculate build costs	Derive gross material quantities, labour/process hours, machine time, tooling and QA from each design; use dated rates or declared engineering assumptions even where no quote exists	Low/base/high factory, delivered, installed and whole-life planning estimates, with confidence and exclusions
6. Plan production and construction	Generate EBOM/MBOM, assembly order, precast and vehicle factory throughput, workstations, crews, inspections, haulage, cranes/launchers and site sequence	Capacity, workforce, construction duration and cashflow scenarios
7. Review and execute	Compare equal-service candidates and independently check the preferred design; prepare procurement, production, project, QA and training records in ERPNext/Frappe HR	Reviewed handoffs; purchasing, fabrication, installation and commissioning require human approval
8. Operate and improve	Integrate approved equipment, GoA4 protection, FUXA monitoring, maintenance, serial records and actual measured hours/consumption	Real performance and cost evidence fed back into new design revisions

This is the target workflow, not a claim that every stage is already commissioned. The current generator, engineering Workbench, component registry, numerical models and preliminary manufacturing/cost outputs cover much of stages 2–5. City-specific production schedules, site-validated quantities, physical proof, approved business transactions and operating integration remain incomplete. See the step-by-step cost and delivery implementation plan.

A failed design is not the end of the process. The coordinator should diagnose why it failed, propose a larger change where necessary, and rerun that candidate through weight, simulation, factory work, construction and whole-life economics. Missing prices can trigger calculated estimate ranges; missing safety-critical physical evidence must still block physical approval.

Candidate completion is reported in three separate levels: simulation complete for a named scope with measured or explicitly assumed inputs; engineering validated with credible physical evidence and independent review; and approved for manufacture or operation under the applicable safety, accessibility and release requirements. A completed provisional cost scenario does

not imply an integrated journey simulation, engineering validation or release. The current counts and evidence rules are in the coordinated railway report.

The separate conditional P1 journey study runs low/base/high assumed force maps through the reduced journey model for the 28 coordinated campaign variants and the exact selected v0.6 vehicle. The selected vehicle keeps its physical identity while its P1 route remains a separate evaluation. These cases are assumption-only; they do not change the coordinated report's count of completed physical journeys.

Structural refinement is tracked against the exact candidate, loads, mesh, solver and resource request. Completed, failed and nonconverged executions remain inspectable in the coordinated engineering report; solver completion and mesh agreement alone do not qualify the structure.

The selected vehicle mass reserve sensitivity turns the existing positive mass-account residuals into provisional low/base/high tare scenarios by replacing each residual once. It does not label the vehicle's installed mass complete.

v0.6 development basis

development-baseline.toml selects one reference vehicle, body and running gear, with a passive interface for propulsion comparisons; it does not select an installed propulsion system. Research alternatives receive separate identities and do not silently replace it.

Design choice	Reference basis
Formation	Six 17 m cars; approximately 104.5 m including gaps
Comfort	44 fixed seats + two wheelchair places/car; 276/train; no planned standing
Layout	Four abreast where structure permits, wall benches at doors; two passenger doors on one side
Body	Aluminium/GFRP architecture, sandwich floor/roof, lightweight benches and glazing
Running gear	Twelve unpowered two-axle bogies: 24 axles, 48 wheels
Propulsion interface	Passive aluminium secondary on each car for LIM comparisons; installed architecture and powered length remain candidate-specific
Ordinary guideway	Conventional rails and fastenings on precast concrete; box, twin-beam and U-beam comparisons
Services	Flush solar roof and a separately sized auxiliary battery; HVAC, doors, ATP and emergency loads remain
Automation	Station-led GoA4 architecture, continuous onboard ATP and independent mechanical brakes
Network	Passenger transfers between fixed lines; no mainline switches, with depot exceptions explicit
Reference service case	72 km/h, 0.4 m/s² acceleration, 0.6 m/s² service deceleration, 35 s dwell, 120 s target headway
Capacity arithmetic	8,280 passengers/hour/direction at 30 departures/hour; protected headway remains unqualified

The normal passenger cap is a comfort and load-management choice; it does not remove crowd, evacuation or accidental-load checks. Structural checks retain a 160-person/car crowd-load control case. One-sided doors require compatible platforms, accessible boarding and emergency exits. Any change to propulsion, formation or passenger loading is a separate candidate whose mass, energy, vehicle, civil, factory and service effects must be evaluated together.

Loss of station propulsion leaves the vehicle supported on its wheels. Independent braking must remain available, and a train stopped outside a powered zone needs a protected recovery/towing and evacuation arrangement. Walkway removal or a proposed escape device is not treated as approved.

A separate practical launch comparison applies the same 1.5 km and 2 km analytical route matrix to 12 station-LIM, wheel, roller, cable and gravity concepts. It exposes required effort, peak mechanical power, route bounds and open force-map, thermal, stopping, recovery, grid and headway evidence; it does not produce a physical journey. These comparisons do not choose an active installed propulsion system.

The active baseline selects a wheel-supported vehicle reference, not an installed propulsion system. Propulsion architecture, powered geometry, acceleration, braking and headway are candidate-specific design variables. Neither the former 900 m shared-winding allocation nor the separate 100 m experiment is a prescribed installation.

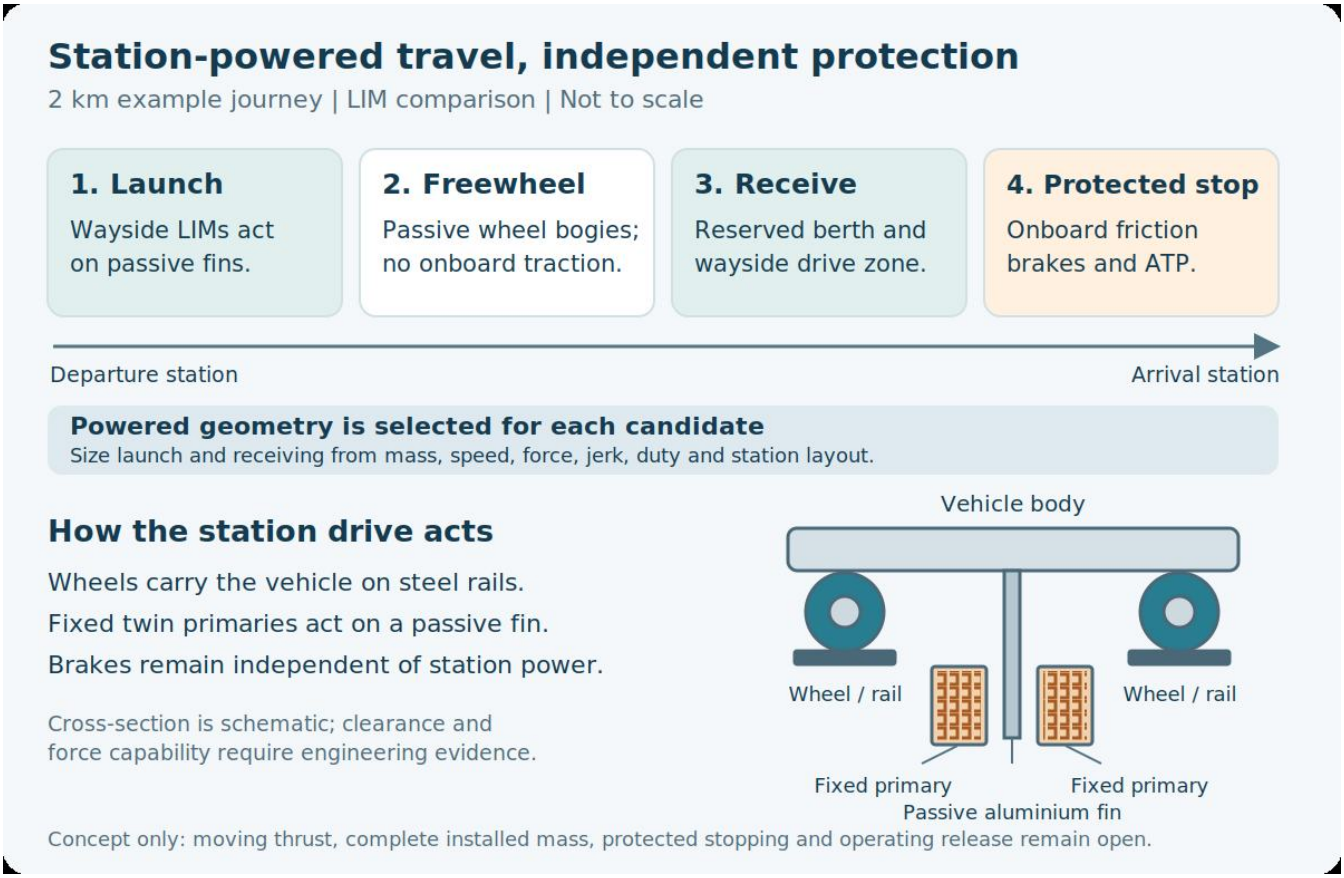
Start with passenger demand, waiting/journey limits, accessible stations and protected operation. Generate and compare compact LIM sections, common-drive rollers, cable/capstan launch, distributed wayside supply, onboard traction, hybrids or another explicitly modelled approach. Each needs its own vehicle/interfaces, installed quantities, force and thermal model, independent stopping/recovery provisions, manufacturing route and whole-life cost. An unfamiliar idea is a valid research proposal; a real physical contradiction requires revision.

Kinematic acceleration distance is not motor length. The candidate-specific layout model requires explicit sections and counts whole modules or declared tails, directions, joints and electrical branches. No missing length becomes zero or falls back to another fixed installation. The 30 km demand-led outline compares calculated concepts under common transport requirements; force-map, structural, thermal, braking and prototype evidence remain separate engineering work.

The current candidate export manifest and coupled delivery index expose candidate-bound CAD/IFC, engineering BOM, manufacturing, procurement, assembly and controls records. `./osr twin export` inspects these outputs. They do not establish complete installed equipment, a physically supported journey, approved production instructions or operating acceptance. The cost and delivery model calculates provisional factory, delivery, installation and lifecycle ranges when supplier quotes are unavailable; assumptions, uncertainty and evidence status remain attached.

The railway in pictures

Passenger-focused lightweight car	Local modular vehicle assembly
	
Proposed v0.6 passenger cabin with seated accommodation, an accessible door zone and wheelchair provision	Workshop assembly concept with body, roof, doors, glazing, seating and passive bogies
Seated comfort, level circulation, wheelchair provision, lighting and HVAC remain part of the vehicle. Detailed seating and accessibility come from the controlled layout.	Modular shells, doors, glazing, service systems and running gear support repeatable offsite assembly, inspection and repair.
Precast guideway construction	Station LIM comparison
	
Crane placing a precast rail beam onto elevated piers beside an urban corridor	Concept cutaway of passive wheel bogies and an aluminium secondary between two wayside LIM primaries
Repeated spans, conventional running rails and planned access aim to simplify construction. Lifting and temporary stability require site-specific engineering.	A passive aluminium fin provides an interface for a LIM alternative. Mechanical launch and other propulsion approaches remain open; the picture selects no installed motor or powered length.

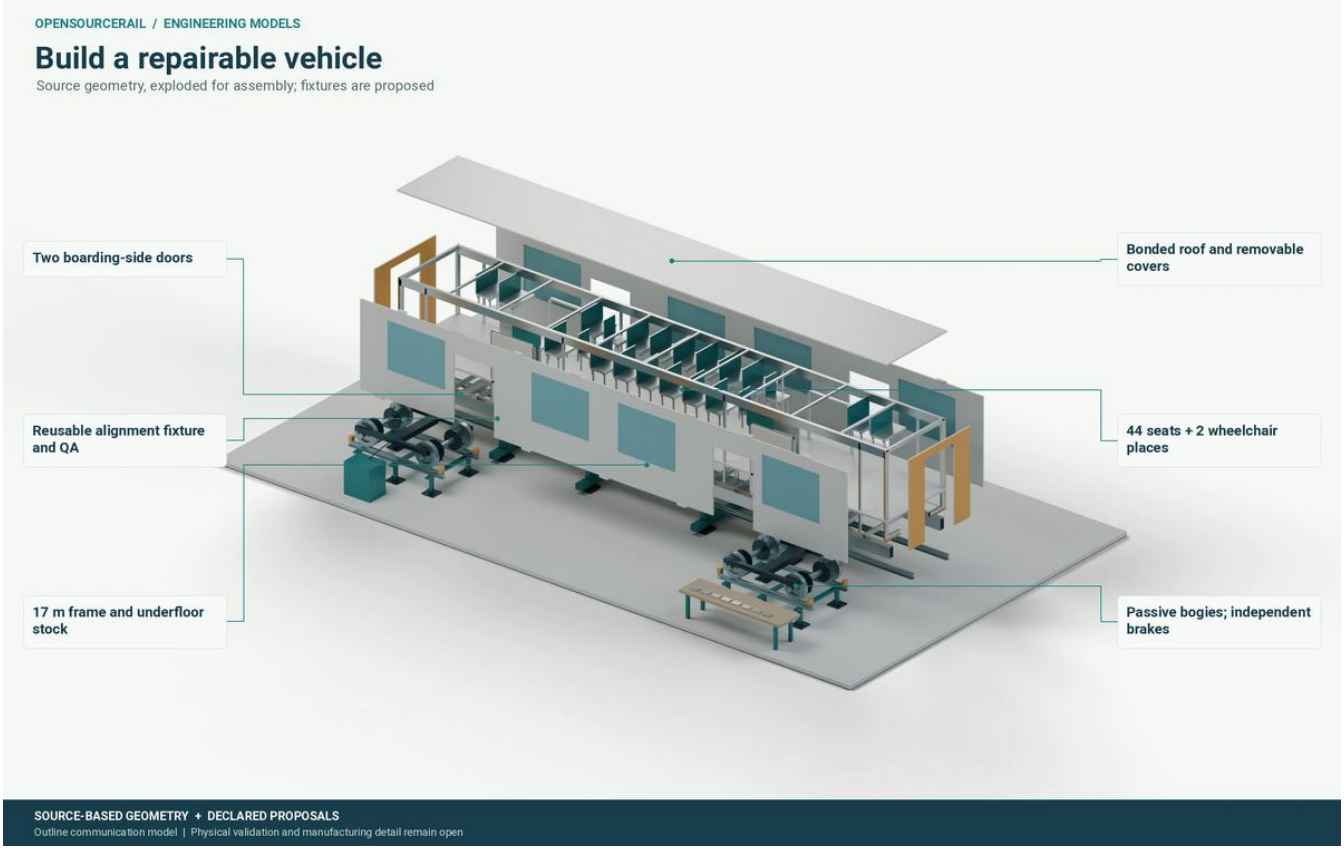


Station LIM comparison showing launch, freewheeling, receiving and independent protected stopping

These original concept renders complement the CAD-based models below. They are illustrations, not accepted engineering geometry. The illustration brief and original files retain their provenance. The schematic explains a LIM comparison; propulsion, speed, powered length, braking and headway remain candidate-specific.

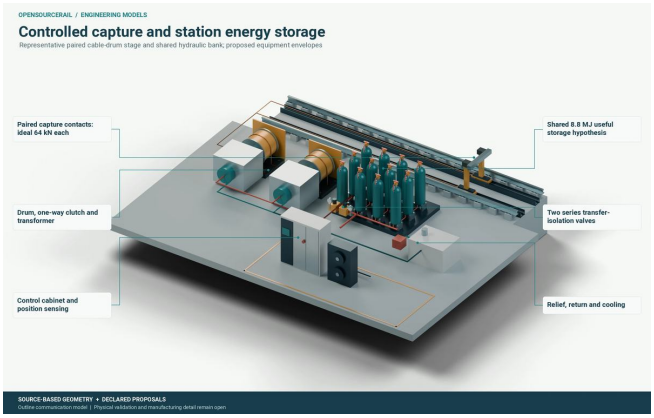
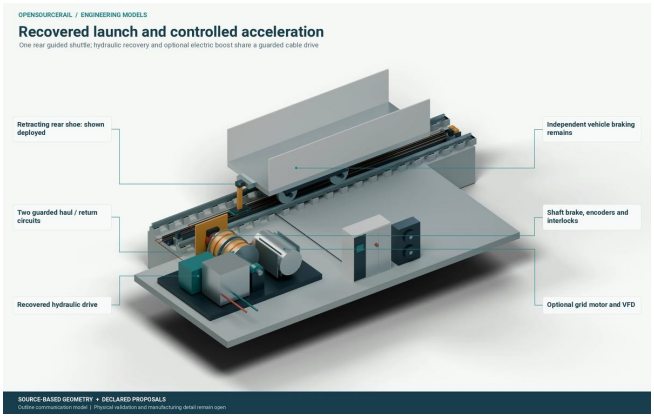
See how it is built and propelled

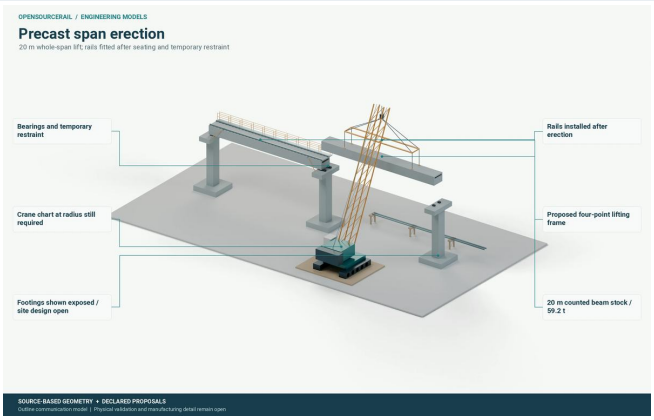
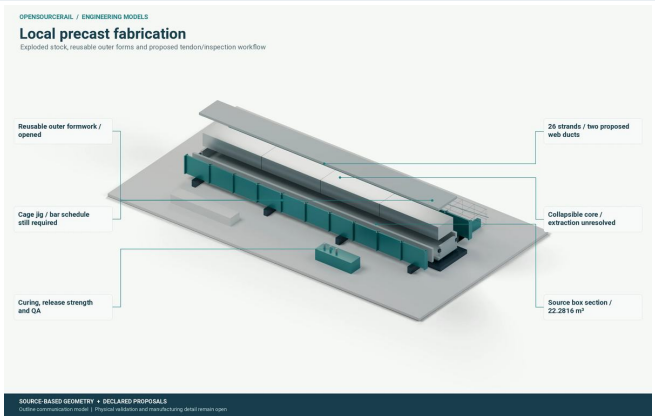
The 3D engineering gallery connects the mechanical layout to construction and fabrication. Each image comes from explicit geometry, with rotatable STL, coloured GLB, editable FreeCAD and STEP models, source identities and export checks.

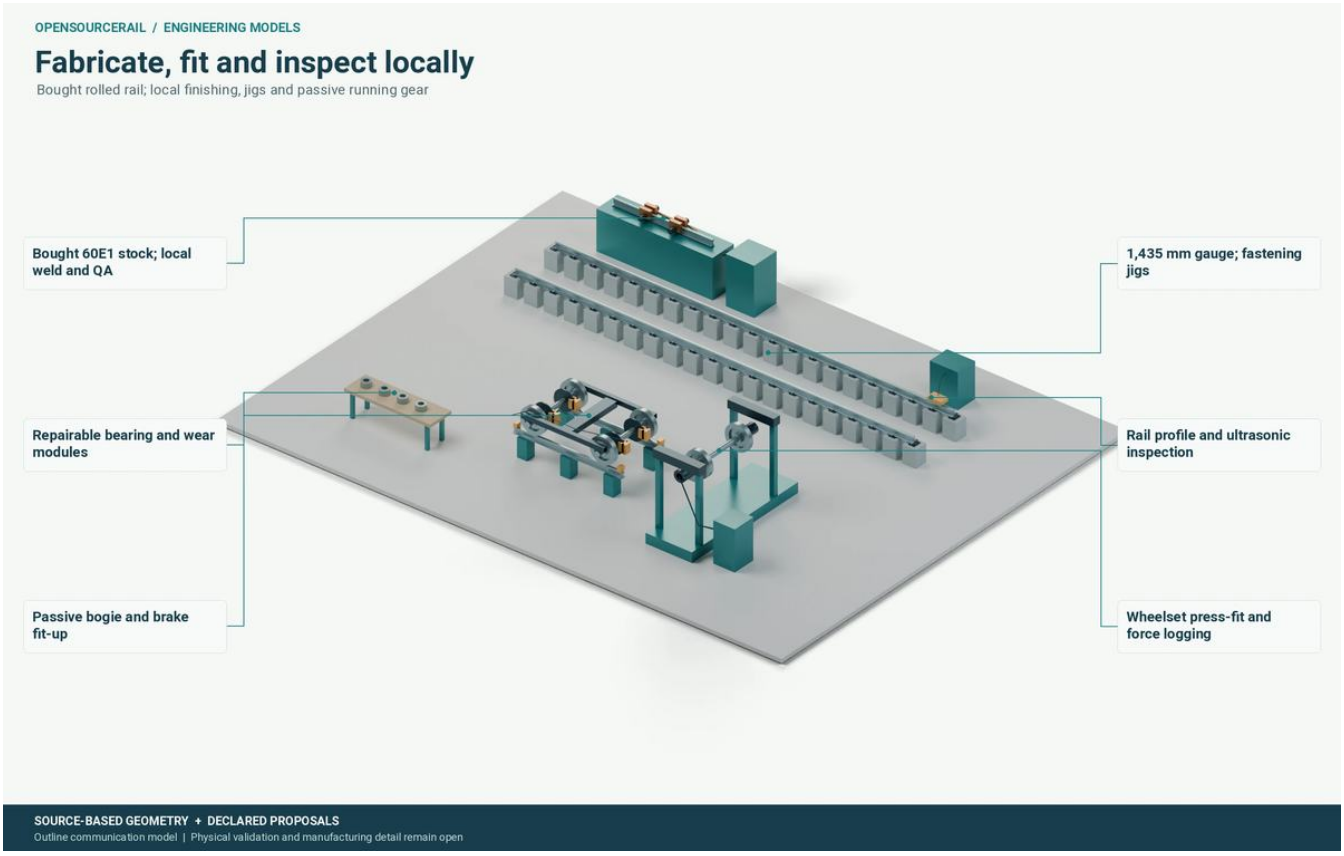


Exploded 3D assembly of the source 17 m vehicle, with roof, seating, doors, passive bogies and proposed shop fixtures

The parametric vehicle model preserves all 2,482 located source parts. Render-only separation reveals the assemblies; downloadable CAD retains source positions. Located stock is distinct from complete installed mass and physical qualification.

Capture braking energy	Launch and accelerate under control
	
Paired cable arrestor drums, hydraulic vessels, isolation, cooling and controls	Guided launch shuttle, coupling interface and optional capstan booster drive
Progressive cable engagement feeds a proposed station-side hydraulic store. Independent train brakes remain.	A controlled rear shuttle returns stored energy; an optional motor supplies further acceleration through the same capstan.
Erect repeatable viaduct spans	Manufacture precast stock locally

Capture braking energy	Launch and accelerate under control
	
Crawler crane, spreader and temporary restraints placing a source 20 m box beam	Exploded concrete box beam with reusable forms, proposed void former, tendon and reinforcement jigs
Whole-span crane erection and an overhead gantry alternative share the same beam. Site access, lifting ratings and temporary stability govern the choice.	Source concrete quantities are preserved. Mould/core extraction, reinforcement, prestress, curing and inspection need detailed production engineering.



Proposed rail-finishing, wheelset assembly, bearing and passive-bogie inspection workstation

Purchase standard rolled rails, then finish, join, inspect and fasten locally. Assemble and check wheelsets and passive bogies using proposed fixtures. The geometry identifies the process; it does not imply qualified tooling or a local rail-rolling mill.

The complete station layout shows five capture stages, the stationary store, launch/boost travel and a separate shuttle runout. Its illustrative hydraulic study uses a 60 km/h departure target. Speed, headway, capture/storage technology and acceleration remain design variables; this gallery selects neither a mandatory LIM length nor a final propulsion system. Equipment envelopes, source geometry and tooling are labelled separately, and engineering/manufacturing/operating releases remain separate gates.

Passenger interior and architectural concepts complement these source-based models.

Delivery records and operations integrations

The v0.6 workflow links candidate revisions to engineering and delivery records. The Workbench exposes current candidate exports and links to the separately configured business and supervision components:

Capability	Where to inspect it	What remains necessary
City geography and planning	266 city design dossiers, source-verified planning workspace	Current map context and route/service proposals; survey, demand, grades, route fit and service evaluation remain open
Engineering Workbench	`./osr workbench`, current UI, coordinated design experiment	Available physical models, complete equipment and independently confirmed designs
CAD, BIM and quantities	Current CAD/IFC and delivery records, candidate body-stock IFC packages	Complete installed geometry, tolerances, full civil federation and surveyed interfaces
ERPNext and Frappe HR	Business deployment and integration app	Configured legal entities, roles, calendars, masters, reviewed imports and production procedures
Equipment supervision	FUXA/gateway deployment, integration source	Authenticated device bindings, current equipment models and deployment-specific commissioning
Manufacturing and procurement	Reference manufacturing/procurement exports, coupled EBOM/MBOM and assembly records	Supplier configurations, prices, tooling, labour, QA, inspection and release
Maintenance and evidence	ERPNext lifecycle app, FUXA and gateway integration	Real serials, measurements, approved procedures and independent handback
Train protection and supervision	Rust crates, v0.6 control adapter	Physical braking, localisation, timing, hardware and operating assurance

The ERPNext app provides draft project, procurement, stock, workforce, manufacturing, quality and maintenance workflows when deployed and configured. FUXA and the gateway provide observation, trends, alarms and review-linked lifecycle integration when connected to authenticated equipment. Business case closure, an alarm reset or a completed repair cannot approve manufactured hardware or grant movement authority.

The current v0.6 candidate exports carry candidate-bound CAD/IFC, engineering BOM, manufacturing, procurement, assembly, civil and controls records. The manufacturing and assembly exports also carry low/base/high process-hour work packages linked to selected cost-line identities, quantities and source/occurrence records. Hours remain provisional; they are not timed work instructions, crew assignments or qualified production capacity, and broad assembly operations remain unallocated where no direct mapping exists. Unlocated equipment, compatible supplier prices, complete installed quantities and physical acceptance remain open. ERPNext/Frappe HR and FUXA/gateway handoffs use their own configured permissions; generated business records remain disabled drafts until reviewed.

Candidate delivery records follow the work

```
transport requirements + city/site inputs
                    ↓
candidate geometry + materials + installed equipment
                    ↓
physics / interfaces / quantities / manufacturing evidence
                    ↓
EBOM + MBOM → suppliers + assembly + inspection tasks
                    ↓
project schedule + resource plan + local/import cash requirements
                    ↓
reviewed business records + serial installation + maintenance history
```

The current v0.6 coordinator binds the selected candidate through engineering, inventory and delivery drafts. Planning handoffs carry route, stop, service and OD proposals only; they are verified with `./osr plan verify`. Priced outline revisions generate provisional construction programmes and local business-system drafts. Verified physical performance, approved production data and measured operating records remain separate release work.

The vehicle mass and support sensitivity shows the effect of assumed equipment positions without claiming installed mass or axle loads. A known terminal-recharge calculation issue and its pending source-epoch migration are recorded separately.

The repeated-duty thermal sensitivity explores two cooling assumptions using a synthetic 500 m propulsion trace. Its results do not establish actual motor thrust or cooling performance.

From a verified planning handoff, `./osr twin factory-draft compile build/my-plan --output build/factory-plan` creates a provisional six-car factory work package; `./osr twin factory-draft verify build/my-plan build/factory-plan` checks its source and candidate identities. The draft carries modelled production hours and low/base/high cost estimates under the declared two 8-hour production shifts. It does not release procurement or manufacture, and it does not complete site civil schedules or operating acceptance.

Manufacturing, civil works and affordability

Cost a design from its parts and production process

OpenSourceRail calculates provisional prices for panels, chassis members, bogie parts, electrical equipment, LIM banks, precast beams and installation activities even when no finished product or supplier quotation exists. The candidate-bound cost model connects material and supplier evidence to manufacturing, civil, delivery, operations and lifecycle estimates. Each scope records its inputs, assumptions, uncertainty and exclusions.

A simple first estimate is (gross material quantity × material rate + direct labour hours × hourly wage) × 1.20. The 20% is a declared illustrative allowance for a quick fabrication screen, not an approved market margin. A fuller estimate adds separately counted machine hours, energy, mould/tooling amortisation, consumables, inspection, scrap and rework, with any overhead applied once to a stated cost base. Delivery, cranes/erection, foundations, wiring/connection and testing are then added to obtain installed cost; fleet, staffing, electricity, maintenance, replacements and downtime form the whole-life model.

For example, a GFRP panel should be priced from ply areas, glass/resin/core quantities, inserts and fasteners, mould preparation, layup/infusion, curing, trimming, assembly and inspection hours. A 20 m precast span should include concrete, reinforcement, prestress, mould cycles, curing, crew, transport, lifting, bearings, rail equipment, piers and foundations. Changing the part shape, material, section or method should regenerate both its cost and its engineering checks.

Cost records need explicit low/base/high scenarios, quantities, source dates, currency, locality, labour and yield assumptions, and one of three statuses: model-cost complete (internally estimated), procurement ready (reviewed real purchasing basis), or physically accepted (separate engineering and safety evidence). A Class C estimate is sufficient for provisional economic ranking, not for a supplier commitment, a certified design or passenger operation. Missing parts and costs are never silently valued at zero. See the detailed implementation and regression plan.

The manufacturing objective is a repeatable system a local builder can fabricate, assemble, inspect and repair. The design search compares frames, sandwich panels, honeycombs, material arrangements, joints, inserts, floors, roofs and integrated service supports. Joining, fire/thermal behaviour, tooling, access and replacement matter alongside stock mass.

Breadth-first exploration retains different physical families before detailed refinement. The body portfolio compares 54 stock compositions; the 20 m civil competition generates 18 box, ribbed, varying-depth, folded and hybrid geometries with provisional manufacturing and delivery costs. The material-distribution study generates cut-steel web layouts through filtered topology optimization, then checks their explicit binary stock on two meshes and propagates manufacture, construction and renewal costs through the same corridor scenario. Its explicit revisions connect or remove disconnected stock while preserving every parent and failed attempt. All twelve revisions have unresolved mesh sensitivity; large calculated displacements also require a more suitable physical model. A separate flanged-girder study compares three steel-girder and concrete-deck configurations using a preliminary beam model and scoped per-span costs. Its same-service system scenarios use common rates and cover fourteen provisional cost scopes. The reference beam changes with each mass case; separate matched-renewal research compares bounded alternatives without selecting a railway. An implicit-geometry experiment generates a three-dimensional cellular volume and reproducible voxel export; its use in a complete civil design remains open. These research models do not establish structural strength, buckling, fatigue or installed feasibility.

The Workbench's Design Portfolio lets you inspect physical alternatives, scoped costs, assumptions and remaining engineering tasks. Separate quantity and conditional-mechanics archives preserve diverse next experiments without selecting a replacement railway. Optional pymoo/pyribs adapters independently check existing archives; the topology study uses its own bounded numerical method.

Civil alternatives must include rails and fastenings, beam stiffness and torsion, bearings, piers, foundations, containment, drainage, emergency access, transport and erection. Common precast parts and offsite inspection should reduce site operations; minimum concrete volume alone does not identify the least-cost structure.

The economic comparison covers rolling stock + guideway + stations and power + factories and construction + energy + maintenance and renewals + service availability. Passenger capacity, headway, fleet size and travel time belong in the same comparison. Local/foreign-currency procurement and financing sensitivity matter, but financing terms are separate from physical resource cost.

The reference vehicle retains a calculated factory/assembly estimate. Active railway budgets come from explicit candidate studies: geometry and service drive material, labour, production, civil, energy, maintenance and thirty-year cost calculations. Supplier quotations improve confidence; their absence does not block a transparent parametric outline. Earlier fixed-service system calculations remain comparison controls and are not the active hardware or city budget.

Current engineering and the next working-system milestone

The priced city and design outlines cover 266 mapped city networks and 1565 lines, plus 112 railway bundles, 54 bodies and 18 civil spans. The Workbench compares original, straighter and rounded square-ring proposals, joins close stop markers into operational complexes, and exposes fleet, manufacturing, construction and fourteen whole-life cost scopes. The local workflow edits and re-costs an outline, then exports a resource-leveled delivery plan and disabled business-system drafts. These are completed calculated outlines under explicit demand, site and rate assumptions; physical approval remains separate.

The city design dossiers show each city's map, network, service, fleet, energy, civil/factory resources, costs and explicit finance scenarios directly in its README. Baghdad and Samawah provide examples. The city railway atlas connects those calculations to selectable routes, station/platform and access layers, journey inspection, line capacity and energy, civil quantities, full cost ledgers and explicit fare scenarios. Completed recalculations replace the map and its figures together; geographic context and every displayed study remain source-bound.

The demand-led outline workflow connects a passenger brief to conditional service screening, complete provisional costs, production/resource quantities and an ‘outline accepted’ result. The Workbench explores service choices and three station-launch concepts under selectable pricing bases. Its illustrative 4,000-passenger/hour/direction project-volume case uses a 200-second service and costs a central US$15.03m/km; this differs from the higher-capacity benchmark below. Supplier quotations and prototype validation remain later evidence requirements. Other approaches remain open under the transport-goal policy.

The 30 km / 12 station cost benchmark preserves the original generic-rate comparison for equal-service design analysis: a double-track elevated line, 72 km/h, 1.0 m/s², 120-second peak headway and 276 places/train. Its Class C central estimate is USD 738 million, or USD 24.6 million per route-km, including 52 six-car trains, stations, civil works, compact launch/power equipment, depot, production/tooling, testing and contingency. The versioned criteria and analysis selector retain the assumptions, full ledger and broad sensitivities. Cable launch is a provisional budget anchor; the selected hardware, engineering acceptance and physical release remain separate. Future comparisons must preserve transport requirements and use the same rate basis; different formations, seating, bogies and propulsion arrangements need their own complete quantities and service calculations.

The project-volume supply comparison reprices those same 30 km, 12 stations and 52 trains using Chinese wholesale/manufacturer observations, landed costs and calculated manufacturing alternatives. Its current Class C central installed estimate is US$548.1m, or US$18.27m per route-km. Fixed factories, shared premises, freight, QA, duties and future replacement setup are included once. Lower purchasing prices do not change physical acceptance or establish a globally best transport architecture.

The current dossier records the selected candidate and its evidence. The coordinated railway campaign passes one candidate identity through geometry, mass, analysis, delivery and costing. A research scenario may use explicit assumptions to complete a calculation; physical validation and release remain separate gates.

Workstream	Current capability	Open evidence
Candidate design	Composable body and system alternatives with ancestry and candidate-bound downstream evaluations	Broader validated evaluators and independently confirmed Pareto selections
Installed inventory	Component identities, geometry-linked stock and explicit mass reserves	Located equipment, complete installed mass, centre of gravity, inertia and interfaces
Propulsion and journey	Candidate-specific LIM, braking, electrical and conditional journey calculations	Available motor force and thermal maps, validated stopping and a physically supported protected journey
Production and civil works	Candidate CAD/IFC, BOM, process, assembly, procurement and provisional construction quantities	Complete installation scope, site data, qualified processes and supplier evidence
Cost and service	Research-bundle Class C scenarios plus selected-vehicle and synthetic-corridor Class C estimates	City-specific site quantities, compatible prices, operating performance and independently reviewed equal-service comparison
Business and supervision	Candidate-linked draft ERPNext/Frappe HR and FUXA/gateway handoffs	Configured organisations, reviewed records, authenticated devices and commissioned procedures
Verification	Source-bound publications, independent numerical checks and software workflows	Prototype correlation and physical engineering, manufacturing and operating acceptance

Current cost scenarios use two 8-hour production shifts and two 8-hour civil shifts. Operating costs cover two shifts for 16/18-hour service and a three-shift 24-hour case. These staffing and shift premiums are planning assumptions, not an approved local roster.

The selected candidate now has internally calculated factory-to-lifecycle cost coverage for one declared synthetic corridor, with visible assumptions and separate physical-validation holds. The next economic work is to replace generic account allowances with supported account-specific estimates and to replace synthetic corridor and site proxies with candidate-specific survey, service and quantity inputs. The software labels calculated, benchmarked and supplier-confirmed values separately.

Cities, service and deployment

All 266 city geographies remain available as planning inputs. The workspace uses indexed routes, station and transfer locations, plus source-verified road, water, building, land-use and place context where supplied. It supports local route, stop, service and origin–destination proposals. `./osr plan compile edits.json --output build/my-plan` creates an immutable handoff bound to the selected v0.6 candidate; `./osr plan verify build/my-plan` checks its candidate, source and payload identities. The handoff contains proposals only. It excludes vehicle, propulsion, fleet, civil quantity, cost, performance and cashflow results; those require candidate-specific evaluation. Planning revisions do not grant movement authority.

Each corridor still needs its own demand, terrain, climate, geotechnical, utility, accessibility, transfer, power, rescue and procurement assessment. A common vehicle and precast family can reduce manufacturing complexity without proving that every city needs the same formation or service pattern.

Run and reproduce

```
./install.sh
./osr status
./osr readiness
./osr workbench
./osr plan --help
./osr twin export
./osr design railway plan
./osr design railway run --max-bundles 2 --native-jobs 1
./osr design railway publish
./osr design railway verify
./osr build
./osr verify
./osr book --check
./osr test
./osr sim baghdad
```

The Workbench shows the factory draft inputs and open holds. Without a verified planning handoff, its compiler reports a hold and creates no package.

The coordinated run is bounded and restartable; it preserves pending work and failed executions. Native FEM/CAD workers need the relevant installed tools and declared resources. The old `./osr build` dossier path is held at its source-binding mismatch and must be regenerated only with compatible source evidence. `./osr book` builds the maintained platform reader independently. The local review UI has no live train-control authority.

Find your way

I want to…	Start here
Understand the whole project	Overview, architecture, first adoption path
Review the selected train and open engineering	Current dossier, implementation, roadmap
Generate and compare new designs	Coupled experiment, design engine
Inspect city geography or planning proposals	Current city register, city catalogue, planning UI
Review CAD, BIM, BOM and manufacturing	Reference exports, candidate delivery packages
Follow the end-to-end workflow and calculated costing	Workflow above, cost formula above, implementation plan
Inspect current candidate delivery exports	Current export manifest, candidate delivery packages, `./osr twin export`
Configure business and people workflows	ERPNext/Frappe HR deployment
Inspect supervision and maintenance integration	FUXA/gateway deployment, ERPNext lifecycle app
Work on controls or electronics	Rust packages, v0.6 control interfaces
Set up tools or contribute	Tools, CONTRIBUTING, GOVERNANCE, project instructions

About the author

Hayder Aziz initiated OpenSourceRail. His published work on engineering design systems dates back more than 20 years. Beginning in the early 2000s, he researched how product data management, manufacturing knowledge, open standards and collaborative software could be joined into a coherent design process, enabling engineering teams to assess manufacturability and exchange design knowledge from the conceptual stage rather than treating design and production as separate activities.

Among his research publications are New Technologies and Standards for Design Knowledge Management in Collaborative Conceptual Design (2003, with James Gao, Paul Maropoulos and Wai Ming Cheung), Open standard, open source and peer-to-peer tools and methods for collaborative product development (2005, Computers in Industry), and Engineering design knowledge management in de-centralised virtual enterprises (2008, Computer-Aided Design and Applications). These works investigated reusable engineering knowledge, distributed product-development systems and the integration and automation of design-related workflows.

OpenSourceRail carries that research direction into an ambitious open transport-system project. The aim is to connect requirements, alternative generation, material selection, CAD/BIM, weight and cost calculations, supplier evidence, assembly methods, operational simulation and safety analysis in a traceable toolchain. This is the project's design philosophy and research ambition; the current railway candidates remain in development, without completed physical validation or certification for passenger operation.

Sources, revisions and contributions

Concern	Source to change
Selected v0.6 requirements and assembly	Development selector, station-rail inputs and generators
Functional remit, design grammar and campaign budgets	Design-search source, campaign definitions
City inputs and geographic proposals	Each city's `city.json` under cities/catalogue/; current service outputs remain held
Controls and simulation behaviour	Rust crates, their tests and candidate-specific physical adapters
Business and equipment integration	ERPNext app, supervision deployment, integration service
System boundaries and remaining work	Architecture, roadmap, engineering instructions

Regenerate derived artifacts from their current sources. Physical candidate, evaluation and selection identities are distinct; a source hash identifies bytes and does not approve engineering. `./osr verify` checks the current source-bound publication, and `./osr twin export` verifies the selected candidate's delivery exports. Neither command grants engineering, manufacturing or operating acceptance.

Contributions can improve supplier/material evidence, structural and motor models, complete equipment, manufacturing and repair methods, city data, controls, tests and documentation. Follow CONTRIBUTING.md and GOVERNANCE.md. The documentation index is a search aid; it is not another engineering authority.

Software uses Apache 2.0, open physical designs use CERN-OHL-S v2, and documentation uses CC-BY-SA 4.0, subject to more-specific and upstream notices. See LICENSE.md and LICENSES/.

1.3 Architecture and evidence boundaries

docs/ARCHITECTURE.md

OpenSourceRail connects three products: a design and delivery platform, a reference railway and manufacturing system, and open GoA4 control software. The delivery lifecycle covers planning, design, procurement, manufacture, construction, commissioning, operation, maintenance and renewal. Local design authority, repairability and whole-life affordability are objectives that each candidate must demonstrate.

Selected v0.6 vehicle reference

`development-baseline.toml` selects six 17 m comfort cars with 44 fixed seats and two wheelchair places per car, no planned standing. Passive steel-wheel bogies run on conventional rails carried by concrete. The vehicle retains a passive aluminium secondary interface; this does not select an installed propulsion system. The reference vehicles have no normal-service traction motors, traction batteries or permanent magnets. Independent braking, service power, protection and rescue provisions remain required.

The selector's `propulsion_design` policy requires candidate-specific selection of architecture, installed powered length, acceleration, braking and headway. Neither 900 m nor 100 m is a prescribed installation. The 72 km/h launch cap, 120-second headway and 35-second dwell form a named retained service control, rather than selected operating requirements.

The selected candidate is a development reference. It has not been physically qualified or released for manufacture or passenger operation. The 8,280 passengers/hour/direction figure is capacity arithmetic, not a demonstrated service. A reference records the purpose of selection; an alternative records a comparison role. An evaluated result names the calculation and evidence scope. Prototype validation requires its own physical evidence and remains undemonstrated.

Connected platform

The planning workspace verifies current city route and map-context sources, displays the available geographic layers, and supports route, stop, service and OD editing. Run `./osr plan compile edits.json --output build/my-plan` to create an immutable proposal handoff bound to the selected candidate, then verify it with `./osr plan verify build/my-plan`. The handoff carries proposals only; it excludes vehicle, propulsion, fleet, civil quantity, cost, performance and cashflow results. It grants no movement authority.

The coordinator carries one candidate bundle through geometry, mass, analysis, BOM, CAD/IFC, procurement, assembly, construction and controls. The current engineering exports and coupled delivery index provide candidate-linked research records. They do not imply complete installed equipment, qualified manufacturing instructions or approved business records.

ERPNext and Frappe HR provide draft business, workforce, procurement, production, QA and maintenance workflows. FUXA and the gateway support observation, alarm, history and review records. Company and site masters, permissions, authenticated device bindings and reviewed imports are deployment inputs. Neither integration supplies movement authority or accepts engineering evidence automatically.

Identity and evidence boundaries

The physical candidate identity describes the vehicle, support and propulsion configuration. Evaluation identity binds models, sources and scenarios. Selection identity records which candidate is used for a stated purpose. A different model hypothesis is a different evaluation; a physical revision gets a new candidate identity. Mixed identities fail closed, and downstream results must be regenerated when upstream evidence changes.

Unknown installed mass, force, heat, cost, origin or qualification remains unknown or bounded. Calculated assumptions can complete a research scenario but cannot establish a physical result. Software execution, numerical validity, engineering screening, evidence maturity, selection purpose and physical acceptance remain separate records.

Coupled engineering domains

Domain	Required coupling
Vehicle	Complete fitout, bogies, service supply, thermal equipment, load paths, mass and inertia
Propulsion candidate	Mechanism-specific force, power, losses, thermal duty, interfaces and handoff; electrical and magnetic maps where applicable
Braking and journey	Same candidate mass and geometry, independent brakes, service energy, station power and protection
Rail and civil	Moving axle reactions, rails and fasteners, alignment, deflection, torsion, piers, foundations and erection
Delivery	Candidate-matched BOM, manufacturing processes, procurement, assembly, CAD/IFC and maintenance records
Economics	Equal passenger service, routes, reliability, design life, installed scope and price uncertainty

The held v0.6 reference dossier and its candidate twin exports retain their source identities. The platform reader summarises the published software and current v0.7 work without treating that dossier as supported release evidence. The CLI and Workbench show source holds. Candidate-specific city performance publication remains held until propulsion, braking, route, civil and economic evidence supports it. Physical qualification and independent review remain open.

1.4 Capability register

docs/capabilities/README.md

OpenSourceRail connects three products: an open railway design and engineering system, a manufacturing/delivery and maintenance platform, and city planning and service tools. The aim is affordable whole-life transport supported by local design, manufacture and repair knowledge. Planning, engineering, business and equipment-observation records have distinct identities and review gates.

Reference and research configurations

The development baseline remains a six-car, wheel-supported passive reference: six 17 m cars, 44 fixed seats and two wheelchair places per car, with independently sized service batteries and brakes. It has no selected onboard traction installation. Speed, headway, propulsion architecture and powered length are candidate decisions; neither a 900 m LIM nor the historical 120-second schedule is mandatory.

The reconciled research pilot and no-store control are newer, separately identified powered-wheel research candidates. They join counted body parts, explicit external drives and reserves, axle loads, civil geometry, concurrent-train simulations, station storage alternatives and fourteen-scope cost ledgers. Several declared nominal timetables complete 16/16 model arrivals. This is conditional simulation evidence; it neither changes the passive reference nor selects an installed architecture for physical deployment.

Body mass subtotals, including a provisional installation reserve, remain separate from external motors, gearboxes, drive BOP and recovery fittings. Whole-train mass and cost must be read from the matching integrated evaluation. Proposed component positions, CAD volume comparisons and solver agreement do not establish measured installed mass, a collision-free assembly or qualified hardware. Native results apply only to their exact source, load, mesh and candidate; the no-store alternative does not inherit recovery-candidate FEM.

Available workflows

Capability	Current entry point and boundary
Coordinated candidate engineering	`./osr design railway`; preserves component/evaluation identities and open evidence gates
Reconciled research assembly and service	Recovery pilot and no-store control; conditional model-level arrivals, economics and fault cases
Native CAD and structural comparison	SolidCraft/FieldCraft pilot, viaduct alternatives and reconciled pilot; independent FreeCAD/CalculiX comparisons where recorded
Counted vehicle manufacturing definition	Vehicle parts catalogue and pilot installed-body records; materials, processes and stock quantities with detailed-drawing holds
Candidate-bound pilot manufacturing and installation proposals	Delivery compiler; local draft parts, source assembly routes, inspection holds, external ownership and existing cost-scope references
Selected reference mass account	Mass reconciliation and roof-finish distribution estimate; distinct from powered-wheel research mass
Reference CAD/IFC, BOM, procurement, assembly and control records	Current export manifest and delivery package index
City geography, maps and route/service proposals	Planning guide, workspace; `./osr plan compile` / `verify`; route-specific engineering remains separate
Selected-candidate factory work package	Current Workbench; `./osr twin factory-draft compile` / `verify`; planning draft from the selected Class C ledger
Calculated manufacturing and lifecycle scenarios	Selected 6 km system estimate, vehicle estimate, cost and delivery method, and separate pilot ledgers
Business and workforce workflows	ERPNext/Frappe HR; retained integration contracts and native business-platform acceptance scenario, with reviewed conversion required for the new pilot
Equipment observation and maintenance review	FUXA/gateway; observations and reviewed maintenance handoffs, without safety movement authority
Open control software	Rust workspace; software tests remain distinct from physical qualification

The pilot delivery compiler preserves part definitions, engineering revisions, occurrence patterns and future serial/lot allocation separately. Its ready-for-planning-review tasks support material/route review and preparation of supplier information requests. Production, installation, procurement release and ERP conversion remain held for actual controlled drawings, reviewed native Item mappings, inspection references and engineering evidence. No accepted serial installation or approved work order is generated. Its process-hour ranges are declared capacity hypotheses, not measured cycle times or a second labour charge on existing direct prices.

The retained factory-draft command creates a six-car work package from a verified planning handoff and the selected candidate's Class C ledger, under two 8-hour production shifts. Its schedule represents one serialized lane and does not infer crews, parallel capacity or dates. It is not an ERPNext import, purchase order or manufacturing release. The planning handoff itself contains route, stop, service and OD proposals; vehicle, propulsion, fleet, civil, financial and performance evaluations remain separate candidate-specific records.

The selected reference ledger retains a 10.090 t/car reference or unlocated residual. Its separate roof-finish estimate distributes already-accounted 47.809 kg/car over 45.05 m² of source surface; it adds no mass or CAD occurrence. Those reference-scope figures must not be combined with research-pilot totals.

Preserved platform and open evidence

The v0.4 operating/manufacturing release and v0.5.1 maglev research remain distinct historical foundations. ERPNext/Frappe HR, FUXA/gateway, revision-controlled assets, city planning, workforce and maintenance are retained platform capabilities, not replaced by a new mechanical candidate. The source-pinned historical manifest records preserved engineering evidence. Historical results do not qualify a new wheel-supported design.

Calculated outlines can be complete under stated assumptions. Installed interfaces, actual force/duty and thermal behaviour, independent braking, structural joints/fatigue, site-specific works, controlled production drawings, physical prototypes and independent acceptance remain open gates. The new pilot's drafts do not yet constitute a seamless approved ERPNext project or an operational commissioning workflow. Software functionality grants no manufacturing release, physical acceptance or movement authority.

II. Published software and adoption

What v0.6.1 ships, what remains open and how a user starts.

2.1 Published v0.6.1 release

docs/releases/v0.6.1/README.md

Published on 12 November 2026: v0.6.1 release. The immutable tag identifies commit `5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea` and tree `58ba4411d502a166b3a14000bc1e09b733b65cbd`.

Start with the published INSTALL-v0.6.1.md. Download the release assets and verify them against SHA256SUMS. The quickstart follows that tagged installation route; native CAD/FEM/GIS and business-system services are optional for the Samawah demo.

Document	Purpose
Quickstart	Install the published tag and run the Samawah journey
Demonstrations	Samawah, Baghdad and candidate research walkthroughs
Feature maturity	Available, preview, held and planned functions
Release notes	Published scope and download links
Validation	Exact-tag CI evidence and clean-install handover procedure
Known limitations	Current dossier, native GIS and engineering boundaries

The software release includes planning, engineering research and delivery tools. Its Samawah journey calculates and exports a fresh case, reuses the identical request, and gives changed inputs a distinct evaluation. Baghdad and candidate examples use retained source-pinned packets. The current integrated dossier and portfolio selector remain visibly held and excluded from supported results; the native GIS round trip is v0.7 work.

The older design-closure campaign is superseded where it controlled software publication. Its seven open/partial engineering findings remain in the engineering ledger and feed the v0.7 roadmap. The retained authenticated-review profile keeps its truthful pending records; it does not describe whether v0.6.1 is published, and no passing review was fabricated.

The earlier annotated `v0.6.0` development tag remains at `61080cafc6998e587772810707dc552524df5306`; its historical preparation documents are retained. Neither release tag is moved by this handover. Physical, manufacturing, movement and operating acceptance remain separate from software publication.

The published installation verification records the completed clean installation and two Samawah journeys against the tag.

2.2 Installation and Samawah demonstration

docs/releases/v0.6.1/quickstart.md

v0.6.1 is published. The authoritative release installation asset is INSTALL-v0.6.1.md. Use the attached SHA256SUMS to verify downloads before installation. This guide uses the same Linux route. Native CAD, GIS, FEA, SUMO and business-system services are optional for the Samawah software demonstration.

Get the tagged software

```
git clone --filter=blob:none --sparse --branch v0.6.1 \
  https://github.com/modernecotech/OpenSourceRail.git
cd OpenSourceRail
test "$(git rev-parse HEAD)" = 5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea
test "$(git rev-parse 'HEAD^{tree}')" = 58ba4411d502a166b3a14000bc1e09b733b65cbd
git sparse-checkout set --no-cone \
  /.github/release-smoke-tests.txt \
  /tools/automation/prepare-v06-release-checkout.py \
  /docs/design-engine/outline-completion/manifest.json \
  /docs/design-engine/city-atlas/manifest.json \
  /docs/design-engine/coupled-railway/manifest.json \
  /docs/workbench/city-atlas-assets.json \
  /docs/design-engine/v06-design-closure/compact-drive-source-replay-01/compact-737-h100/body.json.gz
python3 tools/automation/prepare-v06-release-checkout.py
```

The helper expands the clone to the selected release sources, the outline and city-atlas inputs, Samawah's data, the browser verifier and the source paths needed by the pinned Baghdad and candidate examples. It omits the unrelated source paths in the held global current-dossier manifest, optional native solvers and unrelated research archives. Git fetches required blobs on demand; the helper reports the allocated checkout size and remaining free disk.

The global current dossier remains explicitly held at this reviewed source epoch. The Workbench reports that hold and does not use its stale global costs or portfolios. It is excluded from supported v0.6 results and release evidence until its exact source provenance is resolved. The Samawah calculation has a separate source and evaluation identity and does not require a full 266-city publication refresh.

Install the small software-demo environment

```
python3 -m venv .venv
.venv/bin/python -m pip install pytest==9.2.0 scipy==1.16.3
npm ci
npx playwright install chromium
```

The recorded local profile used Python 3.13.16, SciPy 1.16.3, Node 24.21.0, npm 11.19.0 and Playwright 1.62.1 from the lockfile. Pytest is needed for the release smoke tests; the Workbench itself only needs the Python numerical dependency shown above. Install optional native tools separately only when running their declared methods. See the broader project setup guide.

Run the browser demonstration

With the project environment and browser dependencies installed, use a new output directory outside the checkout. Run the script twice as a complete demonstration, including an identical warm request inside each run:

```
run_root="$(mktemp -d /tmp/osr-v061-demo.XXXXXX)"
node tools/automation/tests/verify-v06-software-journey-a.mjs \
  "$run_root/journey-a-1"
node tools/automation/tests/verify-v06-software-journey-a.mjs \
  "$run_root/journey-a-2"
```

Each run starts a loopback Workbench, calculates the default Samawah case, repeats the same request to check cache reuse, changes peak demand, and verifies that the changed request gets a distinct identity. It writes the report and exports to the supplied directory. It runs no native solver. It requires at least 4 GiB free at job admission and caps each output package at 64 MiB.

Inspect each report's source commit, input and evaluation IDs, 14 cost scopes, resource measurements, cache result and independent replay before using the example in another study. Compare cold and warm wall time, CPU time, peak RSS, cache-store bytes and filesystem free space between runs. The script demonstrates software behavior; it does not validate route survey, physical railway design or cost quotations.

Open the precomputed examples

- Baghdad: open the city atlas, select Baghdad and inspect the published route/service comparison and its source/evaluation identities. Treat it as a read-only, precomputed planning example; no new Baghdad calculation is required for this release. The source-bound result download is available; native RustGIS project export is held by a publication-epoch mismatch.
- Candidate research: the Workbench shows the selected passive-wheel development reference, but its portfolio selector is held at this source commit by a stale current-dossier source binding. Inspect the precomputed coupled-railway packet for separate candidate IDs and read its limits. This packet is verified in its recorded historical source epoch; the current railway verifier holds when that epoch no longer matches. Run `python -m engineering.station_rail.retired_station_reference verify` to check the retained source-pinned packet set. This check reads original Git blobs from the historical source epoch, which the sparse evaluation checkout does not fetch. Run it from a clean full-history checkout of the same commit. This separate historical byte check does not replay the campaign or admit old results as current evaluations or physical evidence. A research alternative does not replace the reference.

The Workbench links to source packets and export manifests. An output marked calculated or replay-verified is digital evidence, not physical acceptance.

2.3 Feature maturity

docs/releases/v0.6.1/feature-maturity.md

Maturity	Capability	Release meaning
Available	City atlas, mapped outlines and planning proposals	Geography and proposals are inspectable; survey and route acceptance remain open
Available	Workbench, source-bound result views and export manifests	Supported digital workflows expose identity, status and holds
Available	Samawah bounded city study	Published exact-tag evidence contains two complete runs: fresh case, identical warm replay, changed-input case and exports
Available	Fourteen-scope cost and delivery drafts	Calculated assumptions and uncertainty are visible; unquoted prices remain provisional
Available	Rust protection/control research and deterministic simulations	Research and refusal checks only; no certified live control
Preview	Baghdad atlas comparison	Published route, service and cost evidence is read-only; native RustGIS export is held because GIS and outline publication epochs differ
Preview	Candidate comparison	Source-pinned alternatives are in the coupled-railway archive; the current Workbench portfolio selector is hidden by its source-binding hold
Preview	ERPNext/Frappe HR and FUXA/gateway adapters	Configurable drafts and integration examples; no live purchasing or railway control
Preview	CAD, IFC, FEM and native solver interfaces	Supported examples vary by tool and model domain; physical qualification is open
Held	Native RustGIS project export from the current city atlas	GIS and outline publication epochs differ at this source commit; no GIS round trip is claimed in v0.6
Excluded (visible hold)	Integrated `docs/design-engine/current/` dossier	Its source binding is unresolved; stale global costs and portfolio results are outside the supported v0.6 release scope
Unsupported	Physical acceptance, manufacturing release and movement/operating authority	No v0.6 software output confers these states
Planned	Full city-specific surveys, installed vehicle closure and controlled drawings	v0.7+ engineering work; tracked by the seven-gate design-closure ledger
Planned	Repository-wide artifact store, history reduction and global compute/disk quotas	Separate storage modernization programme; no release history rewrite is authorized

“Available” describes software behavior, not engineering acceptance. Review the known limitations and the source packet before relying on any calculated result.

The integrated current dossier remains in the repository so its hold can be inspected, but it is excluded from supported v0.6 results and release evidence until its exact source provenance is resolved.

2.4 Known limitations

docs/releases/v0.6.1/known-limitations.md

- The passive-wheel development reference remains a design reference, not a physically accepted installation. The installed propulsion architecture is not selected.
- Vehicle mass, equipment fit, fasteners, joints, cooling, wiring, clearances, fatigue and removal access have unresolved bounds or holds.
- Propulsion force, current, heat, braking and rescue evidence is candidate- and model-domain-specific. Numerical convergence does not establish feasibility.
- City demand, route geometry, gradients, ground, utilities, land and access may be generic or unsurveyed. Cost ranges are not vendor quotations or approved finance.
- The 266-city catalogue is a planning portfolio. It is not 266 independently surveyed and engineered railway designs.
- The integrated current engineering dossier currently has a source-binding hold. The Workbench shows the selected reference and city geography, but keeps global engineering cost and portfolio views held and does not treat the stale dossier as current evidence. The dossier is excluded from supported v0.6 results and release evidence until its exact provenance is resolved. At this source commit, the portfolio hold identifies `.github/workflows/development-baseline.yml` as changed from the dossier's recorded source hash. The Workbench therefore hides its candidate-portfolio selector; use the source-pinned archive described in the demonstrations document for read-only research comparison. The Samawah software journey binds and verifies its own city inputs and calculation.
- The city atlas currently refuses native RustGIS ZIP export because the published GIS and outline epochs differ. The source-bound Baghdad result and its read-only atlas view remain available; a native GIS export/import round trip is not included in v0.6.
- CAD/IFC stock agreement and structural screens do not establish complete installed mass, manufacturability, fatigue life or construction approval.
- ERPNext, Frappe HR, FUXA and gateway workflows are configurable software examples. Real transactions and live railway commands are not authorized by generated drafts.
- Full native CAD/FEM/GIS workflows are optional and limited to declared solver versions, resource requests, inputs and model domains.
- Large historical studies remain in the repository history. Sparse checkout and optional local cleanup reduce active workspace use, but no repository history rewrite or complete artifact migration has been performed.
- Physical prototype validation, independent safety assessment and any manufacturing, movement or operating approval remain separate future work.

2.5 First adoption path

docs/first-adoptable-product.md

OpenSourceRail has three connected products: design and delivery software, a reference railway and manufacturing system, and open GoA4 controls. The current v0.6 front end provides city-planning proposals, calculated city/design outlines, candidate engineering views and delivery records. ERPNext/Frappe HR and FUXA/gateway are separate configurable integrations. Calculated completion is distinct from approved construction and live railway operation.

The v0.6 software-release checklist is tracked separately from engineering closure. Current readiness and its six source-bound software gates are in the software-release ledger.

Who can use it

A public owner can edit geographic, route, stop, service and origin-destination proposals, then hand them to candidate-specific engineering review. A local builder can inspect candidate CAD/IFC, bill of materials, process, assembly and procurement drafts. An organisation can configure ERPNext/Frappe HR and FUXA/gateway workflows for its own project and equipment. Universities and reviewers can reproduce source-bound numerical methods and identify measurements needed for a prototype.

The current v0.6 engineering Workbench displays the selected candidate, source-bound engineering and cost summaries, readiness holds and links to its delivery records. The coordinated experiment carries one candidate identity through geometry, inventory, engineering analysis and delivery drafts.

The planning workspace verifies current city map context and route sources, then supports route, stop, service and origin-destination proposals. Run `./osr plan compile edits.json --output build/my-plan` to create an immutable handoff bound to the selected v0.6 candidate; run `./osr plan verify build/my-plan` to verify its source files and identities. The handoff contains proposals only. It excludes vehicle, propulsion, fleet, civil quantity, cost, performance and cashflow results; those require candidate-specific evaluation. Planning revisions do not grant movement authority.

The priced outline workspace evaluates mapped networks, joined station complexes, straighter corridors and rounded-ring alternatives. It calculates service, fleet, manufacturing, civil quantities, project budgets and thirty-year cost scenarios with visible assumptions. Its completion record distinguishes complete calculated outlines from physical engineering and release. Run the local Workbench with `./osr workbench`, choose a priced city and calculate a scenario. Edit this outline and recalculate opens the exact calculated routes and services in the planning workspace. Stop, geometry and headway edits then recalculate fleet, civil quantities and all cost scopes. A proposal that fails demand or access remains priced with its failed checks visible. The result includes a resource-leveled production/construction programme and disabled ERPNext/Frappe BOM, routing, workstation, work-order, project and staffing drafts. Real calendars, company/warehouse masters and reviewed imports require site configuration; nothing is submitted automatically.

What is available and what needs configuration

Area	Available scope	Deployment or engineering work still needed
Design research	Candidate definitions, source-bound calculations, failed/pending work, comparison and readiness holds	Trustworthy physical evaluators, complete installed equipment and independent confirmation
Manufacturing and delivery records	Candidate CAD/IFC, BOM, provisional process and assembly records, procurement drafts and calculated cost estimates	Exact drawings, joints, tooling, labour, suppliers, inspection, complete installed quantities and production release
ERPNext/Frappe HR	Current OSR app and deployment guide for projects, procurement, workforce, quality and maintenance	Legal entities, calendars, roles, warehouse/item masters, reviewed imports and operating procedures
FUXA/gateway	Integration components for observations, historian, alarms and review-linked lifecycle records	Actual equipment models, authenticated bindings, communications and commissioning
Candidate delivery exports	Candidate-bound CAD/IFC stock, engineering BOM, process and assembly records, and disabled ERPNext drafts	Complete installed geometry, serialised as-built evidence, priced project execution and independent acceptance

The current repository includes ERPNext/Frappe HR deployment, FUXA/gateway deployment, candidate-bound CAD/IFC and delivery exports, and an integration service. These are configurable software components and research handoffs. No v0.6 site has been commissioned, no physical asset has been accepted, and the railway does not have operating release.

A controlled adoption sequence

1 Select a workshop, depot or bounded planning/pilot scope and define its owners and review responsibilities.

2. Use current product/site identities; separate sample records and incomplete data from the proposed deployment.
3. Configure business masters, users, permissions, record retention and recovery through the ERP deployment.
4. Review draft material, process, procurement, assembly and inspection records before creating real business transactions.
5. Configure the supervision gateway with authenticated equipment bindings; keep simulation and physical inputs distinct.
6. Link measured inspections, serial installation, defects, repairs and independent handback to exact product revisions.
7. Keep railway control in simulation or separately authorised development until the physical plant and deployment safety case are accepted.

Business case closure, an alarm reset, a revision hash or a simulated inspection cannot accept hardware, release a railway or authorise movement. The current design runner does not automatically provision ERP sites, import city assets, submit purchasing or commission controllers.

Project introduction · Architecture · Current engineering · Roadmap.

III. City and corridor

A candidate-bound Samawah revision, native GIS and city evidence.

3.1 Samawah project continuity

docs/design-engine/v07-project-continuity/README.md

The first milestone is implemented in the existing Workbench: open the city atlas, select Samawah, and choose Compare Samawah station merge. It recalculates the published network and the midpoint edit using the same explicit-headway evaluator, vehicle inputs and cost policies. The comparison offers both dossiers, the revised delivery draft and a downloadable report.

The two raw markers are 79.976780 m apart. The edit replaces them with one stop at chainage 9,339.173756 m without changing the alignment coordinates. The published example already groups them as one operational complex. The retained compact-drive midpoint study uses another body candidate; its fleet and cost results are not transferred to this comparison.

Measure	Before	Midpoint edit	Change
Raw markers	26	25	−1
Operational station complexes	25	25	0
Route length	57,873.249527 m	57,873.249527 m	0
Total trainsets	81	81	0
Base installed capital	$1,141,490,831.10	$1,141,480,801.60	−$10,029.49
Base whole-life NPV	$2,258,426,041.73	$2,258,655,801.42	+$229,759.68

These are provisional model estimates. Line 1's one-way time increases by 1.494405 seconds; the other four lines are unchanged. All fourteen cost scopes are reported, including zero deltas. The base energy NPV increases by $239,881.94. The existing 79.976780 m access-repair allowance is replaced by a 70.690214 m revised-access hypothesis with its own explicit risk allowance; contingency, design provision and disposal also change. This does not establish a surveyed walking route or a verified local price. The civil quantity identity and vehicle/manufacturing reference inputs remain unchanged.

The report preserves physical candidate, planning revision, city/service/cost evaluation, selection, civil quantity, manifest and delivery identities. Unknown quantities remain unknown. It retains the line-1 coast-reach failure, conditional powered-axle research comparison, missing actual force/braking evidence, transferred cost rates, access/catchment/platform questions and open installed vehicle/manufacturing findings. The passive-wheel reference is not replaced by the research propulsion policy.

Measured execution and verification

The real browser run passed on clean commit `798a2c292acdddde8227c9b9ad49e74bb6e5b561`, comparison `d66f7add2536a0c678ca5bf56f51d9788cedf8ac402f36b7e4456eaf6cbab88e`. Desktop and mobile screenshots were inspected. There were no browser errors, external requests or changes to tracked/untracked source files.

The two numerical studies were produced on `91916d900e` and retained after the browser exposed an integer JSON transport mismatch in the new helper. Correcting that helper did not alter their exact request or evaluator source identities. The successful browser run verified and reused those existing artifacts; its 10.96-second first click is therefore not a cold-calculation measurement. A second unchanged click reused the same verified job identities in 1.62 seconds.

Worker measure	Before	Midpoint edit
Worker elapsed	16.25 s	16.31 s
Evaluation elapsed	13.73 s	13.81 s
Peak RSS	302.33 MiB	419.25 MiB
Artifact payload	24,482,861 bytes	24,477,356 bytes
Native solver / CAD jobs	0 / 0	0 / 0

The compact evidence record contains exact identities, all scope deltas, source hashes, performance, retained failure descriptions and hashes of external raw reports, logs and screenshots. Raw study packages remain outside Git. Identity equality alone is not presented as proof of execution cache reuse; the browser checks stable verified job identities separately.

Focused regression checks passed: 61 tests covering the helper, API cache custody, city jobs and planning adapter. A read-only specialist review reproduced those tests and checked the browser numeric-boundary fix; it was not OS-isolated or a physical design approval. The existing Samawah CI workflow now also runs this browser comparison from a fresh store and uploads its outputs.

Reproduce in an installed development checkout with an external, empty output directory (the normal environment supplies `.venv` and Playwright):

```
node tools/automation/tests/verify-v07-samawah-corridor.mjs /tmp/osr-v07-samawah-new-run
```

`OSR_PYTHON` selects another Python environment. `OSR_CITY_STUDY_STORE` can point to retained external verified jobs; the report records that configuration and does not label it cold execution. Stores inside the source checkout are refused.

Next integrated work

This closes the first comparison milestone, not the v0.7 programme. Subsequent native city cycles, project invalidation, counted door views and performance work are recorded in the execution update. Next bind one installed six-car candidate, representative civil/station assembly and door operator to this project, then complete their detailed interfaces, controlled manufacturing/QA/removal package, native GIS return path, ERPNext/FUXA lifecycle and Baghdad replay. Supplier information and physical tests retain their own open milestones. See the six workstreams.

The local coordinator kept the 20 GiB free-space reserve, removed only its known redundant browser downloads and retained failed and completed study evidence. The worker reports shared-filesystem admission and bounded output monitoring; an aggregate filesystem quota is not enforced.

3.2 Editable city GIS

docs/gis/README.md

OpenSourceRail uses RustGIS 0.5.0 for native city GIS projects and headless map export. The existing interactive city atlas, SVG maps and calculated service, fleet, delivery and cost dossiers remain available. QGIS is not required by any current city generator or by the business-platform artifact reader.

In the local Workbench, select a city and choose Download editable RustGIS project and source tables. Extract the ZIP and open `<city>.rgis` in RustGIS. The download is bound to the exact displayed city packet. A verified local planning study exports its own geometry and source base, so repeated native editing does not revert to the published network.

After editing and saving, choose Import native GIS edits and recalculate in the same city atlas. Name the author and reason, and upload the `.rgis` file. OSR compares embedded route/station data with the authenticated export, then uses the existing planning, service, fleet, civil-quantity, cost and delivery evaluators. The new map, prices and delivery links belong to one revised study; the original published city remains available. Camera or styling changes alone create no railway revision or recalculation.

```
./osr gis install                        # Verified upstream desktop + CLI binaries
./osr gis audit                          # Version, executable pin and cache location
./osr gis export --city baghdad --preview  # Native project, exact sources and PNG
./osr gis open --city baghdad             # Open a separate editable desktop copy
./osr gis export --all                   # Import, save, reopen and check all 266 cities
./osr gis export --all --output /path/to/retained-gis  # Keep every complete ZIP
./osr gis verify --output /path/to/cache/city/request-hash
./osr gis export-study --study LOCAL_JOB_ID
./osr gis import --city samawah --packet-id DISPLAYED_PACKET_ID --project edited.rgis \
  --actor "Planning author" --reason "Declared route revision" --data-crs EPSG:4326
./osr gis reprice --city samawah --packet-id DISPLAYED_PACKET_ID --project edited.rgis \
  --actor "Planning author" --reason "Declared route revision" --data-crs EPSG:4326 \
  --refreshed-output /path/to/fresh-updated-gis
./osr gis verify-revision --output /path/to/retained-import
```

`reprice --refreshed-output` verifies the retained native edit and its canonical request, waits for the bounded city study, then exports and verifies a new RustGIS project bound to that exact completed study. The input receipt remains in `build/rustgis-imports`; the refreshed editable project is written beneath the requested fresh directory. A no-op edit does not create a child study or refreshed export. This returns edited planning geometry and its recalculated outline costs; it does not establish revised railway physics or engineering acceptance.

The installer checks the pinned release archive and both executable SHA256s. Linux x86_64 is the currently tested installation target. It uses about 170 MiB for the desktop and CLI together, without a Rust compilation or Python GIS runtime. It adds a desktop menu entry and `.rgis` file association support. It does not replace existing commands or remove other applications automatically.

What each city contains

- Native editable geometry and styles, plus portable relative source files.
- The eight existing railway GIS datasets: routes, stations, platforms, original access markers, interchanges, civil segments, access connectors and unlocated requirements. All-null geometry is correctly imported as a table.
- Retained offline roads, buildings, land use, terrain, boundaries, places and water context where present, with original attribution and licensing.
- Exact original compressed GeoJSON, atlas packet, city summary and CSV tables for costs, finance, line service, stations, energy, local prices, make/buy, delivery and workforce. Cost confidence and uncertainty remain unchanged.
- A manifest binding the physical/selection/evaluation identities, original bytes, tool/source versions and actual native round-trip checks.
- A hidden `_osr_export_context` table identifying the trusted published seed or verified local study. Keep this table unchanged. Older exports without the context must be re-exported before using the return path.

Data coordinates are WGS84 longitude/latitude. The saved display uses Web Mercator. Use OSR's declared chainages and quantity models for engineering and costs; screen distances are not a replacement. Water lines and polygon areas are separate native layers, and polygon holes stay attached to their polygons. The adapter does not close open water lines or invent missing site locations.

RustGIS assigns numeric local feature IDs. `_osr_source_index` maps each feature back to its exact original record and string ID. Nested attributes become JSON text in native tables; every value is checked after export. Single/multipart geometry normalisation is permitted only when coordinates and topology agree. The coordinate comparison tolerance is 10^-12 degrees; this checks digital interchange and makes no claim about the accuracy of unsurveyed geography.

Disk, execution and editing

The default cache is `$XDG_CACHE_HOME/opensourcerail/rustgis`, falling back to `~/.cache/opensourcerail/rustgis`. Successful reproducible ZIPs have a 256 MiB cache budget; older ZIPs are evicted and regenerated on demand. Receipts and native logs remain, as do failed/interrupted inputs and outputs. Use `--output` to retain every ZIP outside that automatic cache policy. No bulky city exports need to be checked into Git.

The native RustGIS worker solves no transport physics: it imports geometry, styles, saves, reopens, exports every layer/table, checks coordinates and attributes, and optionally renders a PNG. Each native process has a 60 CPU-second, 75 wall-second and 2 GiB address-space limit. The download endpoint permits one worker and requires loopback client/Host and the exact current city packet. Plugins, AI, saved credentials and external tiles are not enabled by this adapter.

`open` creates an editable copy under `$XDG_DATA_HOME/opensourcerail/gis` (normally `~/.local/share/opensourcerail/gis`). Reopening that same copy preserves edits. Saving the drawing does not automatically reprice or select it. The explicit import operation submits a typed revision to the same bounded city job service used by the planning editor. It grants no engineering, manufacturing or operating approval and performs no business transactions or movement commands.

Editable data are the existing route geometry, route `shape`, `headway_s` and `design_speed_cap_m_s`, station positions and `dwell_s`. New stations need a new string `id`, an existing `line_id`, explicit dwell and a null source index. Retired/merged stations retain their original access-marker consequences. Stops must lie within the existing 20 mm digital consistency tolerance of their declared polyline; ambiguous intersections need an explicit geometry revision. This is not a survey-accuracy or physical-acceptance claim. Original source identities and derived civil/platform/access/cost layers are read-only; they are regenerated by OSR, never imported as approved evidence.

The importer reads embedded data without executing saved jobs, tasks, plugins, workspaces or external source paths. Native uploads are bounded at 32 MiB and retained separately from the 2 MiB canonical city request. Source input, failures, normalized changes and calculation links remain in `build/rustgis-imports` under a 256 MiB evidence-store budget. Oversized or ambiguous revisions fail explicitly; no geometry is silently truncated. The loopback upload requires matching Host and Origin. See the returned import receipt for exact candidate/base/source identities and execution status.

Historical QGIS `.qgz` metadata inspection is retained using Python's standard ZIP/XML readers. No QGIS installation is needed to inspect those archives. An arbitrary external QGIS project, plugin or processing model is not covered by this migration; only the repository's actual current city work is covered.

Upstream release · Pinned backend · City catalogue · Current engineering

3.3 City design dossiers

docs/design-engine/city-dossiers/README.md

Browse all cities · Baghdad · Samawah · Project introduction

Every city has a readable design dossier at its original city README address. The page includes a visible geographic network map, current line/station/service/fleet tables, energy accounting, civil and depot quantities, manufacturing and workforce cases, capital and whole-life budgets, explicit passenger/funding sensitivities, site requirements and downloads. National briefs aggregate standalone city outlines without assuming shared-factory savings.

The interactive Workbench atlas remains available for map inspection and recalculation. A new local calculation carries its own identities; a city page describes its published calculated outline until regenerated.

Data and scope

The dossiers consume the authenticated city atlas and complete calculated outlines. The selected route, operational platform groups, original access markers, explicit interchange references, service, fleet, mass allowances and cost ledger belong to the same candidate configuration. A city research configuration can include onboard traction with its own added mass and cost; it does not change the selected passive reference vehicle.

Maps use source-checked geographic context, current selected coordinates and operational stations. A local geographic projection, rounded display coordinates and background simplification support legible maps; GIS exports preserve source coordinates. Collision-limited labels use line/station reference numbers rather than inventing official station names. Depot and energy requirements without locations have null GIS geometry and explicit planning status. Geographic context is not surveyed clearance or land availability.

Service calculations are conditional outline results. Mechanical traction work, nominal auxiliaries and priced supply-efficiency scenarios are distinguished from physically supported total grid demand. Complete installed equipment, actual propulsion/thermal/braking maps, detailed civil interfaces and protected operating timetables remain separate evidence work.

Prices, resources and finance

Installed capital and thirty-year whole-life low/base/high cases come from the unchanged fourteen-scope ledger. Annual operating provision includes the source annual electricity, operations, maintenance, recovery and reliability accounts; scheduled renewal and disposal capital remain separate. The cases are sensitivity assumptions, not statistical confidence intervals or firm bids.

Factory and civil production use two eight-hour shifts with declared utilisation. Operations retain their service-hour and shift hypotheses. Covered posts, distinct shift positions, productive capacity, paid hours and paid FTE are separate measures. The existing cost already includes payroll/relief and factory capital; displaying workforce does not add those costs again. The civil workforce envelope uses declared labour-budget fractions and a planning delivery window. It is an aggregate resource hypothesis, not a detailed site schedule or an approved employment roster.

The calculated operating posts cover control room and access sites. They are not the whole operating organisation: vehicle/civil/power maintenance, cleaning, security/emergencies, management, HR/procurement, training and independent assurance retain explicit function/competence/authority requirements and unknown staffing where not sized. Unknown allocation is not zero cost or zero people.

The versioned planning policy makes otherwise missing outline assumptions executable. Illustrative paid-trip estimates combine peak-section flow with a daily profile, counter-direction loading, average section/peak loading, average trip length, transfers, collection and demand multipliers. They are not surveyed ridership forecasts. Fare and cost-covering-fare cases preserve adverse results. The separate debt overlay declares loan share, real interest, term and capitalised grace; it neither establishes imported content nor adds finance charges to canonical engineering cost. A real financing case requires local fares, dated capital draws, actual revenue, tax and funding terms.

Files and source ownership

Each city contains `engineering/outline/summary.json`, a static network SVG, line/station/cost/workforce/finance/make-buy CSVs, GIS layers, disabled delivery/business drafts and an output manifest. Each CSV identifies its atlas packet and outline evaluation. The publication manifest binds all city front doors, reports, country briefs, input publications and producer sources.

Physical evidence remains in the unchanged geographic hold summaries under `engineering/current/summary.json`. The `/3` city presentation receipt explicitly owns the enhanced front door and links to its separate outline dossier. The

geographic evaluator and register retain their source epoch; presentation changes do not invalidate unchanged cost calculations or relabel them as physical acceptance. The current coordinated publisher uses this presentation owner and verifier. The low-level geographic producer remains usable on its own for hold-only fixtures.

```
python -m engineering.city_reports.publication publish
python -m engineering.city_reports.publication verify
python tools/automation/generate-city-geography-index.py
```

Publication checks the complete upstream catalogue before rendering. A source/payload-consistent previous publication is reused. The single-writer lock is in ignored `build/`; maps are bounded to authenticated geographic input packets (64 MiB maximum decompressed context). Failed/interrupted generation is preserved as a failed invocation and does not become a completed publication. No CAD, structural, motor or hardware test is executed by the dossier renderer.

Verification

```
python -m pytest -q tests/test_city_dossier_analysis.py tests/test_city_dossier_maps.py tests/
test_city_report_publication.py
python tools/automation/generate-city-geography-index.py --check
python tools/automation/check-readmes.py
```

The analysis tests independently control demand, annuity/debt arithmetic, workforce/source bindings and cost reconciliation. Map controls cover projection, scale, preserved selected geometry, civil segment clipping, source identities and unlocated requirements. Publication tests cover ownership, current-page content, map/CSV source identity, candidate separation, catalogue link resolution and tampered payload rejection. These verify calculated-data consistency; physical release requires the independent engineering workflow.

Source comparison

The restoration was checked against the actual v0.4 city dossiers, including the Samawah native/animated study. Those files retain their own battery, formation, energy, costing and evidence assumptions. They provide a feature/content reference and source history; their numbers are not inputs to the current dossiers.

IV. Railway and engineering

Vehicle, civil, propulsion, manufacture and numerical-method evidence.

4.1 Coordinated railway experiment

docs/design-engine/coupled-railway/README.md

Architecture, adaptive revision planning and physical body generation now share one coordinator, one explicit railway bundle and one dependency graph. A body edit creates new body/railway identities and automatically reevaluates inventory, interfaces, architecture, motor requirements, independent braking, civil loads, manufacture and economic coverage. Each alternative keeps its own candidate identity and evidence; results are not transferred between candidates. New coordinated work starts with `./osr design railway`.

The executed comparison starts with 5 structural recipes, makes 9 physical revisions, and evaluates 112 railway bundles through 784 downstream stage records. Two primary-iron alternatives share unchanged onboard fin stock. Seven architecture families retain their own changed interfaces and unweighed equipment. Class C model-cost estimates cover 112 of 112 executed bundles across 14 cost scopes. Provisional per-stock process routes cover 112 candidate bundles and 109544 located stock occurrences. The process hours remain costing proxies. Installed mass, available operating LIM maps, physical journeys, supplier-priced budgets and a qualified whole-life-cost winner remain open.

Completion levels

Each candidate reports three separate completion levels. The declared cost scenario is simulation-complete for 112 of 112 candidates, using source-linked quantities and explicit Class C assumptions across 14 scopes. This flag covers that cost scenario only; it does not claim an integrated journey simulation. 0 candidates have a completed force-driven journey, 0 are engineering-validated, and 0 are approved for manufacture or operation. Engineering validation requires independent physical evidence and review; release requires a separate approval record.

Closed-loop physical and system changes

The existing adaptive planner diagnoses each declared pin-graph mechanism; the grammar produces a braced physical child instead of changing a pass threshold. A separate assembly-completion operator closes quantified frame/floor/roof gaps using counted metal stock and finite shared-face weld hypotheses. It generates an aluminium frame and an aluminium stressed-skin body, preserving passenger openings and required roof/PV/channel/drain functions. Missing composite laws are not replaced by isotropic GFRP.

Body / physical revision	Known structure stock	Empty-mass account	3D evidence	Architecture cases
aluminium_spaceframe	2108.7 kg	13.006 t	full-body model not admitted	8
gfrp_semimonocoque	2653.8 kg	13.551 t	full-body model not admitted	8
hybrid_structural_panels	2924.8 kg	13.822 t	full-body model not admitted	8
pultruded_gfrp_frame	1687.8 kg	12.585 t	full-body model not admitted	8
modular_segments	3077.5 kg	13.975 t	full-body model not admitted	8
aluminium_spaceframe	3242.5 kg	14.140 t	full-body model not admitted	8
gfrp_semimonocoque	3448.7 kg	14.346 t	full-body model not admitted	8
hybrid_structural_panels	4058.5 kg	14.956 t	full-body model not admitted	8
pultruded_gfrp_frame	2482.7 kg	13.380 t	full-body model not admitted	8
modular_segments	4211.2 kg	15.109 t	full-body model not admitted	8
aluminium_frame	3201.4 kg	14.099 t	computational resource hold	8
aluminium_frame	3201.0 kg	14.099 t	equilibrium/energy checked; maxima refinement reported	8
aluminium_stressed_skin	3473.5 kg	14.371 t	computational resource hold	8
aluminium_stressed_skin	3473.1 kg	14.371 t	equilibrium/energy checked; maxima refinement reported	8

These are exact generated nominal/net stock quantities plus retained source accounts. They are not complete vehicle tare or certified lower bounds: original gross-stock overlaps/cuts and missing installed hardware have different boundaries. The new welded children count net disjoint metal, retained covers, PV rails/spacers, trays/gutter/hangers and unknown weld/joint realization separately. Whole installed CG, inertia, clearances, hardware and repair access remain open.

Three-dimensional structural method

Four independent fixtures check bending/shear refinement, intact finite-face transfer and orthotropic axial/shear constitutive patches. The two new metal bodies have 9 exact native body cases across normal, normal-fine and crowd loading: 9 dispatched in this coordinated run and 0 carried forward from byte-verified prior publications after matching candidate, load, mesh, solver, resources and source hashes. Reuse is provenance-preserving and does not count as a new solve. C3D8I solids resolve the declared actual metal geometry; no invisible gap ties or arbitrary rigid offsets create the load path. Intact autogenous weld/continuous-face transfer is an unqualified numerical hypothesis.

New solid body	Normal 1.7 m target	Finer 0.85 m	Extra 0.425 m	Last-step change
aluminium_frame	15.245 mm	17.283 mm	18.531 mm	6.73%
aluminium_stressed_skin	6.035 mm	6.286 mm	Unknown	3.99%

The source-bound resource revision uses 1.7/0.85 m global subdivision targets; thin physical-feature cuts dominate the conformal grid. All actual mesh counts, resources, force/moment/work/energy residuals and failures are retained. A displacement trend is not strength or operating acceptance. C3D8I torsion limitations, quadratic confirmation, local/global buckling, material/HAZ/joint strength, fatigue, modes, thermal/fire/crash behaviour and physical calibration remain necessary work. Equipment loads are currently lumped over the floor with actual structural selfweight; real component positions and suspension compliance are not resolved.

The first baseline metal-frame job produced 152,004 cells and exceeded its 5 GiB allocation while factoring. It yielded no structural verdict. The generator then standardized and aligned floor/roof extrusion profiles, preserving openings, support pads and actual load paths. These are new physical children with different identities, fewer profile families and roughly 71% fewer conformal cells. The original failed execution, source epoch and CAD remain retained. A 60,000-cell dispatch budget is a computational control, not a railway acceptance rule or proof that the baseline body cannot work.

The frame's first refinement changed the three displacement maxima by about 10-12%. The coordinator therefore requested another unchanged-candidate numerical experiment at a 0.425 m target. Its 61,352-cell estimate justified an explicit computational budget revision to 65,000 cells under the same 5 GiB/150 s limits. Exact unchanged prior solves/CAD are reused across coordinator epochs, with new context records. A 5% refinement trigger requests work; it does not qualify a body. Whole-field, stress, torsion and physical convergence remain separate questions.

The current-source continuation ran the same P1 body and normal-load basis at 0.2125 m under its 90,000-cell budget, followed by the 0.10625 m C3D8I run using 128,382 cells under an 8 GiB/240 s request. Both completed with equilibrium/energy consistency only. The fine result is included as a direct-native package. Matched-point displacement changes remain above the 5% refinement trigger; numerical convergence and physical acceptance remain open.

Original foam/honeycomb and GFRP bodies remain non-admitted to this whole-body solver until their own complete directional material laws, attachment stock and joint models are provided. Orthotropic method patches demonstrate implementation, not purchased composite properties.

Available propulsion and forward motion

Every changed mass automatically reaches the 2 km required-force calculation, nine independent-brake loading/fault hypotheses and three crowd-loaded precast alternatives. Rotor equivalent mass enters acceleration and braking; it is excluded from static axle gravity. P6/P7 remove the old passive-secondary account and preserve unknown replacement mass. A candidate-bound available-map gate checks actual position/speed/current/temperature samples, electrical work, thermal/current/voltage domains and normal-force interfaces. No linear, generic DC or synthetic map is silently admitted as an operating motor.

A new transient A-z/floating-current kernel executed 6 bounded method controls (6 completed). It uses nonlinear reversible H(B), implicit timesteps, actual phase-current excitation, net-secondary-current closure and explicit stored-energy/Joule/mechanical/discrete-time work. Its finite conductor mask stays fixed at a declared frozen velocity. The generic curve, short secondary, incomplete circuit/copper/thermal model, independent steady/mesh/transit confirmation and actual 20 m/s operating domain prevent it supplying an available-force map. Residual closure is reported without relabelling this fixture as the full LIM.

The forward-motion method has independent synthetic tests for constant-force energy, finite powered windows, zero entry force, thermal/bus limits and stopping. Actual railway journeys remain not-run because available propulsion and complete installed inputs are missing. The receiving drive receives no safety credit in independent-braking screens.

Manufacturing, economics and evidence

Each shared bundle produces an EBOM of actual stock, retained accounts and unresolved joints; MBOM cuts/blanks and process groups; assembly/inspection dependencies; material, supplier, access, repair and QA tasks; and source-bound price/lead-time coverage. Missing equipment is neither a box invented for an animation nor a zero-mass part. Quotes require the exact bundle/component/quantity/unit/currency context.

The campaign now also produces 14 low/base/high IQD model-cost scopes per executed candidate: rolling stock, guideway, stations/power, factory/tooling, construction, land/utilities/depot, fleet, energy, operations, maintenance, renewals, recovery, reliability and decommissioning. Each model-cost result includes a candidate-bound provisional process route and low/base/high labor-hour estimate for every located nominal gross body-stock occurrence. Those Class C hours support costing and factory scenarios; they are not measured cycle times or released MBOM routings, and unlocated accounts, joints and final assembly remain separate. Dated July 2026 Iraqi CSO materials/wages are used where matched; the remaining rates and missing site/asset quantities are explicit Class C assumptions. Production capacity and civil scheduling use two 8-hour shifts, with second-shift premiums and per-shift construction productivity included. The low/base operations cases cover 16/18 service hours with two shifts; the high case prices 24-hour service with three shifts, including related staffing, energy, maintenance, reliability and renewal effects. See the model-based cost and delivery plan. These are design-stage estimates with zero supplier quotations, not procurement-ready prices or accepted city budgets.

The two executed new body CADs also compile candidate-bound delivery packages: the same stock IDs feed EBOM/MBOM, disabled research ERPNext Item/BOM drafts, assembly and procurement tasks, and reopened IFC body-stock geometry. Drafts are not posted to ERPNext and cannot release purchasing or manufacture. Unlocated equipment, unknown joint realization and missing commercial/process evidence remain explicit.

Candidate cost ranges are compared only where currency, rate assumptions, corridor length, trackways, station count, headway and offered capacity match. Overlapping low/base/high whole-life scenarios remain incomparable; cost dominance does not select a candidate or change the v0.6 reference. Cost inputs include declared energy, staffing, maintenance, replacement, recovery, reliability and end-of-life scenarios. City alignment/site surveys, tariffs, staffing rules, failures, supplier quotes, tax/finance and physical performance are not established. Civil results count axle envelopes, section/tendon quantities and preliminary foundation/erection hypotheses; geotechnical, torsional and construction qualification remain open.

Immutable stage requests include candidate and upstream evidence IDs. Native cache admission checks source/executable/payload plus exact body, load, mesh, method and resource request. New structural evidence invalidates dependent case evaluations even if geometry is unchanged. Stable experiment identity is separate from invocation cache counters; interrupted/pending work remains in the frontier. Native/CAD packets retain their original receipts.

```
./osr design railway plan
./osr design railway run --max-bundles 2 --native-jobs 1
./osr design railway run --native-jobs 12
./osr design railway publish
./osr design railway verify
```

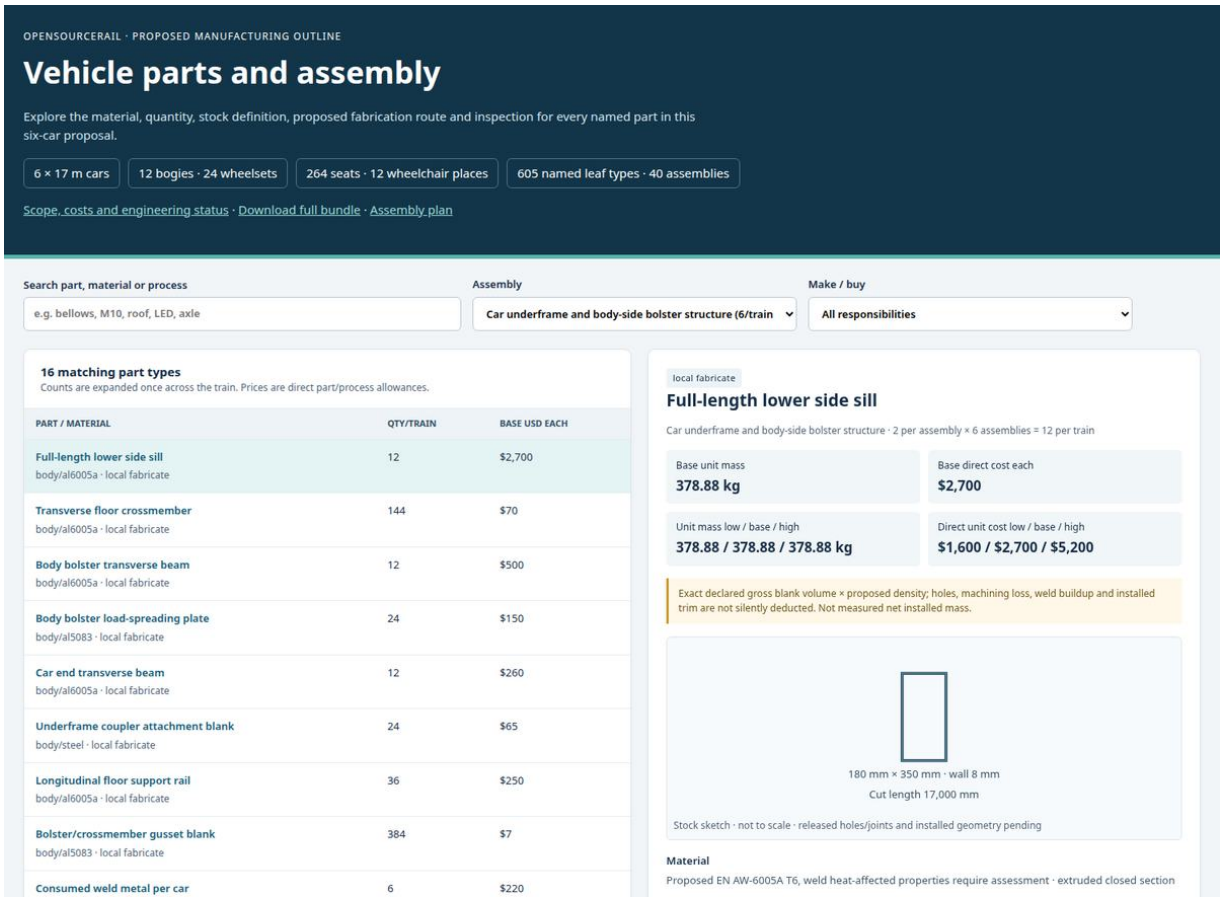
Summary · Body definitions · Canonical bundles · Coupled evaluations · Inventory/manufacturing records · Delivery/IFC/ERP drafts · Native/CAD index · Current refinements · Fine refinement · Ultrafine refinement · Transient method controls · Pending work · Source epochs · Source/payload manifest

Large fields and CAD solids stay local or regenerate with exact sources and indexed inputs; no remote artifact object or complete manufacturing/operating release is claimed. The selected reference and all 266 city performance packages remain unchanged.

4.2 Vehicle parts and assembly

docs/design-engine/vehicle-parts/README.md

The six-car manufacturing proposal now contains 605 named leaf-part types in 40 assemblies. Each part identifies its material, quantity, stock dimensions or bought-component envelope, manufacturing route, fixings/joining, proposed tolerance basis, inspection and maintenance-removal method. This is a useful completed outline breakdown; detailed released drawings and physical qualification remain later gates.



Searchable catalogue with part quantities, materials and stock sketches

Open the searchable part catalogue — select a subsystem, search a part or material, and open its dimensional stock sketch, proposed route and checks. Machine-readable full bundle · Materials · Cut/stock schedule · Assembly plan · Downstream handoff · Disabled business draft.

What is counted

- 12 passive bogies, 24 wheelsets and 48 axleboxes: frame members, wheel/axle forgings, bearings, seals, springs, dampers, emergency support, discs, calipers/actuators, local WSP equipment, guards, pins, bolts, washers and retainers.
- Six car structures: lower sills/bolsters, side/roof members, floor, sandwich-panel skins and cores, adhesive/weld allowances, repairable linings, roof layers, equipment brackets and individually counted attachment hardware.
- 264 fixed chairs and 12 wheelchair places: seat shells, frames, cushion/fabric proposals and attachments; wheelchair places do not acquire extra seats.
- 12 one-sided door openings, 24 moving leaves and 110 panes: mechanical belt/reducer, carriages, rollers, tracks, latches, manual release, closing-edge rubber, separate monitored sensor inserts, motors, controllers, switches, cable carriers, seals and fixings.
- Five inter-car couplers and five separate passenger gangways: drawgear and restraints, floor bridges, articulation/support elements, bellows, rubber/fabric frames, clamp strips, corners, fasteners, air hoses, electrical jumpers and essential lights. Two end/rescue couplings are also counted. Gangways do not replace load-bearing couplers or bogies.
- Service systems: simple LED lighting, battery modules/racks, separate terminal and coolant fittings, active pack cooling, auxiliary conversion, HVAC pressure components, ducts/drains, brake/suspension air, controls, cable routes, optional-interface passive aluminium secondary and the retained roof-PV research function.

The 110,404 expanded BOM units include individual fasteners, cut strips, consumable/charge allowances and bought subcomponents. They are a proposed counted schedule, not a measured count of every internal semiconductor, cell or sealed compressor part. Such bought-unit internals and delivery inclusions require supplier drawings. Parent assembly weights, original vehicle category allowances and cells already inside battery modules are not added again.

Simple LED lighting

Each car has two 12 m rows made from 16 replaceable 1.5 m, 12 W LED modules, plus four 6 W vestibule lamps. Normal output demand is 216 W/car; two separate 150 W drivers feed four protected half-circuits. Four 3 W emergency fixtures/car and two 3 W fixtures/gangway use independent essential feeds. No inter-car HV or low-voltage power buses are paralleled.

At the declared 90% driver efficiency, total train lighting input is approximately 1.55 kW. The 2.5 mm², 7.5 m half-route hypothesis gives about 0.292 V drop at 2.25 A with the stated 80°C copper model. This is necessary electrical sizing arithmetic; illumination, glare, EMC, protection coordination, hot derating and emergency tests remain open. The lighting estimate does not resize the retained 82.8 kWh/car service-energy hypothesis.

Materials and local manufacture

Prefer repeatable cut/folded/extruded members, accessible positive fixings, replaceable panel/door/light modules and controlled jigs. Each material has a proposed grade/form and explicit qualification dependency. Examples include weldable aluminium frame sections, 5052-H32 folded service enclosures, structural steel bogie stock, wheel/axle/bearing components requiring certification, composite panel skins/cores, railway glazing and fire/smoke-qualified elastomers. A generic rubber or household diffuser is not evidence of railway fire performance.

The make/buy assignment distinguishes local fabrication, standard purchased hardware and qualified bought equipment. It does not assume that a new local factory is automatically cheaper: shared tooling, final assembly hours, production rate, yields and capital remain explicitly outside this subtotal. No invented Chinese or repeat bulk discount is applied. Current component quotations can replace individual allowance rows without changing the physical geometry; their evidence and the resulting evaluation identity must change.

Quantities and estimated direct cost

The six-car sum of declared stock masses and bounded component allowances is 78.49 / 89.81 / 106.60 t (low/base/high). These are not verified installed masses, CGs or inertia. Gross stock and estimated bought masses are labelled individually. The old 19.22 t/car category account is comparison-only; the difference is not an established weight saving.

Declared direct part/process subtotals are $507,494 / $1,064,002 / $2,350,849 across six cars. They are engineering allowances, not quotations or a complete vehicle factory price. Final car assembly/commissioning, shared plant/tooling, taxes/duties, delivery, qualification, warranty and complete lifecycle costs remain outside the stated scope. The low/base/high sums are bounding scenarios, not statistical confidence intervals.

Assembly	Instances/train	Leaf types	Base scoped kg/train	Base direct parts/process USD/train
Passive two-axle bogie frame	12	13	12,457.4	$43,303
Wheelset with two monobloc wheels and one axle	24	3	15,360.5	$46,416
Axlebox and sealed bearing installation	48	7	2,464.6	$48,465
Primary coil spring and axlebox guidance	48	10	1,695.7	$11,177
Bolster, air springs and secondary restraint	12	21	5,044.5	$32,629
Replaceable bogie damping hardware	12	7	1,089.8	$12,066
Independent bogie disc brake mechanical equipment	12	16	4,865.5	$48,208
Bogie monitoring, WSP valves and local wiring	12	16	156.1	$9,713
Bogie protection, stops and local grounding	12	6	226.8	$1,469
Semi-permanent intercar mechanical coupling and local air bridges	5	20	1,239.6	$9,112
End/rescue mechanical coupling	2	17	313.9	$2,616

Assembly	Instances/train	Leaf types	Base scoped kg/train	Base direct parts/process USD/train
Flexible intercar passenger gangway and floor bridge	5	44	635.6	$8,703
Car underframe and body-side bolster structure	6	16	9,939.7	$71,667
Sidewall frame, door pillars and cantrails	6	10	3,332.6	$31,142
Roof bows, panels, cantrail joints and equipment mounts	6	12	2,207.7	$15,413
Bonded modular floor and wear surface	6	10	3,283.8	$21,125
Opaque side skins and end trim	6	9	2,056.1	$10,353
Gangway body end	10	8	479.0	$5,338
Terminal body end	2	8	95.8	$1,068
Bi-parting external sliding door mechanical operator	12	51	497.2	$13,383
Passenger door leaf without glazing	24	12	394.1	$5,861
Fixed side glazing	6	7	2,076.6	$14,001
Door leaf glazing	24	7	181.0	$2,274
Terminal windscreen	2	7	109.0	$990
Fixed passenger chair	264	14	2,660.3	$44,183
Wheelchair-space mechanical hardware	12	7	87.1	$1,710
Interior lining, insulation and access panels	6	14	2,658.0	$18,312
Handholds, luggage racks and mechanical protective trims	6	9	231.8	$4,723
Replaceable simple LED cabin lighting	6	14	91.3	$3,667
Passenger door electrical drive and protection	12	23	108.0	$11,742
Service battery25-series×3-parallel module racks	6	60	8,275.0	$365,113
Auxiliary power distribution and charging interface	6	12	586.9	$56,575
Removable roof HVAC module	12	31	1,590.1	$31,037
Ventilation ducts, diffusers and condensate drains	6	9	399.2	$4,959
Brake and suspension air supply	6	19	560.0	$10,134
Vehicle protection, communication and observations	6	19	354.3	$26,058
Intercar flexible signal/data and bonding jumpers	5	11	28.7	$3,041
Passive aluminium station-propulsion interface	6	9	1,671.0	$6,434
Retained roof photovoltaic service proposal	6	8	298.6	$9,363
Gangway essential LED lamps and flexible feeds	5	9	5.1	$460

Assemble and inspect

The assembly plan starts with a datum-controlled underframe, then the body frame, roof/floor, bogies, equipment, panels, doors/glazing, passenger fitout and inter-car connections. Each subassembly has its own ordered operations, identified part lists, tools, inspection and removal method. Stock sketches show the declared blank or envelope; hole patterns, weld/bond geometry and mating fits still need controlled manufacturing drawings. No part list grants manufacturing release.

Several interfaces were reconciled during the breakdown: body roof pads are flat water/dielectric layers, while services owns distinct vibration mounts; electrical door sensitive-edge inserts fit the separately counted body rubber profile. The battery bill

includes 72 series links/car, 150 external terminal fixings/car, 15 cold plates/car, explicit headers/hoses/ports and active chilling. A radiator alone cannot hold a 45°C module environment below a 50°C ambient.

Formation and engineering handoff

This is a new detail proposal, descended from the current reference vehicle. Its proposed 650 mm inter-car gap gives 105.25 m body-plus-gap length, compared with 104.50 m at the parent 500 mm gap. End coupler projection, stopping tolerance and platform margin are still unknown. The handoff carries this envelope and the same candidate ID into inventory, interfaces, propulsion, braking, civil, manufacture and cost. The current city fleet, platform and cost results are not silently overwritten.

Complete installed coordinates/CG/inertia, finite door/gangway/bogie motion, the service pack's underfloor fit, roof occupancy and load paths require one installed CAD model. Critical joints, fatigue, crash, wheel/rail dynamics, independent braking, fire/accessibility, hot-weather duty and available propulsion need their appropriate analyses and evidence. Neither candidate-specific propulsion nor manufacturing/operating release is selected here.

Reproduce and revise

```
./osr design parts plan
./osr design parts build --output /path/to/private/fresh-parts-run
./osr design parts verify
```

Canonical editable source is engineering/vehicle_parts. Physical identity covers geometry, materials, counts, topology and mating contracts; verification outcomes, prices and uncertainty remain evaluation context. All exported items carry the same candidate and bundle. The business draft remains disabled and performs no ERPNext, purchasing or control transactions.

Candidate: `907da3680bcbd01ffe1223652443e4e34a9f581e201385917a99edf4150bd90a`
Bundle: `bc43edc029e8d4acf44df281dcdb640c71dccdb8a0f244f9de82644672aef6f1`

4.3 Engineering model gallery

docs/engineering-models/README.md

Project introduction · Current engineering

These are parametric 3D assemblies rendered from the same explicit geometry that is exported to CAD. They explain the station mechanisms, construction and fabrication routes. Dimensions copied from the current vehicle and beam are distinguished from proposed equipment envelopes and shop tooling.

Use: compare layouts, inspect parts and discuss manufacturing. These are outline communication models; they are not released drawings, a lifting plan, a pressure-vessel design or an operating system. The selected railway and its qualification status are unchanged.

Station capture, storage and launch

The illustrated research layout uses five staged cable arrestors, a hydraulic energy store and a controlled rear launch shuttle driven by a common capstan. Stored energy supplies the first part of departure; an optional motor supplies the remaining acceleration. Independent train braking, energy isolation, mechanical retention, guards, sensing, wiring and maintenance access are shown separately.

The source control uses a 160 t effective point mass, arrival capture from 10 m/s and a 60 km/h departure study. Its calculated capture travel is 70.5 m; recovered launch travel is 43.35 m, followed by 128.94 m of boosted travel. These are train-motion distances, not validated equipment lengths. The 8.8 MJ store requirement and proposed vessel envelopes do not establish pressure containment, installed storage mass or achievable performance. Neither this mechanism nor its speed is a mandatory design choice.

The source six-car formation occupies 104.5 m, including five 0.5 m coupling gaps. Local close-up coordinates are declared in each recipe. The launch view includes a separate proposed shuttle runout; coupling loads, cart disengagement, train compression, jerk and failure behaviour require analysis.

Construction and fabrication

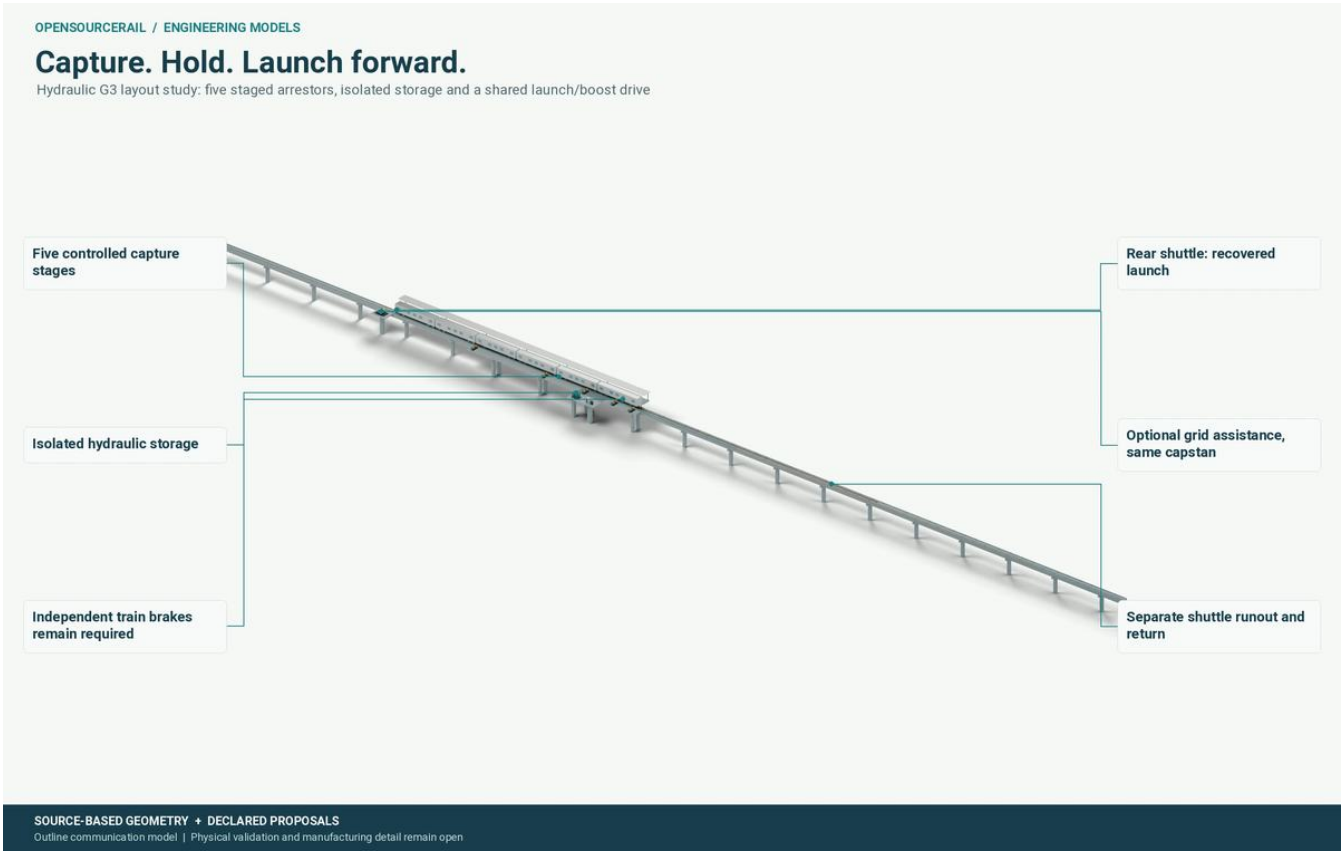
- Whole-span crane erection: surveyed supports and bearings, delivery trailer, crawler crane, spreader and temporary restraint. The source bare lifting stock is about 59.18 t, excluding rigging and unresolved attachments. Crane capacity, working radius, ground bearing and temporary stability must be engineered for the site.
- Overhead gantry alternative: deliver and lower the same 20 m span from a launching gantry. This is an alternative erection scheme; its steelwork, support reactions and launch sequence are proposals. The illustrated seats give a proposed 19.4 m bearing-centre spacing beneath 20 m stock; the source 20 m analytical result is not inherited by that layout.
- Precasting: reusable outer forms, a proposed collapsible core, cage jig, two proposed tendon ducts and inspection specimens. The six disjoint source concrete solids total 22.2816 m³. The core extraction, bar schedule, tendon/anchorage detailing, curing and release strength remain to be designed.
- Vehicle assembly: all 2,482 located source parts are preserved, including drilled floor panels. Their 9,131.80 kg is located stock, not the complete car tare. Render-only explosion separates roof, seating, covers, doors and bogies; downloadable geometry retains source positions.
- Rails and running gear: purchase standard rolled rail, then finish, join, inspect and fasten locally; assemble and inspect wheelsets, bearings and passive bogies. Tooling is proposed. The 60E1 display contour follows a manufacturer drawing and nominal dimensions; its approximate area is not a qualified rolling/contact profile or a replacement for the nominal rail mass.

An outline wheelset route is: receive traceable wheel/axle stock; finish and inspect interfaces; record the specified press-fit operation; inspect the assembled wheelset; mount bearings/axleboxes; then check the bogie and brake installation. Lucchini's production sequence and GHH-BONATRANS inspection and maintenance scope provide industrial context. The displayed press, gauges and inspection positions are proposals, not qualified machines or process specifications. Two 8-hour factory/civil shifts remain planning inputs; a workstation picture does not establish cycle time or annual capacity.

Inspect and download

STL opens in GitHub's 3D viewer. GLB keeps colours and part metadata; FCStd is the native FreeCAD solid assembly grouped by subsystem, with part IDs and SI recipes. Display colours are separate from source material tags; neither is a material certificate. STEP is gzip-compressed: decompress before opening it in CAD. GLB uses metres; STL, STEP and FCStd use millimetres. No presentation explosion is applied to downloaded models.

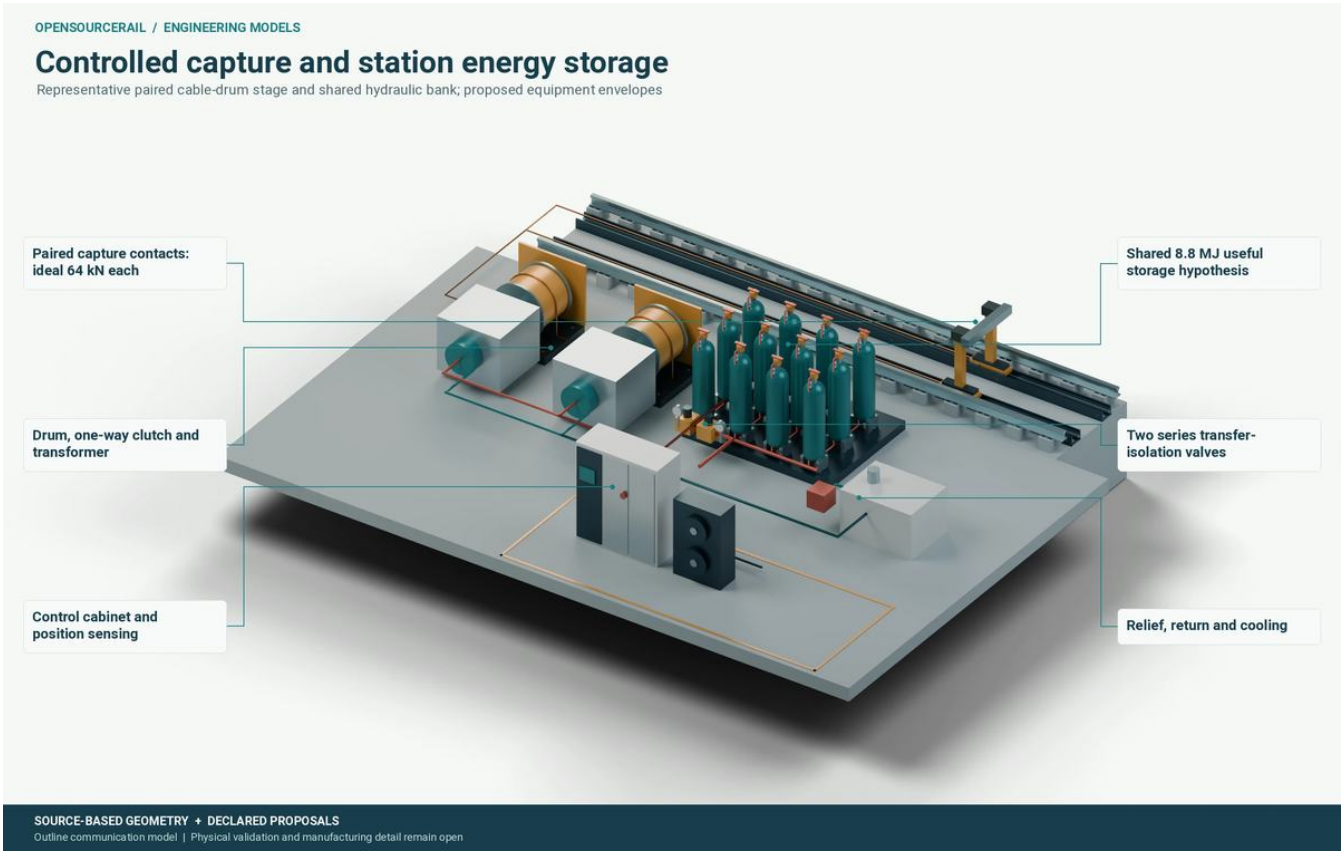
Capture. Hold. Launch forward.



Capture. Hold. Launch forward.: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

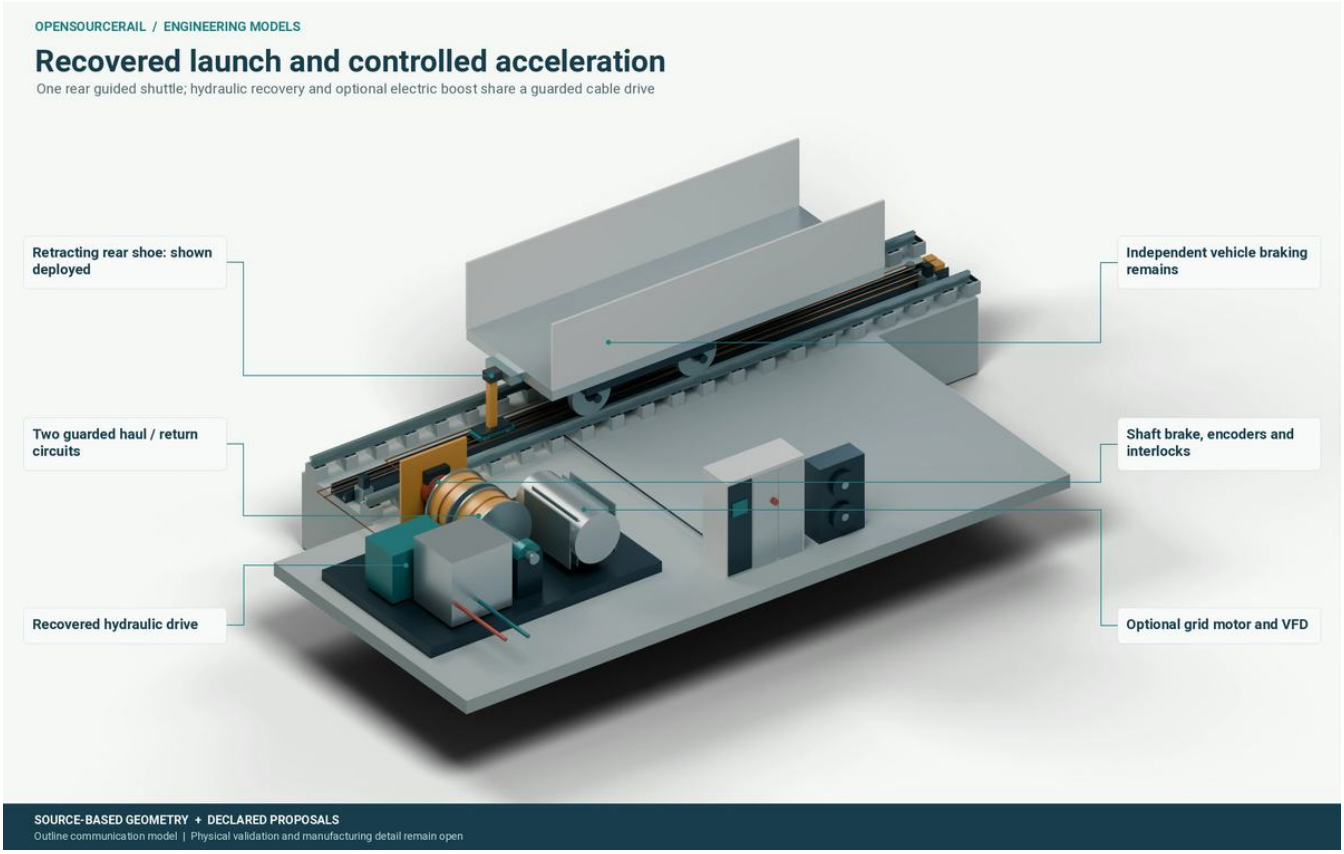
Controlled capture and station energy storage



Controlled capture and station energy storage: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

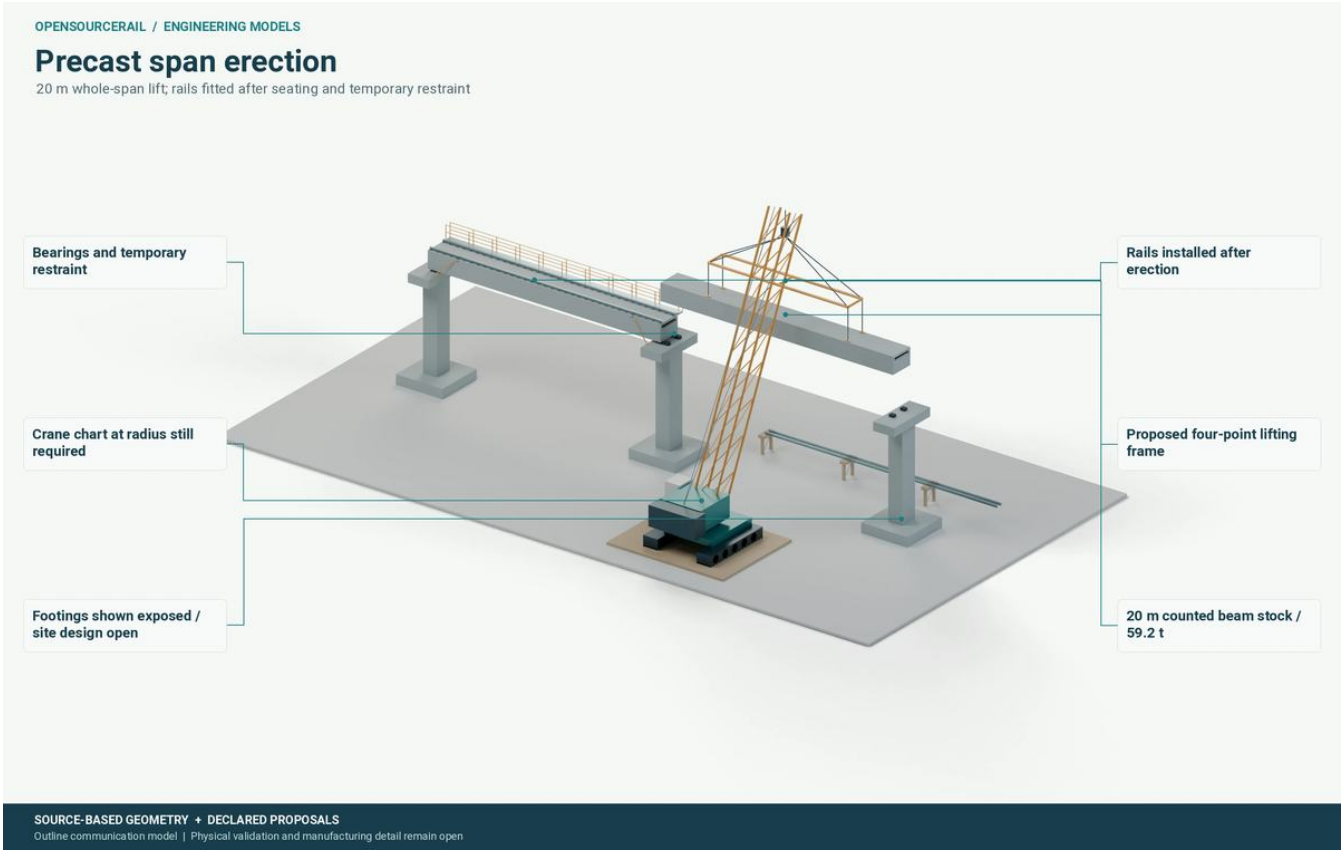
Recovered launch and controlled acceleration



Recovered launch and controlled acceleration: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

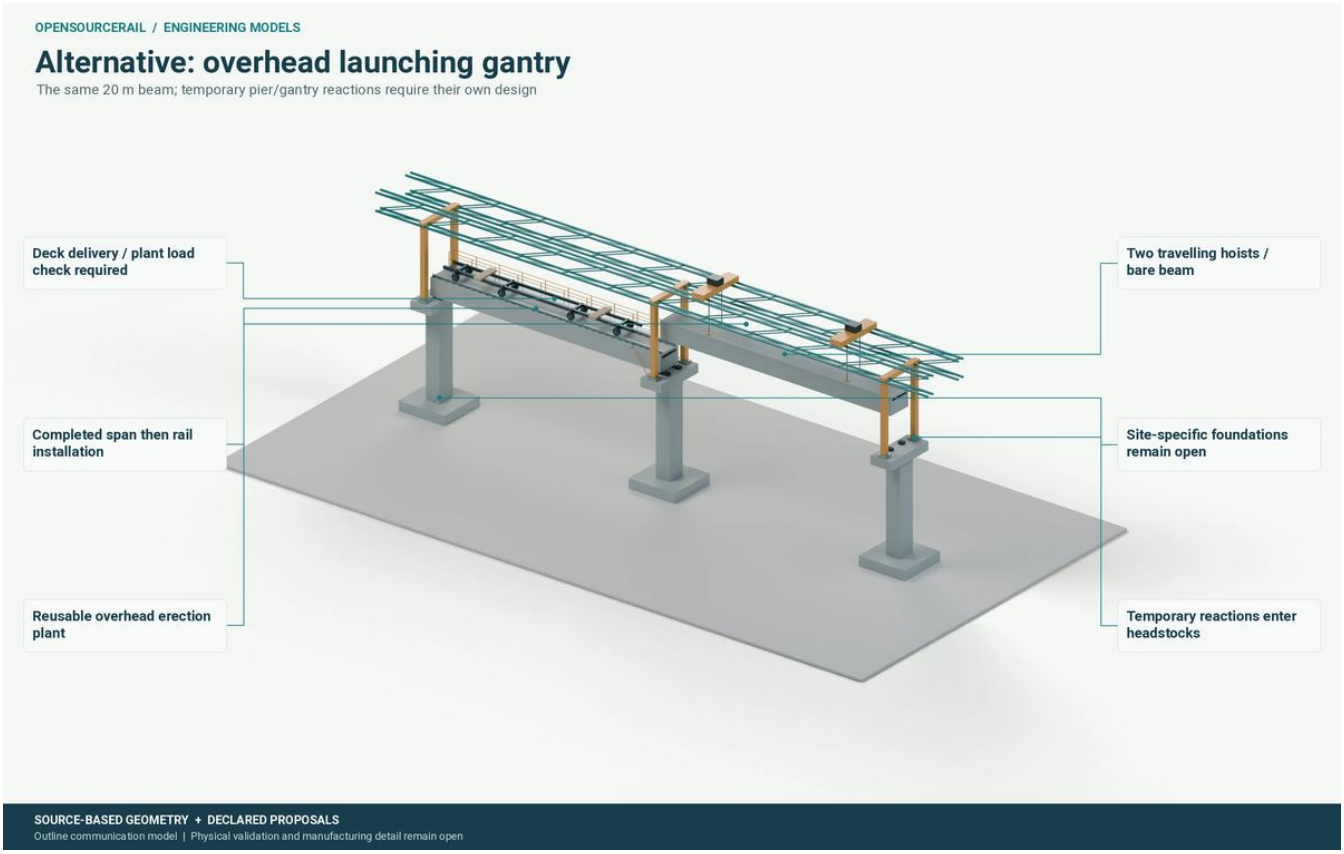
Precast span erection



Precast span erection: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

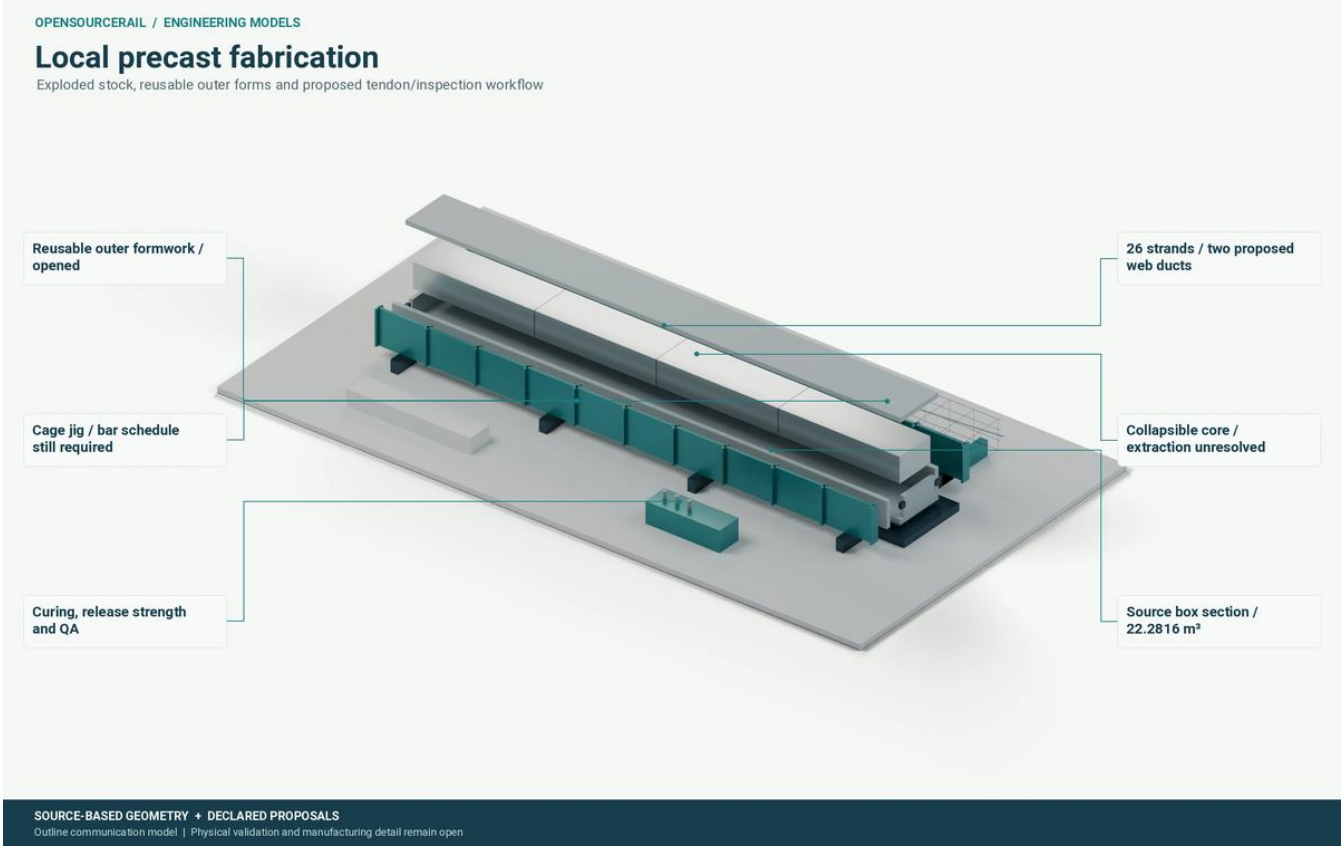
Alternative: overhead launching gantry



Alternative: overhead launching gantry: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

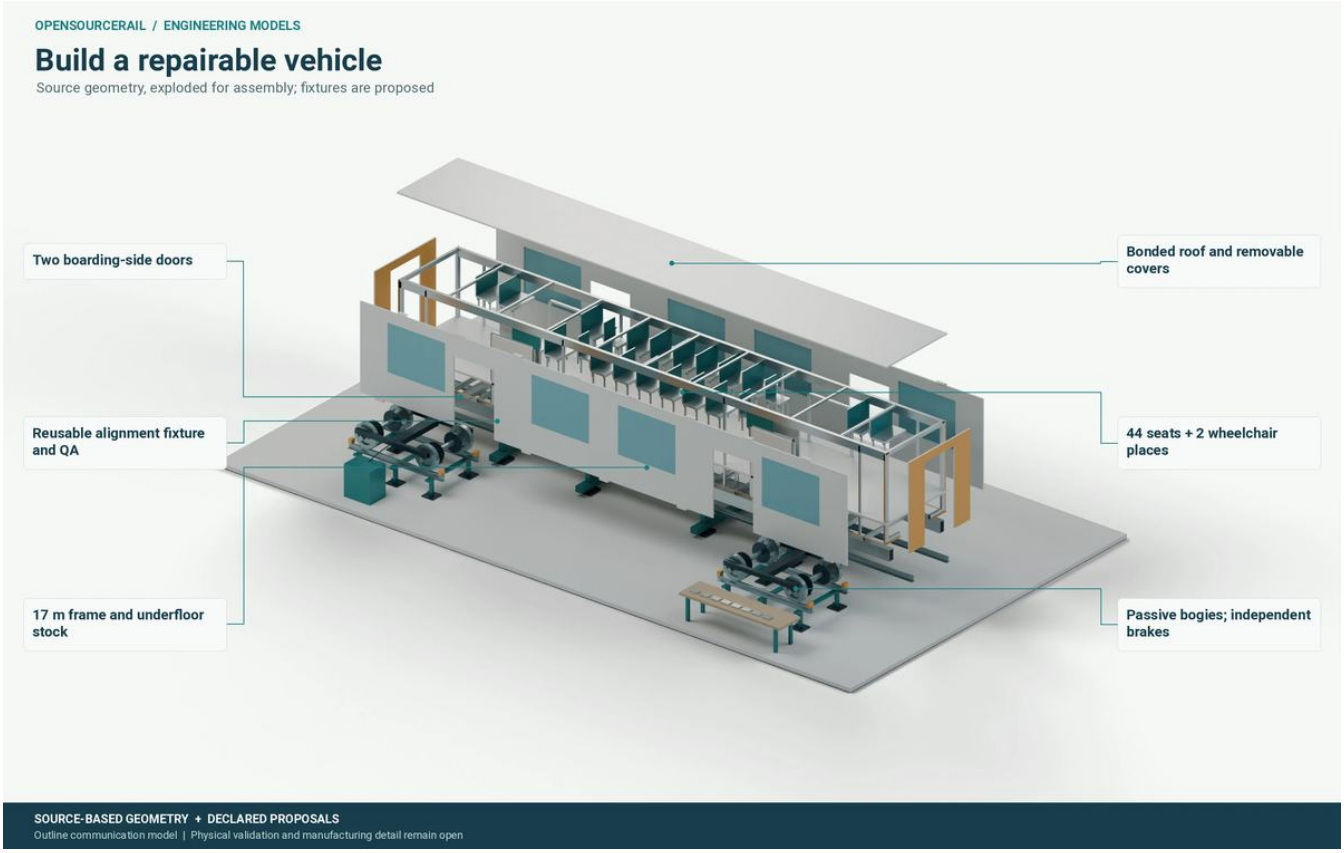
Local precast fabrication



Local precast fabrication: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

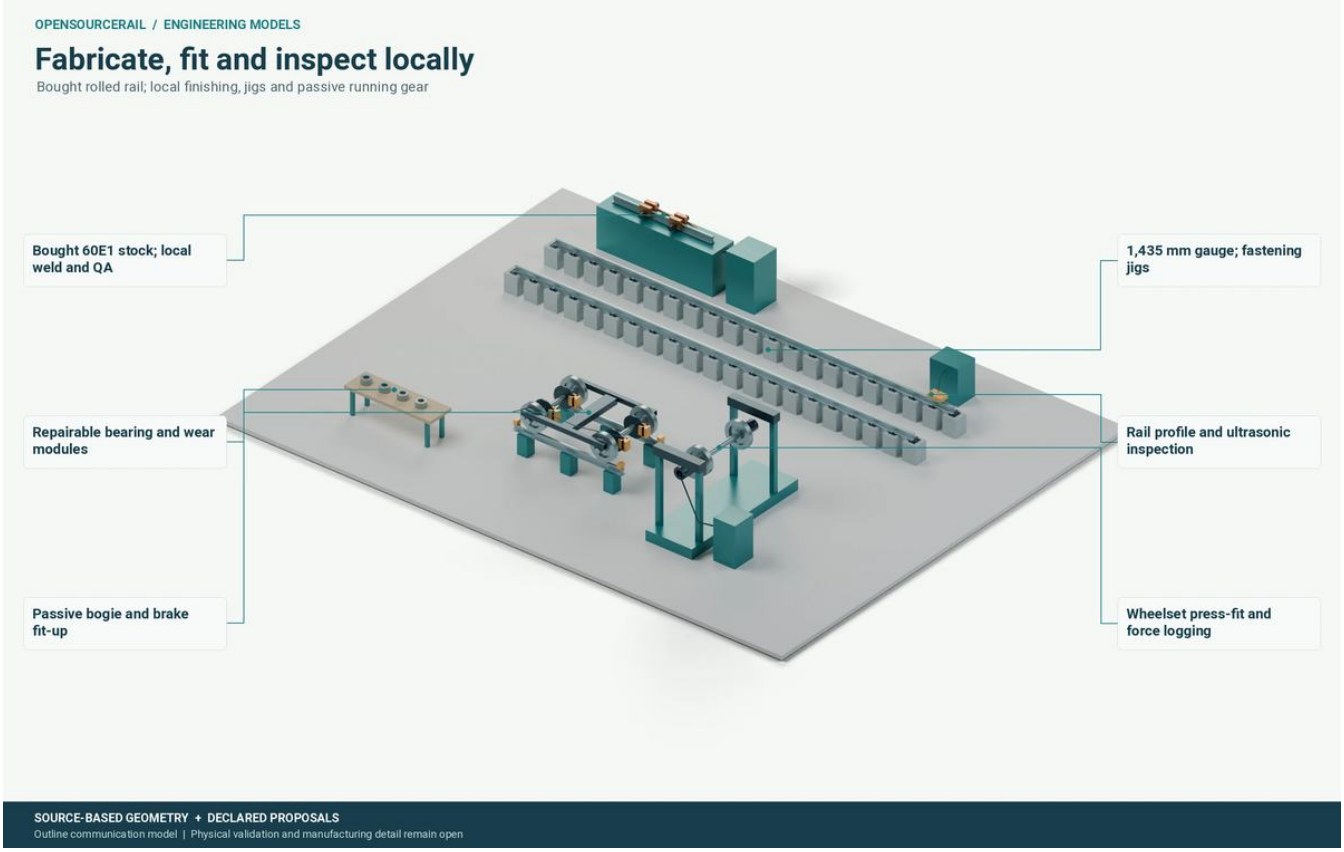
Build a repairable vehicle



Build a repairable vehicle: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

Fabricate, fit and inspect locally



Fabricate, fit and inspect locally: source geometry and proposed layout

Rotate STL · Coloured GLB · FreeCAD solids · STEP solids (.gz) · Exact recipe (.json.gz)

Sources, checks and reproduction

The manifest binds each visual-study identity to source files, geometry recipes, native execution receipts and output hashes. Visual-study identities are separate from physical candidate, selection and evaluation identities. The station mechanism is a new layout proposal; it does not inherit engineering acceptance or the installed price of its source motion example.

The source snapshot preserves the exact producer and input epoch. Verification replays that snapshot and checks its vehicle geometry against the current reference. A price or unrelated evaluator update does not require rendering unchanged geometry again. Native workers are re-executed only when explicitly requested; stored receipts are not new physical evidence.

FreeCAD exports were reopened as both FCStd and STEP. Every native shape was checked for validity and positive volume; source primitives were compared with analytical volumes. GLB was imported again to check all part IDs and bounds. Binary STL was read back to check scale and bounds. These are digital integrity checks, not structural, manufacturing or safety validation. Summed solid volumes include tooling and overlapping equipment envelopes and are not installed material takeoffs.

Primary references: FHWA prefabricated bridge assembly, Caltrans construction guidance, Freyssinet erection methods, voestalpine 60E1 drawing and ArcelorMittal nominal rail data. They support the methods and reference dimensions; none approves these proposed assemblies.

The editable generators are in `engineering/visual_models/`. Blender 5.2 and FreeCAD 1.1 were used through their installed Flatpak applications. Pillow and Roboto fonts compose the callouts. Runs use four Blender CPU threads and one native CAD thread. Raw images, logs and intermediate source snapshots stay in ignored `build/` directories.

```
python -m engineering.visual_models definitions --output build/model-definitions
python -m engineering.visual_models render --definitions build/model-definitions \
+  --assembly arrestor-storage --output build/model-run/render/arrestor-storage
python -m engineering.visual_models cad --definitions build/model-definitions \
+  --assembly arrestor-storage --output build/model-run/cad/arrestor-storage
# Repeat render and cad for the eight IDs in model-definitions/manifest.json.
python -m engineering.visual_models.publication publish --definitions build/model-definitions \
+  --runs build/model-run --output build/model-publication
python -m engineering.visual_models.publication verify --output build/model-publication
```

Publishing requires fresh output paths and all eight completed runs. Native workers execute as ordinary user processes; source hashes and before/after checks detect drift, but are not an OS sandbox or independent authenticated measurements.

4.4 Current engineering interfaces

docs/design-engine/engineering-closure/README.md

This is the evidence-enabled continuation of the research twin. It preserves the six earlier source-bound packages and selected reference. The current configuration keeps six passive 17 m cars, five short wayside launch banks and independent braking. No onboard traction or traction store is added.

The new executable adapters can assess supplied installed equipment, structural results, force/time brake curves, site geometry/geotechnical inputs and complete cost records. They calculate values and residuals instead of permanently returning unknown for every input. Missing, unsupported, failed and authenticated evidence remain different states. None grants manufacturing or operating release.

Run

```
./osr design railway plan
./osr-complete plan
./osr-complete motor-run --state build/current-motors --max-new-runs 8
./osr-complete assess --output build/current-closure --motor-state build/current-motors
./osr-complete verify build/current-closure
./osr-complete verify build/current-closure --require-complete
./osr-complete publish build/current-closure
./osr-complete verify-publication
./osr-complete motor-publish build/current-motors
./osr-complete motor-verify-publication
./osr-complete variants-publish
./osr-complete variants-verify-publication
```

The motor campaign is bounded, locked and restartable; it preserves interrupted, failed and pending requests. Source and numerical-tool changes create new execution identities. Published motor records can be inspected with recorded-execution integrity checks on another machine; fresh native reuse/replay still requires the exact current numerical toolchain. Recorded integrity does not certify local replay or a usable force map.

The strict command returns exit code 3 while the engineering frontier is open. `assess --input-root DIR --input-manifest FILE` imports owned, versioned record files. A trusted calling application may supply the external authentication and independent numerical replay callbacks plus their context identities. The CLI has no self-approval switch, imported executable authenticator or local trust anchor. Without the actual external verifier, signed claims remain held.

Implemented interfaces

- Evidence: current T01–T10 subjects, actual referenced bytes, exact candidate/ source/article bindings, calibrated channels and typed records/events, complete cases/repetitions, pre-approved procedures and guarded limits, derived reviews, independent witness, prerequisites, NCRs and execution times. Numerical and supplier records do not become measured physical tests.
- Equipment: net primitive geometry, source blank/cut reconciliation, complete replacement-account mapping, rigid placement, mass intervals, CG/inertia, hardware/service flexibility, lifting and conservative swept access. A changed physical assembly becomes an explicit revision proposal requiring canonical rematerialization and downstream reevaluation; parent results are not inherited.
- Structures/brakes: typed load cases, mesh/refinement and uncertainty margins; force-driven stopping, wheel-adhesion limits, energy balance and repeated lumped brake heat. Material/joint correlation, fade, WSP and safety floors remain open.
- Civil works: all counted stationary stock, including the wayside motor, enters beam/support reactions and bending diagnostics. Known unlocated span accounts have explicit placement hypotheses. Surveyed footing inputs receive bearing, sliding and elastic-settlement checks. Actual full substructure, torsion, seismic, flood, pile and erection design still require their own work.
- Economics: every canonical occurrence and required whole-system scope needs traced prices/quantities; disjoint cashflows and design-life/discount assumptions produce conditional present costs. Missing installed scope is not free, and no protected-service or economic winner is automatically selected.
- Operating map: sampled voltage magnitude/current, energy, Cartesian/domain coverage, zero-current/off-window behavior, explicit window boundaries, exact installed phase/gap/history context and per-component periodic RC duty. Every scoped support record must be externally authenticated and independently replayed against the exact source/tool/request/output. Four-dimensional map claims do not silently cover other frequencies, gaps or transient histories.

- Journey method: a separate current derivative preserves the retained point-train equations, rejects invalid voltage and reports an initial stall without querying negative route positions. It does not invent an initial brake-hold/ramp controller, safety curve or usable motor force.

Actual 20 m motor method

The new, separately source-bound field/circuit kernel supports 40 periods, ten parallel paths per phase and eight series coils per path. The retained 25-period method remains unchanged. Native workers have explicit node, memory and runtime limits; source/tool identities, exact inputs, actual mesh, traces, losses and discrete power/current residuals are retained, including failures.

Low-voltage zero, forward/reverse phase and moving-entry controls are component experiments. A small positive at-rest force or numerical convergence does not establish operating thrust, converter control, finite-height returns, cooling, section handoff or a train-level operating map. No force is scaled from overlap or multiplied by the number of cars/banks.

Before a rated field solve, a conditional sinusoidal power screen compares one active bank with the train demand and counted copper loss. The reference 375 V peak/140 A peak envelope is insufficient for the approximately 102 t accounted normal-load GFRP train at 20 m/s and 0.4 m/s². A separate rewound hypothesis counts 45 mm² conductors, deeper slots and thicker backiron, with a 230 A peak ceiling while retaining the 4 A/mm² RMS limit. Its optimistic electrical headroom is a necessary bound only. It adds wayside material and invalidates mounting, civil, converter and cost evidence. The coarse/fine-time moving experiments do not establish a converged operating map.

`variant_design.evaluate()` propagates reference, wider-return and rewound research candidates through inventory, full stock geometry ledger, BOM, station dead loads, braking/motion contexts, civil overlays and disabled procurement/manufacturing drafts. Every changed motor remains a new physical candidate. This does not resize the piers or qualify the new winding.

The completion campaign and implementation expose these input contracts. The project front door retains the wider local manufacturing, planning, delivery, ERPNext/Frappe HR, supervision and controls remit. Executed validation records the retained experiment and current closure identities, exact checks and unresolved physical work. This remains an unreleased research/prototype design with an explicit frontier.

4.5 CAD and FEA backends

docs/design-engine/craft-backends/README.md

OpenSourceRail can now construct source-bound component geometry in SolidCraft and run matched mechanical method controls in FieldCraft. This is an opt-in backend comparison, integrated with the existing candidate and delivery records. FreeCAD, CalculiX, Gmsh/GetDP, IFC/Bonsai, Chrono and SUMO remain available.

```
./osr design backends plan
./osr design backends install solidcraft fieldcraft
./osr design backends run --output /path/to/private/fresh-run
./osr design backends verify --output /path/to/private/fresh-run
# Optional newer-format / tagged-surface compatibility test:
./osr design backends install solidcraft-rc
./osr design backends run --include-rc --output /path/to/another/fresh-run
```

Linux requires bubblewrap, GNU time, CalculiX (`ccx`), and the existing `org.freecad.FreeCAD` Flatpak. Installation downloads only the selected pinned upstream release, verifies its archive and executable digests, and retains its licence notices. There is no automatic unconfined execution fallback.

What is compared

Case	Source occurrences	Scope
Selected 20 m beam	6	Concrete walls/flanges and two end diaphragms; reinforcement, tendons and anchorage excluded
Selected 17 m side sill	4	Aluminium material walls in installed coordinates; no exploded-display translation
Research launch shuttle	14	Separate research layout, with unknown equipment/rubber mass retained
Asymmetric unit control	1	Unequal dimensions and nonzero translations expose mm/m and placement errors

Each occurrence retains its source ID, physical candidate and component geometry identity. Changing a provider changes the evaluation context, not the physical candidate. The selected geometry subsets refer to their existing BOM, manufacturing, procurement, cost and IFC packages; their costs are not added a second time. The shuttle does not inherit the selected train's costs or approval.

SolidCraft scripts convert metres to millimetres explicitly. Each occurrence is a separate solid, with explicit material-density hypotheses where available. The provider's default material and mass are never installed-mass evidence. Generated, reopened native, reopened STEP and independently imported STEP solids are compared by per-part volume, centroid and tight bounds. Curved tessellation measurements are diagnostics; FreeCAD/OpenCASCADE supplies the independent exact B-rep comparison. No Boolean-union volume or complete equipment CG is claimed.

The mechanics controls are a one-metre isotropic cantilever on two hex20 meshes, and an axial rod. FieldCraft receives explicit `m-kg-s-N-Pa` model units. Named supports/loads survive remeshing; a named geometry face set is also checked on design reopening. Independently generated CalculiX decks and analytical theory provide comparisons. These are method controls, not complete beam, vehicle or arrestor structural analyses. The next admission tasks remain full assembly loads/connections, torsion, buckling, contact, fatigue and applicable material models. GetDP remains the electromagnetic route and IFC remains the BIM route.

Verified compatibility boundaries

- The pinned compatible pair is SolidCraft 0.30.1 / FieldCraft 0.7.5. FieldCraft's convenience `open` rejects `.sc`; the adapter uses its tested `geometry.open` command without renaming the source file.
- SolidCraft 0.40.0-rc.2 writes format 50, which FieldCraft 0.7.5 (format 48) rejects. The optional probe records this failure and tries face-tagged 3MF. A tagged surface is not an exact solid, editable parametric history or a complete transfer of analysis/material definitions.
- SolidCraft native/STEP reopening retains body names. FreeCAD's STEP importer exposes product labels instead. The pilot reports this name-retention hold. An unambiguous volume/centroid/bounds bijection verifies geometry separately; it does not pretend the importer preserved source IDs.
- SolidCraft density assignments do not create FieldCraft structural materials. Analysis materials, loads and constraints must be declared independently.

Source releases, commits, asset hashes, executable hashes and licences are in `tools.json`. The pilot policy bounds 25 parts, one sequential worker, at most 32 native processes, 120 seconds and 2 GiB address space per native process. The admitted full workload uses at most 29 process slots with the optional compatibility probe. This is not an aggregate memory

cgroup or independently authenticated physical evidence.

Evidence, reuse and promotion

Each fresh private run retains the exact SI cases, JSON commands, source and tool pins, input/output hashes, stdout/stderr, wall/CPU/RSS measurements, comparisons, failed imports and delivery sidecars. Files are never overwritten. Bubblewrap sessions hide home directories, expose designated inputs read-only, allow writes only to the native session, and have no network or graphics devices. FreeCAD uses its separately recorded Flatpak permissions and observed native limits.

`verify` checks the complete inventory, request context and source-bound receipts, then recomputes comparisons from stored readbacks. It performs no fresh solve; this distinction is explicit in the result. Geometry, assumptions, code, tool or resource changes invalidate reuse. Tool path scopes supplement OS isolation; they are not a claim that upstream path handling has no races.

`software_pilot_passed` reports these bounded comparisons only. Native naming, material and unsupported-physics limitations remain visible. It does not select the tools globally, admit arbitrary geometries to native evaluators, or grant engineering, manufacturing or operating acceptance. Promotion should follow a larger matched CAD/FEA campaign with complete interfaces and independent review.

Upstream references: SolidCraft commands, SolidCraft source, FieldCraft source.

Detailed launch assembly follow-on

The R3 launch-shuttle dossier contains dimensioned parts, a counted assembly, native CAD/STEP checks, an actual clevis-ear bearing-load FEM comparison, and a bounded fabrication estimate. It preserves the earlier pin-bending failure and distinguishes local component results from full-shuttle or propulsion qualification.

Extended same-candidate viaduct workflow

The 12-span CAD, mechanics and manufacturing comparison now exercises reusable CAD, analysis and delivery interfaces. Download the actual SolidCraft/STEP/STL models, inspect independently measured geometry, compare two coarse/fine FieldCraft/CalculiX studies, and follow each candidate into its quantities, scoped costs and provisional production route. Original failed executions and the separate box memory-budget revision remain explicit.

```
./osr design backends providers
./osr design backends viaduct plan
./osr design backends viaduct verify-publication
```

This is an opt-in research path with bounded eligibility. It does not promote new tools globally or establish bearing/contact, reinforced-concrete resistance, manufacturing release or complete installed bridge cost.

4.6 Retained v0.6 engineering findings

docs/design-engine/v06-design-closure/README.md

v0.6.1 is published. The old campaign is superseded only where it controlled software publication. All six open engineering gates and the partial Workbench gate remain visible below and feed the v0.7 Samawah integration milestone. No finding or engineering evidence is closed by this documentation handover.

Vehicle manufacturing-document coverage draft

Three candidate-matched bogie/coupler source-stock drawings

Door-system inventory and assembly-gap audit

Compact-drive current-source regeneration and delivery holds

Samawah Chinese-bulk vs project make/buy costing packet

Compact-drive corridor attempt and retained evidence

Installed-vehicle review and compact-drive geometry results

Compact-drive split-lane support-family trials and independent review

Native GIS source-only import prototype and remaining gate work

Run07 native GIS readiness audit

Current run07 disk and memory preflight

Post-cleanup run07 resource preflight

Latest post-cleanup disk and memory sample (not execution authorization)

Samawah historical recalculation and its current-source body/map recalculation, with the corrected source-bound candidate cost bridge and fourteen-scope cost audit

Samawah acceleration sensitivity

Samawah close-station consolidation study

Samawah per-line demand and whole-city cost sensitivity

Erratum for the sealed all-city audit packet

The machine-readable gate ledger is status.json. Run `python3 tools/verify-v06-release-closure.py` to check source-bound evidence and engineering design-closure readiness. It currently reports seven open/partial design gates. This ledger does not decide whether software is published. The separate `./osr release software readiness` command inspects the retained authenticated review profile; its pending state does not undo the published v0.6.1 release. The verifier checks ledger completeness and hash integrity, not the engineering meaning of a closure claim.

This is a supplementary, source-linked audit. It does not modify the frozen reconciled-pilot publication packet. The audit checks controlled references in that packet's delivery mapping; “missing” does not mean a repository-wide search. It records the currently held document coverage and does not authorize manufacturing or ERP release.

Regenerate the report from the repository root with:

```
python3 tools/vehicle_document_coverage.py
```

Engineering gate status

The campaign retains six candidate-integration gates plus an all-city outline reconciliation gate. These are engineering continuation requirements for v0.7, not prerequisites that withhold the published software release. The complete city catalogue and all open findings remain retained. Physical qualification and operating approval remain separate from closure of this digital work.

Required work	Current state	Evidence / remaining closure
Installed vehicle	Open	The detailed 9f6d research candidate still lacks complete installed mass, verified supplier-fit interfaces, measured or verified installed inertia, tool/removal paths and downstream force/civil consequences. Five child geometry configurations have been screened, including two corrected split-lane support variants; all remain on HOLD with thousands of nominal/interface

Required work	Current state	Evidence / remaining closure
		findings unresolved, and no matched journey/civil study closes the parent. The LED, inter-car, coupler and gangway audit finds no verified articulation envelope or full installed service fit. The original 9f6d archive has now been replayed as a corrected child evaluation: physical candidate `9f6d2230…` is unchanged, evaluation `a7a9ee09…` corrects all 24 door-motion vectors, and the frozen parent remains unchanged. This corrects evaluation data only; the geometry screen is still HOLD and supplier fit, installed mass and downstream loads remain open. A separate current-bundle installation run for source body 907da… corrects the same coordinate representation for its distinct candidate 17ce. The six-car passive-wheel development reference in `engineering/development-baseline.toml` remains unchanged. See the split-lane support-family record, vehicle subassembly audit, 9f6d door-motion correction and 907da door-motion correction.
Controlled research documents	Open	The supplementary audit of the retained 668-part delivery map finds 41 assemblies with 82 missing assembly-document slots and 4,676 missing part-document slots; all delivery document references are null. Three source-bound gross-stock SVGs now cover a bogie seat blank, bolster RHS blank and coupler clevis blank, but they have no finished geometry, inspection limits or process times and are not controlled part documents. The lighting, flexible inter-car services, coupler and gangway parts also lack controlled drawings, inspection limits and complete assembly evidence. Whole-vehicle document coverage and manufacturing release remain open.
Mapped corridors	Open	Baghdad's revised 205 m maximum-deviation geometry pass is still on the older candidate and has no candidate-specific service or price. Samawah has a five-line, 57.873 km quantity and conditional fleet outline bound to the current-source package-07 body; 0.4/0.6/0.8 m/s² acceleration hypotheses remain separate calculations. A new current-source midpoint alternative merges one 79.977 m line-1 marker pair, preserves route length, reduces markers from 26 to 25 and line-1's kinematic request from 2,041.8 to 1,985.3 seconds, and removes one 120 m platform-envelope definition failure. Base fleet and counted vehicle BOM/process subtotal are unchanged; high fleet falls from 88 to 87. It is unselected, with station access, transfer paths, catchment and site suitability unverified; the alternate has no complete city scope totals, installed CAPEX or NPV. Force/braking/grade and detailed civil loads remain unknown. The earlier fourteen-scope China-bulk/project make-buy sensitivity belongs to physical candidate `756a6887…` / body `9489e4f2…`. The current `9f37c5d8…` city / `75075df9…` body has only a candidate-linked body BOM/process subtotal plus five quantity-matched Class C infrastructure-rate fragments; no complete scope totals, installed CAPEX or NPV. See the current-source station alternative, corrected current-source cost bridge and current calculation report.
Required work	**Current state**	**Evidence / remaining closure**
Native GIS return path	Open	The source now has a single `reprice --refreshed-output` CLI path that verifies the retained edit/request, runs the bounded city job, then exports and verifies a study-bound RustGIS project. Synthetic CLI and importer tests pass (29 total). No fresh native edit city recalculation export has been run on this source epoch. The latest post-cleanup resource sample narrowly passes the 8 GiB disk reserve, but run07 remains

Required work	Current state	Evidence / remaining closure
		source-stale and unadmitted; no current-source run08 plan, supervisor, independent review or authorization exists. The gate stays open pending a separately reviewed current-source round trip. See the CLI refresh-loop audit, source-only importer prototype, and run07 readiness audit.
Candidate business lifecycle	Open	A sealed delivery proposal exists for the 9f6d body and is identity-matched to that body. The distinct 75075 compact-drive child cannot inherit it and was correctly refused as a mixed identity; it still needs its own canonical downstream bundle. The 9f6d proposal has no complete controlled document set/review pack or native lifecycle evidence. See the candidate lifecycle audit.
Unified Workbench	Partial	A source-bound view covers the reconciled 1.5 km pilot, with its own candidate identity. It does not yet cover the closure campaign's full city-brief-to-compatible-alternatives path, same-candidate journey/cost/delivery, drawing/BOM/manufacturing review or reconciled downloadable dossier. See `campaigns/v06-design-integration-closure.toml`.
All-city outline reconciliation	Open	The evaluator accepts exact per-line demand inputs, and a source-bound five-line Samawah sensitivity now propagates those inputs through fleet and fourteen-scope costs. Its 1,200 pphpd perturbations are planning sensitivities, not surveyed demand. The published 266-city catalogue still uses the shared 4,000 pphpd / 200 s study control and transferred reference rates; city forecasts, final alignments, China-volume versus local-manufacture cases and same-candidate recomputation across the catalogue remain open. See the Samawah sensitivity.

The all-city audit confirmed the catalogue and v0.4 front-door field coverage, not the correctness of local demand, rates or physical alignments. The mapped-corridor and native-GIS execution artifacts are retained under the private, source-bound run record. Their limitations and identities must be carried into any new plan. A geometry pass is not an operating route, and a UI upload alone would not close the GIS round-trip gate. The refreshed CLI workflow is documented separately; the complete native execution remains unrun on the current source epoch.

Bounded vehicle stock-drawing packet

Three A3 research SVGs have now been generated from exact source-stock geometry for body candidate `9f6d2230…`, engineering revision `dfb48ca3…`, and its identity-matched delivery proposal. The packet records all source and output hashes, the command/resource limits, and the execution boundary. It is explicitly a gross-stock geometry draft: the source marks all three finished geometries unverified, and no tolerances, mating interfaces, inspection limits, or measured routing times were invented. The generator consequently reports zero documented part IDs and leaves the controlled-document gate open. See the packet and individual SVGs.

Vehicle services, couplers and articulation

The 9f6d body proposal has counted LED lighting, five inter-car couplers, five gangways and flexible electrical services, with allowance-based mass and direct-cost rows. It does not yet have verified LED loads, rated/flex-tested inter-car wiring, a moving articulation envelope, complete SolidCraft/FEM subassemblies, or controlled manufacturing and inspection documents. The source-pinned subassembly audit records its bounded BOM totals and exact closure work.

All-city baseline audit

The read-only audit was bound to closure commit `ebeb3b769a9b416180c64124debc1f23317ca277`. It found 266 city outlines, 1,565 lines, 13,580 station-table rows, 14 cost scopes and priced energy cases for all 266 cities. The v0.4 front-door categories—maps, line/service, energy, capital/funding, local-evidence and regeneration headings—are present in all 266 city entries. The audit checked 5,586 declared payload hashes (492,192,548 bytes) and found no manifest identity/hash failures.

Those are coverage and integrity results, not proof that the estimates are locally accurate. All 1,565 lines share the same 4,000 passengers/hour/direction, 200-second headway and 276 normal places/train study controls; all-city local rates are `transferred-reference-estimates`. The city studies remain resource-envelope outlines, not executable schedules or dated calendars. All 266 are linked to reference vehicle candidate

`756a68870162f7559b052b05c9b201728c9f448e3bb61114f287848c02f2d3a3` and body `9489e4f2666fd3b3c73232c86df3f12adf42a5531eb27373834cf3fecb1a3117`; none is bound to the detailed `9f6d…` research body. No supplier quote, surveyed city demand, site geotechnics or physical acceptance is implied.

Audit anchors: city-dossier manifest `a9337870a1a5db059a65b2b083011d66ec1ee56f69f2c0cec76f53c172443893`, dossier index `8661b6456bb4931fa0f67e6060eec6e37be5738fc50eca092523a381fa74b1ab`, city-atlas manifest `6f814ccc64a572921aece16a219eb231fbf8ecadfb453a3b92742200956d4c8c`, outline manifest `f53a8b924834679b3937ad4a2da24bdb89b65c26ee5861ae6c65b77986d2e694`, and ordered city-manifest-set digest `a6f85d8f4ee021261962db872ca30bdb458ce6f0885840a8288276800384c402`.

Research images

The Samawah acceleration sensitivity keeps the 0.4 m/s² reference control and evaluates 0.6 and 0.8 m/s² as unselected hypotheses. The larger values raise the calculated inertial force and power terms; no available drive map or achieved-acceleration evidence is present. See the separate calculation note.

The root README already includes renderings from the SolidCraft and FieldCraft engineering work: vehicle and bogie assemblies, launch shuttle and coupling, viaduct assembly and a FieldCraft displacement result, plus manufacturable component details. Those images support inspection of the research models; they are not manufacturing drawings or qualification evidence.

The current-source Samawah station-consolidation alternative recalculates the close 79.977 m stop pair using the current compact-drive body and preserves the mapped route. Its result is a planning screen only: it reduces one marker-envelope definition failure, while access, catchment, transfer paths, site feasibility and full corridor cost remain open. Attempt 02 is retained with an explicit warning because its manifest omitted nested input snapshots.

A stop-preserving route screen from the unmerged map tests 25, 50 and 100 m simplification tolerances. It shortens the map from 57.873 km to 54.518–53.993 km, but retains the close-stop platform-envelope hold (56.552 m gap). A separate chained station-merge plus route screen starts from the midpoint alternative with 25 markers and no envelope-screen failure, then tests the same tolerances: 54.514/54.173/53.963 km, with conditional total fleet requests 39/57/84, 39/56/84 and 39/56/84. Its unpriced guideway screen reduces reference 20 m beam units 5,7945,456/5,422/5,402, shared bent positions 2,9022,733/2,716/2,706, running rail 231,493218,055/216,693/215,853 m and fastening sets 385,825363,427/361,157/359,756. The combined options are unselected. Their four shared evaluator holds are absent exact-body propulsion/braking/thermal/charging/service evidence, unevaluated civil and station assemblies, unknown full costs/NPV, and an unresolved roster for the 18-hour operating day. Civil classes are proportionally transferred as a planning assumption; site/access feasibility and physical civil design remain open.

Native GIS prototype status

The source-only importer and bootstrap regressions are now part of the shared software workflow, with local and independent checks recorded in the prototype report. This is code-level evidence only. There is no run-07 GIS authorization or execution, and the native-GIS return-path gate remains open until the complete project edit and recalculation round trip is generated, independently checked and reconciled to the same candidate. The retained browser-upload proposal is not a substitute: it is stale against the closure source tree, has no admitted supervisor, and does not exercise the native editor. See the readiness audit.

Door-system inventory audit

A candidate-matched read-only review covers 93 proposed part types and 10,296 listed pieces across the door openings, leaves, glazing and electrical operator sets. Its counted six-car allowance is 1,180.332 kg and $33,260 base direct parts/process cost, with substantial low/high bounds. Finished geometry is unverified for every item; the existing GLB is static presentation geometry. The installation proposal contains finite travel and swept-box hypotheses. The original 9f6d archive preserves its historical vector mismatch, while a separate source-bound child evaluation corrects all 24 door travel vectors without changing the physical candidate or rewriting the old receipt. The motion data is internally consistent, but the result is not a kinematic CAD assembly or an integrated collision, clearance, supplier-fit, service/removal or door-safety proof. A distinct 907da source-body evaluation is also corrected and remains separately identified. See the part-level audit and source hashes. This advances component accounting while leaving the installed-vehicle and controlled-document gates open.

Candidate-specific procurement sensitivity

A source-matched Samawah five-line scenario compares project make/buy and Chinese bulk purchase across fourteen cost scopes. Under unquoted Class C rate, freight, duty, QA and factory assumptions, the China-bulk case is +$5.224m base CAPEX and +$9.448m base NPV relative to project make/buy. No project volume discount or binding supplier offer supports a lower-price claim. The scenario uses planning demand and cannot be transferred to the remaining city studies. See the complete source-pinned packet.

A regenerated compact-drive body reproduces the 75075 physical candidate ID under current source pins with a distinct evaluation ID. The current-source city adapter accepts that exact evaluation and emits a held Samawah outline. The corrected cost bridge prices candidate BOM/guideway fragments and separate auxiliary-energy and corrective/disruption sensitivities. The fourteen-scope audit records what remains unpriced. An earlier bridge source map pointed to a historical body; that packet and its correction trail remain preserved. None of this closes service, installed mass, civil loads, complete delivery or lifecycle pricing. The geometry screen remains HOLD. The earlier city packet and its source-refusal log remain preserved as historical evidence. See the body replay, current-source city report, corrected current-source cost bridge and retained earlier delivery audit.

V. Cost and lifecycle

From candidate quantities to production, delivery and bounded business integration.

5.1 Model-based cost and delivery

docs/design-engine/model-based-cost-and-delivery.md

Implementation status — 1 November 2026. `engineering/design_search/model_cost.py` prices the coupled railway research bundles, and `engineering/development_baseline/system_cost.py` now adapts the exact selected v0.6 vehicle into a separate candidate-bound system scenario. The selected vehicle remains six passive-wheel cars with its passive aluminium secondary interface. Installed propulsion remains unselected; powered length, acceleration, speed and headway are candidate-specific design variables. Historical station-LIM studies retain their original assumptions and results. Research alternatives retain their own physical/evaluation/selection identities. Safety, accessibility and engineering qualification are never waived by a cost estimate.

First executable model-cost milestone

For each executed research bundle, the campaign prices occurrence-level body stock and retained equipment accounts, unknown joints and architecture packages, propulsion primary/conversion/support/cooling and installation, generated civil span quantities, bounded foundations/piers, transport, erection, rail installation, stations and service-battery chargers, utilities/depot, fleet, factory tooling/cells, electricity, operating staff, maintenance, renewals, recovery, reliability and end-of-life decommissioning. It emits low/base/high IQD estimates, cost-line identities, source evidence, a production-cell/tooling plan and construction-duration estimate. Production capacity and civil schedules assume two 8-hour shifts per day. Operations cost low/base cases cover 16/18-hour service with two shifts; the high case models 24-hour operation with three shifts. Electrical service includes 35/35/50 kW-per-car hotel-load assumptions and declared battery charging losses. Seven dated July 2026 Iraqi CSO material/wage records are used where their units match; remaining rates and site/asset quantities are explicit Class C assumptions.

The cost result also carries a candidate-bound provisional manufacturing route for every located nominal gross body-stock occurrence. Each route records the selected process hypothesis, material/form, nominal mass, manufacturing-group and operation identities, low/base/high labor-hour proxies, and the matching factory-gate cost-line ID. Hours use the declared process-hours-per-kg range plus setup-hours-per-stock, then scale by the candidate fleet. The route coverage is one-to-one for located body stock only; inherited unlocated equipment, unknown joints, propulsion, final assembly and civil work remain separately estimated or held. These are Class C costing proxies under the stated two-shift production schedule, not measured cycle times, completed shop routings, equipment validation or released manufacturing instructions. The source MBOM's `labor_h` remains unknown until supported by process evidence.

The research campaign is model-cost complete for each declared synthetic corridor, not for a city budget. The selected reference also has a source-bound system estimate across all 14 scopes for a separate 6 km periodic corridor, three stations, 2 km spacing and two physical trackways. Its low/base/high installed CAPEX is IQD 59.16 / 286.49 / 1,721.96 billion; its 30-year whole-life NPV is IQD 123.14 / 672.06 / 5,806.45 billion. The costing allowance is six active plus two reserve trainsets. These are wide coupled Class C scenarios, not confidence intervals, quotes, a city plan or a demonstrated economic advantage. The source-bound selected system record keeps the physical vehicle ID separate from the route evaluation and cost scenario identities.

The selected six-car vehicle separately has an estimate for 2,482 geometry occurrences and ten positive residual mass accounts: IQD 1.500 / 6.518 / 29.497 billion low/base/high for factory production, assembly and a 50 km reference delivery. The 10.090 t/car residual remains unlocated and possible overlap is unknown; no overlap credit is applied. The system cost does not add that separate vehicle total: it prices the selected geometry and residual-account rows as fleet units.

The system estimate includes provisional support/cooling and a charger package for every station/direction platform. At 200 kW/car and 35 seconds dwell, the declared normal hotel-load case has +0.074 kWh/car per-cycle estimated margin; the 50 kW/car adverse case has a −0.686 kWh/car shortfall. Battery charge acceptance, solar yield, converter/grid capacity and repeated duty remain unmeasured. City route quantities, geotechnical/site data, supplier prices, complete installed mass, force/thermal maps, physically supported journeys and safety acceptance remain open. Estimates from other candidate bundles do not transfer to the selected reference.

Provisional factory resource envelope

The selected manufacturing export and immutable factory draft now calculate 27 aggregate resource scenarios: batches of 1/6/12 six-car trains over 60/125/250 working days, each with the cost ledger's low/base/high process-hour case. For this sensitivity only, one process hour is interpreted as one person-hour. At 75% productive utilisation each worker supplies six productive hours in one eight-hour shift; two shifts need two distinct sets of staffed positions. Pooled labour gives an arithmetic capacity bound, while separate process-family staffing shows the effect of keeping those resources apart. Every input package retains its cost-line, source and occurrence links, and delivery hours are excluded.

For one train over 250 working days, the 21,428.373-hour base case requires 7.143 pooled positions per shift, rounded up to eight positions on each of two shifts (16 shift positions before relief). This is a declared staffing scenario. Actual crew skills,

relief, workstation/machine capacity, concurrent working, assembly/cure precedence, procurement lead times and dated calendars remain open. The envelope neither supplies measured production labour nor releases a shop routing, staffing roster or factory throughput commitment.

Why calculated costs are enough for design exploration

A complete planning scenario does not require an actual vendor quotation for every manufactured panel, chassis, beam, pier or factory process. Given a dimensioned part, material, production route, quantity and explicitly assumed rates, the platform can calculate a transparent low/base/high engineering estimate and compare alternatives. A third-party quote improves confidence when procurement is imminent; it is not a prerequisite for calculating a provisional design cost.

Three separate completion levels

OpenSourceRail records completion by scope and keeps three decisions independent:

1. Simulation complete: every required calculation in the declared study scope has run, and each input is measured, benchmark-supported, or an explicitly identified bounded assumption. A cost scenario can reach this level without a supplier quote. The record names its scope so a complete cost simulation cannot be mistaken for a complete propulsion journey or vehicle model.
2. Engineering validated: the relevant physical models and limits are supported by credible evidence, results meet their supported engineering requirements, and an independent review has accepted the evidence. Provisional assumptions can support exploration but cannot satisfy this level.
3. Approved for manufacture or operation: the candidate has passed the separate safety, accessibility, manufacturing or operating requirements for the specific release, with the required independent approvals. Manufacturing and operating approvals remain separately scoped.

For the current coordinated comparison, the research Class C cost scenario is complete for 112/112 evaluated bundles across 14 cost scopes. The selected reference has a separate source-bound vehicle estimate and a selected-candidate 14-scope synthetic-corridor estimate. Both price account allowances without closing installed mass or possible account overlap. Integrated force-driven journeys, site-specific city cost, engineering validation and manufacture/operation approval remain open.

The existing work includes a component/material master (engineering/component_registry), gross/net takeoffs (engineering/sourcing/manufacturing.py), candidate-bound Iraqi material-plus-labour screens with a configurable uplift (engineering/sourcing/fabrication.py), geometric production hypotheses (engineering/railway_progress/production.py), conditional A/B/C manufacturing and lifecycle costs (engineering/reference_cell), and ERPNext/Frappe handoff components (services/integration/osr_integration). The selected-corridor adapter closes a declared synthetic financial scope; these tools still do not form a site-validated, consistently priced city-delivery ledger. Integration must preserve scope and evidence distinctions.

There must be three distinct completion flags:

- Model-cost complete: every required scope has a quantified bill of quantities, explicit assumptions, bounded scenario costs, scope coverage and no unclassified missing items. Class C scenarios are permitted.
- Procurement ready: quantities/specifications have been checked for purchase, supplier or manufacturer offers and logistics/terms are reviewed, and approvals are granted. This is not implied by a model-cost-complete result.
- Physically/safety accepted: relevant vehicle, guideway, traction, braking, accessibility, control and site evidence has passed the project's independent engineering/assurance process. No economic model can set this flag.

Canonical stage-by-stage process

Stage	Canonical input	Generated output	Next decision / feedback
1. Define remit	City geography, passenger demand, climate/site assumptions, service/comfort, safety/accessibility, local production	Revision-bound requirements, scenario ranges and hard vs revisable constraints	Reject incomplete/contradictory scope
2. Generate architecture	Requirements, typologies, material and assembly grammar	Separately identified rolling-stock, propulsion, civil, station and production candidates	Explore alternative families, not just dimensions
3. Detail parts	Candidate, material forms, CAD/IFC and interfaces	Occurrence-level EBOM, geometries, installed positions, mass/CG/inertia and unresolved account ledger	Design missing support, HVAC, cable, fastener, joint and access items
4. Evaluate function	Same candidate, loads, motor/brake map, site/environment envelopes	Model-domain/fault results, force/heat/strength/clearance margins and explicit blockers	Revise geometry/material/topology or request a better model/test
5. Plan production	Approved research configuration, part stock/blank and assembly graph	Gross takeoffs, MBOM, cut lists, operations, equipment, tooling, workstations, crews, yield and inspection plans	Revise methods if production time or access is poor
6. Calculate costs	EBOM+MBOM, process routes, reference rates, amortisation and uncertainty	Low/base/high factory-gate, delivered, installed and lifecycle costs, coverage and source classes	Prefer cheaper Pareto options; surface uncertainty
7. Plan the city	Route-specific civil, stations, fleet, depot, factory output, construction methodology	Production capacity, two-shift schedules, cranes, logistics, CAPEX cashflow, OPEX/renewal and passenger service	Recalculate fleet, schedule, plant scale and capital requirements
8. Prepare execution	Reviewed candidate and approved data	ERPNext/Frappe HR Item, BOM, routing, workforce, purchasing, QA and project draft records	Human authorisation before ordering or issuing site work
9. Build, test and operate	Fabricated serial parts, site measurements, commissioning and independent approvals	As-built revisions, inspection/test records, FUXA monitoring, maintenance/repair data	Feed true material usage, hours, failures and costs back into new estimates

The existing ./osr design railway, ./osr-twin, ./osr-cell and ./osr design progress commands cover parts of stages 2–6. City planning, ERPNext and FUXA are retained but not one commissioned end-to-end execution flow. Stage completion must identify what is executable, estimated, held and independently accepted.

One cost grammar for every manufactured and installed item

Use a canonical cost record with: candidate/evaluation identity; physical part, occurrence and production batch; quantity and units; material grade/form/source/origin; geometry net and purchase-gross takeoff; scrap/yield; process route and setup/cycle times; crew rates; machine, power, consumable, mould/tooling and QA allocations; packing, transport, lift, installation and rework; currency/date/location; overhead allowance and its base; low/base/high uncertainty; and a link to the resulting line item. Distinguish a finished purchased assembly from raw inputs so that labour is not counted twice.

Proposed calculations, with each term separately stored:

1. Materials: sum of gross purchase quantities multiplied by dated or explicitly assumed unit rates. Gross takeoff comes from nesting, dimensional blank, concrete recipe or process yield; it is not necessarily the finished installed mass.
2. Labour: sum of setup hours divided by batch quantity, operation hours per item, handling, assembly, inspection and expected rework hours, each multiplied by the corresponding loaded hourly rate (or a clearly labelled base-wage proxy).
3. Equipment/process: machine hours × machine rate plus electricity, tooling/mould amortisation, maintenance, consumables, factory handling and dedicated QA not already included elsewhere.
4. Factory-gate: materials + labour + equipment/process + any explicit overhead/contingency provision. A quick Class C fallback is (material + direct labour) × 1.20, clearly named a partial factory screen. If equipment/overhead are itemised, the 20% must only apply to the declared non-overlapping base.
5. Delivered: factory-gate + packing + transport + insurance/customs where applicable + unloading.
6. Installed: delivered + lifts/rigging + site labour + connections + temporary works + equipment testing + interfaces and civil/site preparation. Foundations, utilities and rails are separate scopes where not already in the product BOM.
7. Lifecycle: initial installed system + fleet/energy/staff + spares/maintenance/renewals + downtime/reliability + disposal, as a time-dependent scenario with consistent service assumptions and explicit discounting.

Make the default design-stage assumption three cases (low/base/high), including wage and material-rate sensitivity and alternative production volumes. A 20% uplift is a declared scenario, not a universal supplier margin, a warranty or proof that a finished design is affordable. Scenario results must stay linked to their assumptions.

Worked route: a composite carbody panel

CAD determines actual area, number of laminate plies, core volume and insert/fastener count. The process generator selects one of hand lay-up, vacuum-assisted infusion, press/tool, or purchased panel with feasible applicable material and workshop bounds; it calculates glass, resin/hardener, core, adhesive, consumables, gross stock, mould life, cure/trim hours, crew, QA and scrap. The cost grammar prices this as a conditional factory-gate range, then transport, positioning, joining, sealing and installed inspection. If stiffness, heat, fire or assembly constraints fail, design-search changes the layup, material, shape, joint or production method and recalculates the same downstream installed/economic scopes. The manufacturing cost can be provisional even while physical qualification remains held.

Worked route: a 20 m precast guideway span

Use the candidate's counted concrete volume, reinforcement, prestress and anchors, rail/fasteners and support interfaces. Generate work instructions for mould preparation, reinforcement/tendon placement, casting, curing, stressing, testing, demoulding, storage, loading, haulage, erection, bearings and rail completion. Calculate crew/equipment hours and plant utilisation. Include piers/foundations, cranage or launcher, traffic management, access and utility interfaces as separate but linked site scopes. Never call the current concrete+rebar+base-wage screen a complete beam, installed span or route-kilometre cost.

Bought specialist subsystems

For a wheelset, brake actuator, inverter, HVAC, lift, safety controller or cell without a matched quote, use a separately identified benchmark/analogue or bottom-up engineering estimate with bounded mass, quantity, specifications and installation. Label the evidence Class C/B, unknown subassemblies and uncertainty; do not invent compatibility, fatigue life, railway certification or service duty. Procurement can seek quotations later. When a finished-equipment price is selected, suppress duplicated raw manufacturing charges but retain its supports, installation, testing and lifetime maintenance.

Feed every failure back into design

The coordinator should translate each blocker into an owned action:

- Missing stock/weight -> generate component, allocate approved source account, resolve collision and recompute CG/axle reactions.
- Missing unit price -> compute Class C recipe/benchmark range and request a stronger rate only if it changes Pareto ranking.
- Failed structural or thermal margin -> propose physical shape/grade/layup/joint/support alternatives, not weaker acceptance limits.
- Infeasible station launch -> explore winding length, number of sections, converter duty, train mass and separate alternate propulsion architectures under new candidate identities.
- Excessive beam/site cost -> compare span section, prestressing, foundation, factory throughput, erection method and traffic impact.
- Unbuildable install -> redesign joints, modularity, access and sequence, then regenerate construction/workforce tasks.
- Unsupported physical proof -> retain the numerical result but create a simulation refinement, bench-test or independent review task; do not mark it passed.

Require a bounded number of alternatives and a continuation frontier; archive all evaluations, including failures. Prioritise by expected improvement in whole-life cost and passenger service, feasibility and the value of reducing uncertainty. Only an independently checked selection may advance to the prototype gate.

Implementation increments and regression gates

Increment	Integration work	Test for completion
C1. One cost API	Connect engineering/sourcing/fabrication.py and engineering/reference_cell/manufacturing.py through one current-candidate scope/quantity/evidence record while preserving source-bound output identities	The same panel/span recipe prices identically across the current Workbench, BOM and economic report
C2. Complete occurrence mapping	Bind engineering/component_registry and engineering/mass_ledger_audit.py to every MBOM and cost line; handle assemblies/unknown account balances explicitly	Every installed item is counted once; unmatched masses and scopes are named, not zero
C3. Rich process routes	Extend engineering/railway_progress/production.py with mould/cure, welding, cutting, transport/erection, labour resource and equipment amortisation	Price a carbody panel, frame, bogie, LIM bank and box/twin/U span with alternate methods
C4. Scenario/uncertainty engine	Use dated Iraqi material/wage references where available; add transparent assumed ranges and 10/20/30% sensitivity where applicable	Show low/base/high model-cost completeness even with zero compatible supplier offers
C5. Factory/city aggregation	Feed engineering/reference_cell/economics.py with batch rates, factory capacity, civil/site installation, fleet and city scope; prevent double counts	Route CAPEX, annual OPEX, lifecycle and construction calendars reconcile by exact candidate
C6. Adaptive feedback	Feed priced method and manufacturing blockers into engineering/design_search campaigns; retain rejected alternatives	Candidate changes recompute geometry, mass, cost, energy, forces and schedule consistently
C7. Business bridge	Generate versioned draft Items/BOMs/Routings/Operations/Projects for ERPNext and configured observation interfaces for FUXA	Explicit human approvals; drafts cannot purchase, commission or move trains
C8. Physical and safety release	Preserve current tests, candidate identities and independent evidence/approval gates	Class C affordability never implies supplier compatibility, safe operation or manufacture release

The cost engine and selected-corridor adapter now provide the declared-scope estimates in C4 and a synthetic instance of C5. C1–C3, site-quantity validation and city-route aggregation remain partial. A cost scope can be complete under assumptions while its physical inputs, site basis or purchasing basis stay unvalidated.

Tests should also prove that price-source changes update estimates without inventing new physical parts; material/geometry changes invalidate dependent engineering; date/FX/currency and origin errors cannot silently mix rates; 20% is not applied twice; purchased assembly and fabricated-material charges cannot overlap; and results are comparable only for equal transport capacity, route and lifecycle service.

Immediate development decision

Next: improve the provisional service-energy supply scenario because the adverse 50 kW/car hotel-load case does not recover the declared energy during a 35-second stop. Compare service-battery capacity/charge acceptance, station charging power, solar contribution and an independently powered service supply under the same candidate identity when only assumptions change; a hardware change must create a new physical candidate. In parallel, replace generic residual-equipment price allowances with account-specific component descriptions where source accounts support them, and replace synthetic route/site quantities with candidate-specific planning inputs. Keep residual mass and overlap unresolved until located evidence exists, and preserve all physical, manufacturing, procurement and operating-release gates.

5.2 Business platform integration

docs/integration/v06-platform.md

The explicit runner uses real ERPNext, installed Frappe HR/HRMS, MariaDB, Redis, background workers, gateway SQLite/outbox and FUXA. Its input is the source-verified current export manifest and exact `retained-comfort/HVAC` BOM occurrence. Candidate, body, evaluation and selection IDs remain distinct. One `Nos` at USD 1 represents a synthetic software kit; it does not establish purchased stock, manufacturing yield, a supplier quote or complete installed BOM.

```
python3 tools/automation/v06-platform-native.py preflight
python3 tools/automation/v06-platform-native.py run --output build/v06-platform-local
python3 tools/automation/v06-platform-native.py verify --output build/v06-platform-local/evidence
python3 -m pytest -q tests/test_v06_platform_native.py
```

`preflight` and `verify` start no services. `run` requires Docker Compose and all five pinned runtime images, Node with the locked npm dependencies, Playwright Chromium and at least 8 GiB free disk. It creates a fresh random `osr-v06-<12 hex digits>` Compose project/site, loopback ports and labeled volumes. Source mounts are read-only. Docker CPU quota support is reported by preflight. A one-CPU quota is requested per container only when supported; otherwise CPU limits are not enforced. BLAS/OpenMP threads are set to one, with two explicit RQ workers. These declarations are not an OS sandbox. Cleanup removes only that run's resources, including its database volumes; existing pilot and operator stacks remain untouched. Private source inputs, credentials, native state and logs remain in ignored `build/`; only the sanitized `evidence/` directory is suitable for sharing. Failures retain an explicit failed execution record and uncovered checks. Success is published only after independent teardown attempts verify that no owned containers, volumes or networks remain; cleanup errors prevent success.

The bounded chain creates an immutable engineering-to-native BOM mapping, submitted native purchase/receipt, rejected incoming Quality Inspection, nonconformance/rework Task and passing reinspection, native Work Order and serial/batch manufacture, serial material issue/installation, then an observed simulated condition. A stopped ERP frontend causes an actual durable gateway outbox retry; gateway restart and ERP recovery must deliver that exact event. Its native Issue drives Asset Repair, stock consumption and a replacement serial. Readback obtains candidate IDs from the immutable native execution mapping, selects the chronological latest submitted inspection, and checks native manufacture and installation serial/batch bundles. The inspector is a named in-process fixture user subject to native ERP permissions. A latest passing native inspection is reviewed by a separate API-authenticated ERP user through a native Task; readback checks the actual modifier and exact candidate/test/serial. The campaign guard refuses prior/failed tests, self review and another candidate. Native HR coverage is installed HRMS plus creation and readback of a Training Program. FUXA must display current valid observation and refuse an operator's project write. The package contains no equipment commands; a movement-authority request must be rejected and no controller work queued.

The review guard belongs to this disposable acceptance fixture. It is not a new production handback endpoint. Business review never changes the native physical handback-required flag or grants movement authority. This run does not measure a part, validate HVAC performance, establish employee competence, verify a production workforce assignment, commission a railway, or qualify a vehicle. The current read-only engineering Workbench is not a transactional operator UI. The campaign does not cover an installed controller or field device, real protection, source-loss quality in FUXA, or full hiring/roster/payroll workflows. Those remain separate work and cannot inherit these software results.

The workflow runs native acceptance on relevant main-branch changes and manual dispatch, including production ERP/gateway and selected-export changes. It builds the pinned native services on a disposable runner after reserving disk space. Pull requests run the deterministic offline input/refusal/receipt boundaries only. Native job build budget is 90 minutes; the local runner's individual service/fixture waits are bounded. A cached-image local run should be allowed 45 minutes and at least 6 GiB available RAM; actual elapsed time and image IDs are retained per execution.

Historical v0.4 acceptance remains at `205e8dc158c83fcdc2dca9ceb8bad0a7e8b3f61e`. The later maglev native runner and its source-bound receipts remain separate. This current campaign neither relabels their candidates nor inherits their native result. No completed calculation or submitted business document supplies physical acceptance evidence.

A local cached-image execution on 6 November 2026 completed the covered chain in 284.834 seconds before teardown, with 21 native document readbacks. Separate `verify` replay passed, as did all 29 offline tests. The retained private artifact is `build/v06-platform-final-source/evidence/`, with manifest SHA-256 `9731d3c7dc473aca7b818226d6421f47fac7759f769eac5976334de0add3aeef`. Its screenshot shows gateway status 5, condition code 0, valid quality and the actual source timestamp. Cleanup verified zero owned containers, volumes and networks, with no cleanup errors. The existing maglev pilot stayed running. Docker reported unavailable CPU CFS quotas; no CPU isolation was enforced. The final source inventory includes native image recipes, Nginx configuration, Frappe module/patch lists and the shared device-isolated Chromium launcher. The prior successful source epochs remain preserved with their own receipts.

Three earlier private attempts remain preserved: host CPU-quota refusal before startup (0.253 s), a missing configured-baseline fixture prerequisite (219.528 s), and a deliberate interruption to incorporate read-only review findings

(103.402 s). They are failed/interrupted executions and provide no completed acceptance. No private credentials, native databases or source-bound historical receipts were committed. The CI workflow is prepared; it has not been dispatched by this local execution.

VI. v0.7 evidence and next work

Measured progress, unresolved interfaces and the delivery programme.

6.1 v0.7 execution and evidence

docs/design-engine/v07-execution/README.md

The task ledger records implementation, research controls and remaining holds separately. v0.7 is in development; these results do not constitute a completed installed railway or a released v0.7. The agent plan remains the scope.

The closing software verification record pins code `8d1833f74c`: 480 selected Python tests pass (four optional skips), and the full wheel/station workflow passes its supported browser and Samawah journey. All five retained-method/shared correction workflows pass `bce9d27892`; earlier Rust, Workbench and reconciled results retain their own source identities, with unchanged production objects verified. R01 now has a reviewed bounded corridor baseline. The isolated source tree is about 0.94 GiB and the closing host reading has 21.71 GiB free; shared Git storage is separate.

The subsequent repository review confirmed main `88f5ecc88b` and the successful Workbench journey on that exact commit. The agent plan now prioritises P01/C01 mated door integration, M01 controlled manufacturing data and L01 actual module custody/persistence. Completed GIS, profiling and reservation work is retained; Rust changes must follow the measured hotspots. Original plan and execution identities remain in the ledger.

Implemented and exercised

The Workbench compares the Samawah station merge, preserves all fourteen cost scopes and exposes a project envelope that rejects mixed candidates and stale upstream stages. Station changes preserve unrelated vehicle/manufacturing results; cost-rate changes invalidate costs, delivery and dossier. Missing installed stages stay visible. A source-bound door panel exposes the 74 part definitions, assembly proposals, inspection gaps and removal/maintenance basis.

City workers now share cooperative CPU, memory and disk reservations, use a resumable bounded queue and admit at most two analytic workers. The server retains a 20 GiB host disk reserve. The controlled comparison observed real two-worker overlap: a fresh pair took 31.02 seconds versus 45.80 seconds with one worker, a 32.27% reduction. Exact numerical metrics and all fourteen cost scopes matched. Warm repeats took about 1.7 seconds. This is one pair per mode on one host, not a general performance guarantee. Reservations are cooperative; they are not an aggregate filesystem quota. The two workers used 37.31 combined worker CPU-seconds versus 32.47 serially, about 14.9% more CPU in that trial. The benefit is lower waiting time. Set `OSR_CITY_STUDY_WORKERS=1` to retain serial admission when CPU use matters more. Individual worker RSS is recorded; simultaneous aggregate RSS was not measured. Nine retained engineering workflow filters now exclude isolated changes to the new force-gap report and execution receipts. Core vehicle/physics changes still trigger their existing jobs; mixed report/core changes are covered by regression checks. No solver job definition was removed.

The profiled delivery scheduling kernel has a source/tool/binary-verified Rust implementation. Twenty-four actual before/after low/base/high comparisons matched canonical output exactly. Measured kernel reductions were 43–59%, including the boundary cost. It remains opt-in; no end-to-end Rust pipeline speedup was claimed for that kernel experiment. The production compiler now offers an explicit Rust backend through the delivery CLI. It verifies an existing current city-study publication before and after compilation, uses a prebuilt source/tool/binary-bound executable and records three bounded helper calls with zero solver jobs. The analytic city workers retain their zero-helper Python path. A fresh Samawah study produced 4,170 tasks with the same physical candidate and configuration. Three full CLI pairs matched all numerical and policy output: median time fell from 12.49 to 10.79 seconds, or 13.64%. This includes input verification, complete compilation, sealing and writing; it excludes building the executable and is not a whole-city speedup. Python remains the default. The comparison record lists the exact source/evaluation metadata exclusions and all additional Rust bindings. The canonical encoder benchmark made 48 exact byte/identity comparisons but gained only 1.72% in the guarded warm batch. That result does not justify moving more production callers to Rust.

Native GIS uses external import custody and verified parent planning snapshots. Samawah and Baghdad each completed a headless native station edit, canonical import, repricing, refreshed native export and second dwell edit. These are programmatic `.rgis` edits reopened by the native application, not GUI tests. The final acceptance used source `3ba09cf9ac` for both cities: eight native calls each, 41.18 seconds for Samawah and 103.67 seconds for Baghdad. Eligible city calculations were reused with their original identities; native executions and export receipts were fresh. All other four Samawah and thirteen Baghdad lines retained exact protected geometry and assumptions. Each record retains its actual source revision. The Samawah failed attempts (dwell projection and parent lookup) and Baghdad terminal-chainage failure remain preserved alongside the fixes and successful runs.

The current-source revalidation then repeated both journeys on `7516359c40`: sixteen fresh native calls, six legitimate new studies and exact offline replay of both final children. Historical baseline numerical controls and all fourteen cost-scope totals matched exactly. The work took 352.11 seconds and retained about 325 MiB. All other complete line records, route coordinates and protected stop IDs/chainages/dwell inputs remained exact. Twelve unedited Baghdad stops on the selected line reconstruct latitude by 1–3 float ULP; the same differences exist in the earlier run. Exact before/after values are recorded, rather than claiming byte-identical derived coordinates or changing a tolerance.

CAD and mechanics controls

The door definition counts 694 pieces per opening and twelve openings per train. Its nominal 50.433 kg stock/allowance subtotal per opening is not installed mass. SolidCraft generated and reopened a representative header stock blank, including STEP export; independent FreeCAD import reproduced the solid geometry. The blank is not a fully detailed, installed door operator. STEP lost material/source metadata, so the source sidecar remains required.

The existing installed-layout generator was also run analytically against current sources. Physical child `17ce7a…` stayed unchanged; evaluation `123896d4014fd00bc11fc755f132aa8769ebfd217f06d6f34b2fe8342d62c4f1` is new. All 74 operator definitions and expanded counts match the manufacturing parent. A compact one-opening attachment retains proposed transforms and instructions. Eleven operator types still occupy the same generic header-centre zone, and a 2 m route envelope does not represent the full 2.9 m guide stock. These are specific assembly-design gaps; copying those coordinates into accepted installed fields would not resolve them. The regeneration took 12.59 seconds, used about 133 MiB peak RSS and executed no native tool.

Civil controls derive 24 axle positions from the six-car reference and keep normal 46-person and crowd 160-person cases distinct. The fresh normal 3D experiment ran seven native processes: one geometry import, two fresh meshes and FieldCraft/CalculiX solves on each mesh. Fine FieldCraft displacement was −9.989130 mm; relative cross-solver disagreement was about 1.01×10^{-8}. The 2.7255% displacement refinement passes the declared 5% diagnostic but does not establish mesh independence. Offline verification exactly reconstructed both mesh/load records and both solvers' raw readbacks.

That FEM control includes a 1.3 dynamic multiplier, an explicit non-CAD reserve and finite bearing patches. Its ±0.6 m load-spreading patches are an unverified hypothesis, not the selected 1.435 m rail interface. It is not a direct numerical comparison with the different static analytical moving-load model. Old crowd mesh/load replay did not match exactly and remains held; no old solve was reused.

The force/braking compiler reuses existing interpolation, counted rotating stock and independent stopping methods. Named 2,000 m controls require approximately 11.10 kN/car normal and 14.73 kN/car crowd launch force. Actual installed thrust, current/voltage limits and repeated-duty temperature remain unknown. Valid synthetic map samples do not establish installed propulsion performance.

The bounded Samawah performance baseline now covers a fresh ordinary worker and three steady API cache hits. Its complete 25,623,859-byte encoded result matches the retained control without exclusions; all 65 source bindings match. Read-only peer review confirmed the ordinary job status flow, untouched function implementations and exact phase accounting. The publication envelope took 16.58 seconds; exclusive delivery, serialization and business-draft time were 4.06, 3.60 and 2.72 seconds. Three post-warmup cache hits took 66.48, 65.44 and 65.50 ms with no additional worker starts. These are API retrievals, not full browser timings or recomputations.

R01 is closed for this bounded corridor inventory. Read/decode and write/orchestration remain combined categories; per-stage RSS/physical I/O and aggregate scratch highwater are not established. Browser/build receipts retain their original epochs. Monitoring overhead is uncalibrated, the OS page cache was uncontrolled, and no optimization or all-workload claim is made. The failed probe initialization is preserved; both attempts used 35.98 seconds and about 24.65 MB, with no builds or native solves.

Door route proxy and retained analytical child

A read-only audit of the existing sealed installation packet found 830 cut-length patterns, including 24 door guide/belt patterns. The guide's 2.9 m stock and belt's 5.8 m cut length each appear as a 2 m straight display proxy. The stock lengths remain in source data, but the proxy does not show that a routed part fits, clears adjacent hardware or can be removed. The belt loop must not be interpreted as a straight 5.8 m part.

The audit receipt verifies the original packet seal, physical candidate `17ce7a724fcd661e5c4f195cde19b68377c0968e4e7d7dbb5d6ac7c8d4074ad4`, evaluation `123896d4014fd00bc11fc755f132aa8769ebfd217f06d6f34b2fe8342d62c4f1`, and all eleven current source bindings. The installed model's original bytes remain in use. Intermediate experiments with changed model sources are retained in the evidence index under their own historical identities; their outputs are not current installation evidence. The mated positions, controlled dimensions, full route fit, installed mass and physical acceptance remain open.

One-opening interface and custody handoff

The candidate-bound door handoff inventories all 74 mechanical and electrical part types for `car-0/door-0`, with their component identities, source stock, make/buy policy and counted quantities. It binds the manufacturing child `907da368…` to the existing analytical installed child `17ce7a…` and keeps the source proposal's `accepted_installed_transform_m` empty. The opening centre at car-local x = −4.5 m and proposed 0.72 m leaf travel establish a layout frame, not an accepted installation datum.

One opening has 694 pieces: 97 locally fabricated, 568 bought to standard and 29 bought to qualification. Thus 597 purchased pieces across 35 part types need receipt and lot/serial custody. The assembly API now has a narrow software contract for receipt-backed, individually serialized purchased children; this has not been exercised on the full door module in native ERPNext. The handoff identifies the header/body, rail/carriage, belt/pulley, motor/reducer, leaf/stop and sensor interfaces that need controlled datums and dimensions. Every part's installed transform, mating datum and approved inspection limit remains unknown. The packet was generated without native CAD or ERP execution. Its source-bound ID is `a2114619675858be7bba0a44655761b1feb45698a25181aa9f2f1ec3744e1908`.

Rebuild it with `python -m tools.automation.door_interface_handoff --attachment <verified-one-opening-attachment> --installation <sealed-installation-packet> --output <new-file>` after relevant source changes. The compiler checks both records against the same sealed installed candidate and current source bindings. It rejects mixed candidate/counts, a broken attachment seal, missing source bindings and an asserted accepted transform. The first next engineering input is a controlled header-to-body datum and guide/carriage/drive mating definition; the next software input is a sealed bought-source module pack and native receipt, QA, consumption and maintenance replay.

Integration still required

The published reference `756a…` and manufacturing child `907d…` differ physically, including intercar gap. The child cannot inherit the reference's fleet, costs, installed vehicle or engineering results. The current analytical child evaluation is recorded above; a complete mated installation, matching civil/station interfaces and lifecycle package are still needed before those stages can be composed into a complete dossier.

Door installed transforms, clearances, mating datums, tolerances, inspection limits and maintenance intervals remain incomplete. Actual drive/brake/thermal data, bearings, foundations, construction stability and a crane chart at the installation radius are also missing. These are specific engineering inputs, not permission to invent zero mass, prices, limits or passing test results.

The existing disposable ERPNext/FUXA fixture exercises a discrete workpiece. It does not yet demonstrate the counted 694-piece module assembly lifecycle. The existing candidate-delivery protocol now supports source-derived assembly groups and nonexploding parent BOM drafts. It preserves all thirteen identity requirements and verifies source counts, context bindings and child-draft provenance. The conversion uses reviewed source estimates, USD-company and precision guards, and leaves assembly process cost unknown. See the API and conversion limits. Twenty-three focused synthetic tests and independent review cover the extension; actual native persistence and the full module lifecycle remain unverified. The typed assembly lifecycle API now adds installation, replacement, condition events and business handback for manufactured, individually serialized children. It checks own assembly documents, native voucher/serial provenance and latest child QA. A different accepted child QA record invalidates the previous handback review. Twenty new assembly tests and 56 compatibility tests passed for the prior manufactured-only scope. The purchased extension checks submitted receipt row, inward serial-and-batch bundle, parent outward consumption, item/serial/batch identity and latest accepted Incoming QA. Eight synthetic cases now use a sealed, reviewed bought-source one-child pack; they and independent source/schema review do not establish native ERP transaction success or pooled batch-only custody. A matching candidate package and controlled document coverage are still required for the actual door run. Controlled research documents can support software tests without claiming physical qualification.

Tool migration and replay

The consumer inventory records pinned tools and source consumers. The operation matrix records supported, unsupported and untested domains, entry points, resource limits, installation size unknowns and individual retirement decisions. The light door CAD path uses SolidCraft; FreeCAD is an optional independent control. RustGIS supplies the demonstrated native return path. QGIS is not required to run that path; retained lightweight project readers remain useful. Gmsh is still required for the demonstrated conformal meshes, and CalculiX supplies an independent mechanics comparison. Other active native consumers have not passed the retirement gates. No global tool uninstall, historical evidence deletion or Git history rewrite has occurred.

Raw numerical outputs stay outside Git. The local evidence index records exact receipt bytes, hashes and locations; it is not an online artifact store. Existing source-bound records retain their original identities. Failed and interrupted attempts are

retained. Repository changes were integrated from isolated worktrees while preserving unrelated user changes in the main checkout.

All three required software workflows passed on `3ba09cf9ac`: 311 Python tests, Rust contracts and scheduler parity, and the clean Workbench installation and journeys. That clean sparse checkout installed SciPy, npm dependencies and Chromium; it ran frontend contracts, two retained Journey A cases and the new station comparison. Its new comparison took 16.84 seconds cold and 0.94 seconds warm on the CI host, which is separate from the controlled local worker comparison. The CI record includes artifact IDs and download digests. This verifies the supported software slice, not the missing module lifecycle or a complete v0.7 release.

The later software checkpoint passed all five required and retained workflows on exact source `952b7d953f`: 387 selected Python tests, Rust contracts, Workbench journeys, shared contracts and reconciled research. The reconciled lane passed 158 tests plus 13 subtests. The subsequent Rust CLI and assembly lifecycle additions have their own checkpoints below; earlier green checks are not relabelled as evidence for those additions.

Checkpoint `a41c280544` then passed 430 selected Python contracts (four optional skips), Rust compilation/parity, Workbench, reconciled research and the retained native business-platform fixture. The latter remains flat-part compatibility, not native assembly validation. Shared CI identified an outdated presentation policy pin to the delivery compiler. The complete `DEFAULTS` syntax tree and every policy value are unchanged; only the current policy's producer hash was refreshed. No historical packet or receipt was rewritten. Sixteen analysis and five routing checks pass. Exact standalone policy/test edits now run in the bounded Python lane; mixed engineering edits still trigger retained workflows. The existing Shared filter excludes edits to its own workflow file. The explicit final-source Shared run `43229288557` passed on `7516359c40`; no duplicate run was launched. The same `7516359c40` checkpoint passed 447 Python tests (four optional skips), Rust contracts, Workbench and reconciled research.

The later `e2a7ef8055` checkpoint passed 470 selected Python contracts (four optional skips), Rust contracts, Workbench, reconciled research and the twin workflow, including actual native placement controls. Its Shared guard caught an unintended edit to the pinned retired job; that entire job is restored byte for byte, preserving its original ordered command hash. The current-checkout job alone uses historical compact/line integrity checks. The twin architecture guard now checks the exact historical commands and separately requires native placement export/reopen tests after the pinned CAD install. All 42 architecture tests pass. A subsequent variant-publication hold is resolved by the explicit same-epoch check below; no original result was rewritten. The production source comparison confirms that the Rust, outline, Workbench, integration and GIS runner objects are unchanged between that checkpoint and `bce9d27892`. Prior CI retains its original commit.

The later wheel-reference job passed its numerical, native-adapter and comparison contracts, then stopped because strict `./osr verify` correctly holds the excluded integrated dossier. Current CI now explicitly checks the documented workflow-source hold, all five false authority flags and absence of a published result. The reader verifies every declared payload before that first source hold; this does not claim currency or authenticate later source bindings. The actual packet check passed, and six unrelated/tampered/current/authority cases were rejected. The strict verifier remains unchanged.

Those checks, documentation and frontend contracts now precede expensive wheel tests. The generated index was rebuilt for 236 maintained documents; the local README check passes for 771 materialized files. Index-only changes run in the bounded lane before dependency installation, using the existing Markdown policy and one batched sparse expansion. A fresh local-Git test used 2,072,576 allocated bytes and 0.40 seconds; no cold-network timing claim is made. Seven routing and 42 architecture tests pass. Production source objects remain unchanged from `e2a7ef8055` through correction `4f0a0ace7a`.

The owned evaluation working tree was then reduced from 8.45 GiB to about 0.91 GiB, retaining project agent instructions, plans, CI contracts and the bound city inputs. Shared Git storage is separate. Reflink sharing kept host free space around 22.4 GiB during that checkpoint. The final owned bytecode cleanup removed 83 generated files (2.22 MiB logical bytes), leaving 21.81 GiB free; this does not claim exclusive physical reclamation for shared/reflink data. In that compact tree, 53 focused tests and the actual Samawah browser replay passed; Baghdad's existing source-bound packet also verified with zero new calculations. The browser reused eligible jobs: its 11.22-second first click is not a cold calculation, and its warm repeat took 1.69 seconds. The compact-checkout record preserves the environment and the initial missing-default-browser-cache failure. The successful run used the existing shared Chromium installation. The grouped/flat delivery and CI routing checks separately passed 60 tests after integration.

Retained-research CI found two source-epoch mismatches after these changes. The older engineering ledger now references exact archived bytes for three implementation files authenticated against published `5cfe894`; all seven gate states, blockers and authority flags are preserved. The original ledger and relocation manifest remain in the source-context archive.

The three reconciled dossiers now have explicit historical verification. It authenticates each complete publication and its source/input union against one original Git commit, then runs the unchanged original checker in a bounded source context. All three checks passed in 52.56, 42.24 and 49.20 seconds, with no new native solver executions. This authenticates source, data and methods; it uses the installed Python/scientific environment and does not restore the historical operating system. Current strict verification remains unchanged and refuses these source-stale dossiers. Their numerical Workbench routes stay held.

In a research checkout with the original publication and pinned Git objects available, the reference dossier can be checked explicitly:

```
./osr design closure verify --retained-source f990fc24ba9c1c2adcd8afb4456ded5c4753209a
```

The result distinguishes `retained_verified` from `current_source_valid`. The wrapper refuses oversized/tampered inputs, replacement Git objects, foreign project imports and native subprocess execution. Its import/process guards are cooperative controls, not an operating-system sandbox.

Three other retained publications bind the earlier braking implementation. Their unchanged original checkers were run against source epoch `946f1772540d07af85b57b0d13cac08a3293b140`: twin analytical/materialisation replay covered six non-native cases; reference-cell replay covered ten cases; motor verification checked the recorded tool/source/payload/trace without a new field solve. All passed in 81.65, 190.27 and 5.58 seconds respectively. Two earlier cell attempts hit their 120/150-second CPU limits and remain preserved. The successful cell request explicitly used 300 CPU/360 wall seconds; no model, tolerance or original input changed.

The same original epoch also verifies the three retained wayside variants. All 73 declared source bindings and three payloads were independently authenticated. The unchanged original variant checker passed in 24.43 seconds; the current strict reader still rejects the stale reference result. There was no field/native solve or new operating-map evidence. Twenty-nine boundary tests pass after integration.

The coordinator lives outside calculation-source discovery at `tools/automation/verify_retained_publications.py`. It authenticates the exact original manifests, declared payloads and source/import closure, stages private copies and invokes the unchanged checker. Unindexed historical twin/native archives remain untouched and outside the claim. Current strict readers still refuse these stale publications; the result records `current_source_valid=false`. An independent review matched 95 source bindings, authenticated all three receipt/manifest pairs and passed 21 boundary tests in 1.56 seconds. It checked resource guards, retained failures and staging cleanup without rerunning the historical methods. No CAD/native/field processes ran. Read the retained packet explicitly with:

```
python -m tools.automation.verify_retained_publications prefetch twin
python -m tools.automation.verify_retained_publications verify twin --state /external/fresh-twin-check
```

In a partial clone, the separate CI prefetch selects object IDs without asking Git for sizes of absent blobs. It fetches only those selected objects explicitly; verification retains its no-network rule and refuses missing selected blobs. A real partial-clone regression reproduced the BAD size failure, fetched the selected objects successfully and left an unrelated payload absent. The three pinned selections total 156.01, 123.22 and 131.36 MiB respectively; their bytes match the prior authenticated checks. The 512 MiB size limit is checked after fetch, not enforced as a network-transfer quota. After integration, 42 scheduler and retained-verifier tests and four workflow-filter tests passed. The refreshed Rust scheduler build receipt binds the formatting-only source correction; old benchmark binaries and receipts remain unchanged.

Use the existing Samawah comparison instructions for the browser entry point. New bounded native runs require a clean source checkout, explicit source/tool/candidate/load/resource identity and a fresh external output directory. The headless GIS runner is `tools/automation/tests/verify-v07-native-gis.py`; force requirements use `python -m engineering.station_rail.force_evidence --help`. Heavy native tools are optional for the analytic browser workflow.

6.2 Roadmap

docs/ROADMAP.md

v0.6.1 — Open Engineering Platform is the published v0.6-series software release at `5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea`. Use its installation route, downloads and known limitations. The existing `v0.6.0` and `v0.6.1` tags stay at their recorded commits.

OpenSourceRail continues to connect railway design and engineering, manufacturing/delivery and maintenance, and city planning and service. Local manufacture, repairability, open knowledge, ERPNext/Frappe HR and FUXA/gateway integration remain part of the platform's purpose.

Finish the v0.6 handover

The handover is complete. A fresh public clone of the published tag followed its installation asset, verified SHA256SUMS and completed the documented Samawah journey twice. Installation results record commands, identities, timings and the environment limits. The retained readiness-browser harness now uses an external temporary output directory; its repaired CI job passed on main `0472a4f3cadb23847e071e693294a418bf54c3b0`.

The older design-closure campaign is superseded only where it controlled software publication. Its seven open/partial engineering findings remain visible in the engineering ledger. The retained authenticated-review profile keeps its truthful pending records; it is not a reason to repeat publication checks or withhold the already published release.

First v0.7 milestone: the Samawah station merge

Open the published Samawah example, apply the retained station-merge intent, and produce a side-by-side report of station count, service assumptions, fleet, civil quantities, fourteen-scope costs and unresolved inputs. Each difference must trace to the edit and its dependent calculation. Reuse unaffected vehicle and manufacturing outputs rather than rerunning them.

The retained engineering case found two station markers 79.977 m apart against a 120 m platform assumption. Its midpoint merge packet provides the before/after edit and provenance. Carry both cases through the new workflow. That packet belongs to its declared candidate/source epoch: applying the edit to the published example must produce and identify the matching new revision, not silently transfer old compact-drive results to another vehicle. Access, catchment, platform fit, route feasibility and unpriced work remain explicit inputs or findings.

The first Workbench comparison is implemented and browser-verified. The published case already groups the close pair operationally: raw markers change from 26 to 25, operational stops remain 25 and fleet remains 81. Both sides retain the same vehicle candidate and unchanged civil quantity identity; the report exposes actual service, access and fourteen-scope cost changes. The remaining workstreams below remain open.

The agent delivery and tool migration plan assigns bounded work to agents and subagents, with Rust migration benchmarks, CAD/FEA/GIS promotion criteria, legacy retirement gates and resource budgets. The execution record now contains the verified two-worker improvement, opt-in Rust scheduler, fresh CAD/FEA controls, native Samawah/Baghdad cycles and clean-install CI evidence. Its task ledger keeps the remaining installed-candidate, detailed module and lifecycle work open.

v0.7 delivery workstreams

Workstream	Required integrated result
Project continuity	One project entry point binds corridor revision, vehicle candidate, civil option, quantities, cost and dossier. A station edit invalidates its dependants; the Workbench shows changed results and verified reuse of unaffected work.
Installed vehicle	Select one six-car candidate for detailed integration. Resolve recurring geometry definitions, door motion, equipment access/removal and mass bounds. Keep the passive reference and propulsion alternatives separate; supplier and measurement dependencies remain explicit.
Civil works and station	Carry one span, support and station interface through bearing assumptions, moving loads, construction sequence and lifting checks. Establish crane capacity at the actual installation radius. Retain SolidCraft/FieldCraft and independent FreeCAD/CalculiX evidence.
GIS and economics	Complete a fresh native GIS edit–export–import–reprice cycle for the selected Samawah revision. Trace station count, fleet, guideway quantities and all fourteen cost scopes; identify transferred rates and unpriced quantities. Check the implementation on one Baghdad corridor afterward.
Manufacture and lifecycle	Use the door operator as the representative module. Bind drawings, tolerances, BOM, assembly/tooling and QA criteria to its geometry and revision; exercise removal/replacement, wiring and motion clearance. Run that actual candidate-linked module through the existing ERPNext/FUXA lifecycle rather than inheriting the synthetic one-unit kit result.
Workbench and release	Guide one corridor edit through service, cost and revised dossier, with links to calculations, drawings and open findings. Open the same module for manufacture planning, QA and maintenance. Preserve before/after outputs and replay the full journey from a clean install.

Delivery order and acceptance evidence

1. Handover and baseline — complete: the published v0.6.1 installation, Samawah before/after comparison, candidate boundaries and bounded performance baseline have recorded source identities and timings.
2. Project continuity — software demonstrated: the station edit flows through service, fleet and fourteen cost scopes with dependency checks and visible missing installed stages. Complete candidate integration remains open.
3. Engineering interfaces — active: establish controlled door mating datums, force/brake/thermal inputs and represented civil support/loading before new native CAD or FEA. The Samawah and Baghdad native GIS two-edit return cycles are recorded; rerun only for changed inputs.
4. Lifecycle and generalisation — active: take the counted door module through actual native ERPNext receipt, QA, assembly, replacement and handback, then compile a same-candidate dossier and replay one Baghdad corridor.
5. Release acceptance: resolve defects exposed by both journeys, run the clean-install v0.7 acceptance case, and publish supported scope and remaining engineering limitations.

The original week ranges were planning estimates and are superseded by the execution ledger. Supplier inputs, physical measurements and unresolved structural assumptions have separate milestones; software can show exactly which conclusions depend on them.

Consolidate existing computation

Use selected-city source binding, verified caches and incremental publication checks already in the repository. Measure wall time by stage at the corridor entry point. If a station edit launches an unchanged vehicle analysis, repair the dependency first. Benchmark an expensive required kernel before extending Rust or vectorisation. Avoid broad rewrites, all-city regeneration and duplicate calculation pipelines for this milestone. Preserve failed, interrupted and out-of-domain evidence and maintain the project disk reserve during execution.

Retained engineering foundation

The reconciled recovery pilot and no-store control connect an installed-body hypothesis, separately owned drive/BOP additions, mass and axle loads, civil design, concurrent trains, station energy allocation and fourteen-scope costs. Several nominal model timetables complete 16/16 arrivals. Source-linked CAD and independent structural comparisons are retained within their declared applicability. These achievements close important software and calculated-outline links; equipment characteristics remain provisional.

The selected development reference remains passive and wheel-supported. The newer powered-wheel pilots are research alternatives, with separate physical and evaluation identities. No installed propulsion architecture is physically selected. Body-plus-reserve subtotals exclude external motors, gearboxes, drive BOP and recovery interfaces; integrated mass and cost must come from the same candidate. A child revision must propagate through all affected evaluations rather than inherit its parent's results.

Coordinated work continues through `./osr design railway`, the cost and delivery method, pilot delivery compiler, ERPNext/FUXA lifecycle and capability register. Retain the v0.4 platform, v0.5.1 research and source-bound archives independently reproducible.

Cost, resource and production assumptions

Keep inexpensive screening ahead of detailed CAD/FEM. Reuse numerical work only for the exact source, tool, physical candidate, load, mesh and resource request. Run independent cases within measured memory budgets, preserve failed runs and benchmark useful candidates per compute-hour. Python remains useful for orchestration and specialist integration.

Current outline scenarios use two 8-hour production shifts and two 8-hour civil shifts. Operating cost uses two shifts for 16/18-hour service and three for the 24-hour case. Utilisation, shift premiums, relief staffing and wages remain explicit hypotheses. Equivalent resource hours do not establish an actual crew, production calendar, construction critical path or approved roster.

Every price retains quantity, source or assumption, scope, uncertainty and exclusions. Factory, delivered, installed and lifecycle totals remain distinct. Direct part/process allowances, external reserves and fourteen-scope accounts must reconcile without counting the delivery decomposition or capacity proxies again. China sourcing, bulk procurement and local manufacture can be compared using explicit production/supply evidence; an automatic discount is not evidence.

Open physical engineering gates

Area	Required next evidence
Vehicle	Installed mounting and movement geometry, remaining equipment, qualified joints/running gear, measured or validated mass/CG/inertia
Propulsion and recovery	Attainable force-speed-duty envelopes, capture/reset/release losses, thermal limits, independent stopping and recovery
Electrical and HVAC	Hot-weather charge/discharge/converter limits, cooling, emergency endurance, fire/isolation and maintenance access
Protected service	Model bounds plus validated braking/WSP, degraded operation, terminal protection, separation, rescue and actual controls
Civil and construction	Prestress/joints/cracking/fatigue, bearing/contact and dynamic behaviour, site ground, temporary works and verified lift/transport configurations
Production and business	Controlled drawings, tolerances, materials/process qualification, QA hold points, reviewed ERP conversion, actual capacity and traceable installation
Economics and site	Equal-service factory-to-installed/lifecycle ranges, renewals/recovery/disposal, local demand, utility/land/ground and supplier evidence
Prototype and release	Authenticated measurements, model correlation, independent review and separate manufacturing/operating approvals

Software completion, numerical convergence and a favourable estimate do not substitute for physical evidence or independent acceptance. Draft tasks and calculated outlines remain useful completed design-stage results.

6.3 Agent delivery and tool migration plan

docs/plans/v0.7-agent-delivery-plan.md

Status: implementation in progress; remaining work reprioritised after review of main `88f5ecc88b052663768b339f3cd3d7244de5dd7b` and the subsequent candidate-bound door and purchased-custody work through `eff8db7c36` on 12 November 2026. The execution ledger retains the original plan identity and records this revision separately. v0.6.1 remains published; full v0.7 acceptance remains open.

Deliver one continuous Workbench journey from a Samawah corridor revision through service, civil quantities, fourteen-scope costs, a vehicle door-operator manufacturing package, QA and maintenance. Repeat the corridor journey on one Baghdad corridor. Increase Rust computation and use SolidCraft, FieldCraft and RustGIS as preferred tools in demonstrated domains. Remove superseded production paths and heavyweight default dependencies as their replacements pass the gates below. Preserve independently useful solvers and historical evidence in optional verification/replay installations.

This is an execution plan, not a declaration that tool migrations or engineering qualification have happened. It extends the roadmap; the published v0.6.1 handover is complete and is not a recurring work package.

Reviewed progress and next priority

The execution ledger and evidence summary are the current status authority. The work-package table below retains the full intended exit conditions; it is not a list of tasks to restart.

Tasks	Reviewed state	Next action
P00, R01	Inventory complete; bounded Samawah phase baseline reviewed	Reuse the baseline; refresh only affected measurements
S01	Shared reservations, bounded queue and recovery implemented and verified	Maintain existing admission and interruption tests
G01, G02	Both cities completed the supported headless native two-edit cycle and offline replay	Integrate changed candidate inputs when available; no routine native rerun
R03	Rust delivery scheduling available as a verified production opt-in	Retain explicit backend selection; assess a specific wider caller separately
R02, R04	Canonical port evaluated without promotion; additional geometry port deferred	No repeat port without a new measured bottleneck or changed boundary design
P01, P02	Project identity and dependency checks implemented; installed child propagation held	Resolve the integration candidate and propagate its actual interfaces
C01, M01	Counted door definition and stock CAD control; mated assembly incomplete	Resolve placements, datums, motion/removal and controlled manufacturing data
F01, C02, E01	Bounded mechanics controls and evidence-gap compiler; installed engineering open	Establish represented rail/support/load and drive/brake/thermal inputs before new solves
L01, W01	Assembly lifecycle APIs, receipt-backed serialized purchased-child software contract and browser panels tested	Replay a sealed bought-source module pack and actual native receipt/QA/consumption; pooled lots and full 694-piece module remain held
T01, D01	Twelve-domain capability matrix; partial slim installation	Close specific consumer/domain gaps before further default or adapter removal
V01	Clean software scope verified	Final integrated module/corridor installation acceptance remains pending

The live GitHub review found the Workbench journey successful on the reviewed main itself: run 43254678158. The 480-test Python lane and full wheel/station run passed `8d1833f74c`; Rust passed `4f0a0ace7a`, and the five retained/shared correction workflows passed `bce9d27892`. Their exact sources and scope remain in the CI record. An older failed browser setup run was repaired; it is not a current blocker. These software results do not close the installed engineering findings.

R03 reduced complete delivery CLI median wall time from 12.49 to 10.79 seconds (13.64%, three paired trials, prebuilt binary). It did not measure a whole-city speedup. R02's 1.72% guarded canonical gain did not justify default promotion. The R01 publication envelope ranks delivery at 4.06 seconds, serialization at 3.60 seconds and business drafts at 2.72 seconds. Geometry is smaller. Further Rust work should examine redundant representation/copying or the business-draft boundary before proposing another geometry kernel; first compare a Python algorithmic/batching improvement, vectorisation where applicable, and Rust with conversion costs included. Keep policy and exact output identities unchanged.

The next milestone is one mated, source-bound door opening on the selected integration child, feeding controlled drawings/BOM and the existing lifecycle protocol. Eleven operator types currently share a generic header-centre zone; the 2 m route envelope does not cover the 2.9 m guide stock. Fix those definitions before expanding native CAD or FEA. The 694-piece count and 50.433 kg gross stock subtotal per opening do not establish installed mass or assembly feasibility.

Starting point and fixed controls

The first Samawah comparison is complete. Its 79.976780 m marker pair becomes one midpoint marker at 9,339.173756 m: raw markers 2625, operational stops 2525, fleet 8181. The current provisional base estimates change by −$10,029.49 capital and +$229,759.68 whole-life NPV. These are regression controls, not a chosen optimum. The fresh CI comparison took 38.52 seconds; repeat reuse took 1.43 seconds. Local and CI numerical metrics and all fourteen scope results agreed. Compare future timings on the same machine and workload; these timings are not a cross-machine speedup claim.

The project initially binds:

Identity	Baseline
Published source	v0.6.1 at `5cfe89488a64549b37d4a7fd3ea9e3d3a8923aea`
Planning seed	`efae2c3a62370f0cc5c6de9ab8191564323629537310 39d8c9a841fc81eb215a`
Physical reference	`756a68870162f7559b052b05c9b201728c9f448e3bb61114f287848c02f2d3a3`
Body	`9489e4f2666fd3b3c73232c86df3f12adf42a5531eb27373834cf3fecb1a3117`
Support component	`3e9aef846e6ec55bac013512163394618b37cf8c74b1100d6f8e5450ae5479d7`
Passive secondary interface	`67e2bfc4ef429a6c49e6d07095bc298c4eb76b8386e4ee316c440ff128c30fa9`

These IDs identify reference inputs, not a qualified installed railway. Preserve six 17 m cars, seating/accessibility policy, passive wheel reference and explicit propulsion research configurations. The conditional line-1 powered-axle research policy is not an adopted onboard propulsion system. The selected module is `body/door-opening`; the civil research control is `reference_box--reference`. P01 must finish installed integration, not repeat this selection. Manufacturing child `907da3680bcbd01ffe1223652443e4e34a9f581e201385917a99edf4150bd90a` changes the reference intercar gap from 0.5 m to 0.65 m. Its fleet, costs, civil loads and installed results must be derived for that child; neither the reference nor the old compact-drive body supplies them by inheritance. Physical changes create child candidates; provider, mesh and numerical implementation changes create distinct evaluations. Selection identity remains separate.

Already implemented and to be extended: selected-city source collection, completed city-request indexing, incremental artifact verification, bounded comparison reuse, deterministic streaming archives, performance/storage commands, Rust canonical encoding, Rust design-search kernels, Rust routing/alignment and the station-pair native kernel. Do not recommission these as new subsystems.

Agent organisation and ownership

Use the existing `.codex/agents/` roles. The coordinator owns the task ledger, interfaces, integration, main and publication. Specialists own bounded source changes; numerical workers produce immutable run packages and never integrate. The existing project configuration allows three concurrent threads per session: plan for coordinator plus two specialists. Subagents consume those same slots; a lead delegates serially or releases its slot. Do not spawn the entire hierarchy at once or create a checkout per parameter sample.

Lead role	Bounded subagent responsibilities	Source ownership
coordinator	Project identity, dependency ledger, release scope	Campaign/task definitions, shared schemas, integration and roadmap
service	Corridor/service/cost continuity; GIS revision loop; Baghdad replay	`engineering/outline/`, `engineering/rustgis/`, existing routing/alignment bridges
mechanical	Installed vehicle; door geometry; motion/access/removal; controlled drawings	`engineering/vehicle_parts/`, new door modules in existing CAD adapters, owned installation adapters
civil	Span/support/station; bearings and load cases; erection/lifting	`engineering/design_search/craft_backends/viaduct_*`, candidate-specific civil records
electrical	Available-force/braking maps; door wiring/power; thermal duty	Existing station-rail/electrical models and allocated door electrical definitions
twin	CAD/BOM/IFC consistency; manufacturing/QA; lifecycle integration	`engineering/pilot_delivery/`, delivery exports and `services/integration/`
verifier	Differential numerical checks, identity/tamper checks, independent solver and clean-install review	Verification fixtures/reports; no ownership of the implementation being reviewed
bounded software specialists	Rust kernels, profiling, Workbench, runtime/storage, packaging	Explicit files allocated by coordinator under the leads above; no blanket ownership of all `engineering/`

File ownership is per assignment, not permanent team territory. For example, service owns `city_workflow.py` integration while a Rust subagent owns its new kernel and bridge; the coordinator applies shared call-site changes after the kernel lands. Every assignment states: “You are not alone in the codebase; preserve other edits and adapt to integrated changes.” Review is separate from authorship; AI peer review does not supply physical measurements or a person's identity/signature.

Task contract and execution ledger

Maintain the existing `campaigns/v07-execution.json`; do not create another ledger or dispatcher. Retain its original plan commit/hash as historical identity and bind subsequent plan revisions separately. Each new bounded assignment uses the following contract, with its measured resource request and exact files:

```
task_id: R02
parent_task: R01
owner_role: service
status: planned
source_commit: EXACT_COMMIT
candidate_ids: {physical: EXACT_ID, component: EXACT_ID}
evaluation_context: {provider: PIN, material: HASH, load: HASH, mesh: HASH}
requires: [R01, P01]
owned_paths: [EXPLICIT_FILES]
fixed_interfaces: [SCHEMA_AND_UNITS]
allowed_variables: [ONE_PROFILED_KERNEL_IMPLEMENTATION]
requirements: [SUPPORTED_REQUIREMENT_IDS]
resource_class: software
resources:
  maximum_memory_mib: 2048
  maximum_scratch_mib: 1024
  maximum_output_mib: 128
  maximum_build_cache_mib: 512
  maximum_wall_seconds: 600
  maximum_processes: 1
  threads_per_process: 1
trial_limit: 1 cold plus 3 paired warm trials
completion: [PARITY, END_TO_END_BENEFIT, REVIEW]
artifacts: [INPUTS, OUTPUTS, LOGS, HASHES, BENCHMARK, REVIEW]
next_if_blocked: NAMED_MEASUREMENT_OR_INTERFACE_TASK
```

Replace the example resource values with measured/requested limits for each assignment before it becomes ready. All resource fields are mandatory, including CAD/FEA disk and build-cache growth. Reserve the combined scratch, output, build and download growth across active jobs above the host reserve; reject an incomplete request rather than inferring zero.

Use planned ready running review integrated. Deferred resource jobs retain their queue position and inputs; failed/nonconverged/model-domain results remain evidence with a next action. Software execution, numerical validity, engineering screen, evidence maturity and physical acceptance are separate fields. After two unsuccessful local revisions, review the model/domain or change family instead of blindly extending a sweep. An agent stops at its trial/runtime budget, returns the package and next experiment, and does not weaken an obligation.

Each handoff contains source commit/tree and dirty-state receipt; exact candidate, request, material/load/mesh/tool/build identities; commands and environment; input/output hashes; stdout/stderr and failure records; wall/CPU/RSS/I/O/scratch measurements; units and numerical comparisons; dependency invalidations; tests; remaining findings and a reproducible next command. Source edits are committed before native runs. No source edits during a benchmark. Root integrates only the owned change after review and preserves unrelated user edits.

Work packages and dependencies

Budget classes are defined below. “Complete” means the stated exit condition, not simply that code exists or a command returned zero.

ID	Assignment and dependencies	Deliverable and exit condition	Budget
P00	Coordinator, first	Record the completed v0.6 handover and Samawah milestone; inventory disk, worktrees, tool consumers, existing caches and current source bindings. Allocate ownership and ledger.	inventory
P01	Service + mechanical + civil, after P00, serial selection	Freeze the Samawah before/after pair, one six-car integration candidate, one door operator and one civil/span/station assembly. Publish interfaces, missing supplier inputs, required loads and ancestry. Unspecified dimensions remain holds.	software

ID	Assignment and dependencies	Deliverable and exit condition	Budget
P02	Service continuity, after P01	Extend existing corridor entry point into a project envelope linking vehicle, civil option, quantities, cost, manufacturing and dossier. Test one station edit, one cost-rate edit and one vehicle change: invalidate only their true dependants; reject mixed identities and stale artifacts.	software
R01	Performance specialist, after P00; shared call sites coordinated with P02	Measure source verification, decoding, geometry, service, civil, cost, delivery, sealing, writing and browser stages. Produce ranked hotspots, cold/warm/build costs and baseline receipts.	benchmark
T01	Tool-interface specialist, after P00	Operation/domain capability matrix, compatible pinned tool sets, schemas, units, headless commands, error classification and installation footprints. Mark supported, unsupported and untested separately.	inventory; bounded controls if needed
R02	Rust serialization specialist, after R01	Extend existing canonical bridge to one measured hotspot. Exact encoded bytes, IDs and rejection behaviour; end-to-end gain including startup/FFI. Preserve a Python oracle for testing.	benchmark
R03	Rust numerical specialist, after R01/P01	Delivered: profile-selected delivery scheduler, exact parity and production opt-in. Retain Python default; any wider caller or new kernel requires its own end-to-end comparison, threshold/unknown preservation and resource identity.	benchmark
R04	Rust geometry specialist + GIS lead, after R01/T01	Reuse `osr-routing/osr-alignment` for measured corridor geometry operations; consistent CRS, chainage, endpoints and stable feature IDs. Explicitly handle unsupported transition geometry.	benchmark
S01	Runtime/storage specialist, after P00/R01	Extend existing admission/reuse with shared reservation, resumable jobs and bounded build/artifact caches where gaps exist. Demonstrate concurrent reservation refusal, interruption recovery and safe cleanup.	software
C01	CAD specialist + mechanical, after P01/T01	SolidCraft door/operator assembly from canonical parts, counted fasteners/wiring/clearances, reopenable native and STEP, exact geometry comparisons, door-motion and removal envelopes. Attach drawings and measured/unknown dimensions.	CAD
F01	FEA specialist + verifier, after T01/P01	Domain-specific controls for the actual material, joint/support/contact and load types needed by C01/C02. Admit only represented physics; analytical controls and independent solver agreement recorded.	mechanics
C02	Civil, after P01/F01 and the E01 load envelope	One actual span/support/station interface with load combinations, moving loads, bearings, foundation assumptions, temporary stability, lifting geometry and crane load chart at installation radius. Refine to declared convergence/function limits or retain a specific hold.	mechanics
ID	**Assignment and dependencies**	**Deliverable and exit condition**	**Budget**
E01	Electrical + mechanical, after P01	Same-candidate available-force/current/thermal maps and independent stopping envelope; door power/wiring, isolation, manual	software then admitted native

ID	Assignment and dependencies	Deliverable and exit condition	Budget
		release, emergency duty andprotection. Supplier or measuredlimits remain traceable blockerswhere missing.	
G01	GIS specialist, after P02/T01	Delivered for supported native scope: external import custody and Samawah edit import reprice refreshed native export second edit, including offline replay. Preserve old receipt readers, CRS/topology/IDs, access and nulls; rerun only affected scope after relevant changes.	GIS
M01	Twin + mechanical/electrical, after C01/F01/E01	Door package: drawings, mating dimensions/tolerances, material/finish, counted BOM, make/buy, assembly/tooling, inspection criteria, serial/lot plan, removal/replacement and maintenance intervals with evidence sources. Unknown acceptance values remain held.	CAD plus software
L01	Twin integration, after M01/P02	Candidate-linked module BOM in disposable ERPNext/FUXA journey: item/BOM revision, work order, stock movement, QA rejection/rework/acceptance states, serial/installation, maintenance replacement and handback. Record unresolved real engineering approvals; no invented physical test result.	integration
W01	Workbench specialist, incremental after P02, then G01/M01/L01/C02	Guided edit-to-dossier and module-to-maintenance journey; visible affected results, drawings and missing evidence. Compact indexed views; download verified packets on demand. Reconnect holds only when sources match.	software/browser
G02	Service/GIS, after G01/W01	Repeat on one explicitly selected Baghdad corridor with its own demand/geography/rates and identities; no Samawah outcomes copied. Preserve all fourteen cost scopes and transferred/unpriced inputs.	GIS/browser
D01	Packaging specialist + verifier, after T01 and relevant C01/F01/G01	Per-capability promotion/deprecation/removal matrix; slim default install, optional verification/replay environment, migrations, notices and rollback. Pass the retirement gates below for each removed path.	integration
V01	Verifier + coordinator, after required tasks above	Clean install of final supported scope; both city journeys, module lifecycle, failure/recovery, numerical parity, provenance and performance/resource regressions. Publish supported scope, open engineering evidence and replay instructions.	integration plus bounded independent checks

G01/G02 supported native cycles are complete. Remaining corridor integration is P01 completed child stage packets P02/W01 V01, reusing unaffected GIS evidence. P01 C01/F01/E01 M01 L01 W01/V01 is now the leading module path; C02 feeds the same dossier. Rust work runs beside these paths and lands in small verified increments. D01 follows demonstrated replacements, rather than blocking every engineering task until all old tools disappear.

Rust migration policy and success measures

Extend `crates/osr-design-search`, `native_canonical.py`, `native_search.py`, `crates/osr-routing`, `crates/osr-alignment` and applicable station-pair kernels. Keep orchestration, policy assembly and readable evidence contracts in Python where that is efficient. Avoid duplicating complete cost/service policy in two languages. New Rust crates require a clear domain boundary, typed errors, versioned input/output schemas, pinned build/ABI receipts and documented callers.

For each port, record total runtime including data conversion, extension loading, serialization, build/install time, peak memory and disk. Use one cold and at least three paired warm trials on identical hardware/inputs. Report distributions and raw trials; increase repeats only if variance prevents a decision. A provisional promotion target is at least 20% improvement in the measured affected stage and no material end-to-end regression; the coordinator may accept a smaller speed benefit for demonstrated memory/maintenance gains with an explicit rationale. These are decision targets, not promised performance improvements. An unhelpful port remains experimental or is removed from production.

Exact identities and canonical bytes must remain exact. Cover negative zero, large integers, bool/int distinctions, Unicode, non-finite values, unknown/None, cycles, JS integral-number transport and unsupported values. Preserve summation, rounding, fleet ceiling and near-limit decisions. Where numerical output cannot remain bit-identical, declare tolerances before testing and a new evaluation implementation identity. Do not reuse historical hashes under changed arithmetic. Vectorisation/SIMD/parallel reductions must pass the same criteria. Start with bounded batches and deterministic reductions; avoid competing nested BLAS/Rayon thread pools. Retain a small reference oracle after removing production fallback.

Change-bound current evidence is regenerated only for affected scopes or shown held. Broad atlas/compiler changes need a measured reason and a costed refresh; a local UI feature does not justify regenerating all 266 cities.

CAD FEA and GIS promotion and legacy retirement

Current repository pins and demonstrations define the starting domain, not the latest upstream feature list. SolidCraft 0.30.1/FieldCraft 0.7.5 is the tested pair; the recorded SolidCraft 0.40.0-rc.2 format-50 import into FieldCraft's format-48 reader failed. RustGIS is pinned at 0.5.0. T01 may test newer versions in an isolated compatibility matrix, with new tool identities and retained failures. No automatic “latest” upgrade.

SolidCraft's command engine exposes JSON commands across its CLI and agent interfaces, which makes it a suitable preferred automation target (upstream command documentation). FieldCraft and RustGIS are Rust projects with broader advertised capabilities (FieldCraft, RustGIS); upstream declarations are inputs to admission tests, not evidence that the pinned OSR adapters support every case.

Tool or path	v0.7 intended disposition	Required evidence before removal/switch
SolidCraft	Preferred CAD for demonstrated door/civil operations	Named part/placement identity, exact volume/centroid/bounds, holes/features, native/STEP reopen, editable history where required, manufacturing exports and motion/removal results
FieldCraft	Preferred mechanics for admitted material/load/joint domains	Analytical controls, mesh sensitivity/convergence, force/moment/energy residuals, representative assembly and independent comparison; unsupported cases explicitly routed/held
RustGIS	Preferred native corridor edit/reprice interface	Fresh two-edit Samawah cycle, Baghdad replay, CRS/topology/ID preservation, malicious/stale input rejection, bounded headless execution
FreeCAD	Remove from default migrated CAD installation; retain optional verification/replay initially	All current consumers of removed path migrated; independent exact B-rep checks remain available; drawings and editable exports meet requirements
CalculiX	Remove from default migrated mechanics installation; retain independent optional solver initially	Equivalent admitted scope and independent check route demonstrated; compare the same loads/materials/mesh meaning, not just similarly named cases
QGIS	Remove obsolete install/runtime requirements where any remain	It is already not required by current generation. Preserve lightweight historical `.qgz` ZIP/XML readers and source geography
Gmsh/OpenCASCADE meshing	Retain where currently needed	Current successful viaduct route uses conformal tet10 after native shared-prism hex meshing failed. Remove only after equivalent conformal mesh/connectivity/quality controls succeed
GetDP	Retain electromagnetic specialist scope	No equivalent validated electromagnetic replacement is established by the static mechanics pilot
IfcOpenShell/Bonsai	Keep IFC interoperability; make optional where supported	CAD replacement does not replace BIM relationships, georeferencing or IFC export validation
Chrono/SUMO	Inventory and isolate specialist usage	Multibody/network simulation needs independent replacement evidence; do not infer replacement from static FEA or a city cost model
Duplicate launchers, unconditional installers, stale UI branches	First deletion candidates	No active consumer, tested redirect or replacement, documented migration and historical replay path

The existing 12-span CAD comparison and two coarse/fine mechanics cases are a starting point. Homogeneous uncracked concrete, perfect bonding and bilateral restraints do not establish bearing/contact, reinforcement/prestress, buckling, fatigue or actual erection behaviour. Treat each required material/physics domain as a separate admission task. Compare stresses at equivalent locations and recovery definitions; a nodal sample is not a global integration-point maximum. Two mesh levels can reveal error without establishing convergence.

Each retirement progresses through inventory compatible controls actual assembly/journey preferred opt-in default promotion remove default install remove unused active adapter. For each path, retain a registry entry with consumers, model domain, replacement, parity evidence, performance/installation footprint, migration instruction, replay location, rollback and final status. Default removal and source deletion are separate decisions. A heavy GUI can leave normal installs while an isolated command-line verifier remains useful.

Before deleting active code, verify all callers, generators, imports, schemas, CI installers and historical readers; test with the removed tool genuinely absent. Missing backends produce a precise unsupported-operation message, never a silent change of solver or physics. Keep original evidence bytes/hashes and pinned replay instructions; archives need not remain normal checkout payloads. No Git history rewrite is part of v0.7. Removing an adapter is not permission to uninstall a user's unrelated system installation.

Resources storage and isolation

Run `./osr perf storage` and admission checks at session start, before native jobs, after each completed batch and before publication. Extend existing `measure`, `compare` and cleanup facilities rather than introducing another profiler. Use sparse worktrees for code/docs and shared read-only source artifacts; no full checkout per agent or numerical sample. Keep private run directories outside source. Limit Rust target caches per pinned toolchain; set bounded build threads.

Maintain 20 GiB host free reserve plus all outstanding job reservations. The closing evidence recorded 21.71 GiB free, and this review again found about 22 GiB rounded by `df`, leaving less than 2 GiB above reserve. Remeasure bytes before every admission; a 2 GiB benchmark request is not automatically admissible. Count worktree, build, download, archive and output growth, not just solver files. Parallel jobs are limited by CPU, RAM and disk after reserves; start with one native job.

Budget class	Initial assignment limit
inventory	Read-only discovery, no installs/solves; 15 minutes per bounded audit
software	One owned change/review slice; at most two failed local iterations before reassessment; relevant tests under a declared 10-minute allowance; 1 GiB scratch request
benchmark	One hotspot; one cold plus three paired warm trials; 10-minute total run budget, 2 GiB scratch, one numerical worker
CAD	One assembly, one bounded revision; initial 25-part pilot ceiling or explicit counted expansion; one process at a time, 120 seconds/2 GiB address space per pilot process; predeclare total process count
mechanics	One admitted case at two mesh levels and one independent comparison; initial application slice at most 8 native processes/30 minutes total; per-process budgets from its existing recipe, revised only with a new resource identity
GIS	One selected city at a time; retain current 60 CPU/75 wall seconds and 2 GiB address-space bounds per native process, 32 MiB upload, 2 MiB request and 256 MiB successful ZIP cache
integration	One fresh disposable environment and one named journey at a time; inventory image/dependency sizes first and defer until the measured reserve fits; 15-minute browser-job ceiling initially

The full existing backend pilot has its own ≤32-process contract; do not squeeze it into the 8-process application slice or silently enlarge either. A fine box already required a separately recorded 3 GiB resource revision after a 2 GiB failure. New larger models need explicit measured budgets, not repeated retries. Thread budgets do not establish process-tree containment. Record bubblewrap, Flatpak, cgroup/quota and monitoring actually applied. GNU time, sampled directory size and RLIMIT_FSIZE do not enforce aggregate disk quotas.

Cleanup automatically only owned reproducible scratch/cache whose jobs are terminal. Inventory dirty worktrees and ownership before proposing cleanup. Preserve failed/nonconverged/interrupted evidence. For large research payloads, copy to the chosen artifact backend, verify original and stored bytes, create source-bound retrieval manifests, test offline/online replay, then retire redundant working copies. Choose the backend and retention/backup responsibility in S01; do not pretend an archive service already exists. Quarantine uncertain material and request a concrete deletion decision rather than guessing ownership.

Delivery waves and completion

Sequence by evidence readiness rather than restarting the original calendar.

Remaining wave	Work and exit condition
A — Candidate and mating definition	P01/C01: resolve the reference/child propagation boundary and one opening's real transforms, datums, route envelopes and missing inputs. Publish a reviewed interface packet before native assembly work.
B — Admitted engineering	C01/F01/E01/C02: model the actual assembly and rail/support/load interfaces; resolve available electrical and civil inputs. New solves require represented physics and declared limits; unavailable data creates a specific evidence request.
C — Controlled module lifecycle	M01/L01: produce the controlled module package, support its actual make/buy and serial/lot custody, and exercise native ERPNext/FUXA rejection, rework, installation, replacement and handback in a disposable environment.
D — Integrated journey and packaging	P02/W01/D01/V01: propagate the selected child through both corridor dossiers, test supported installation with optional tools absent, remove only proven redundant paths and verify the final supported release scope.

A bounded performance or packaging slice may run beside these waves when the second specialist slot and resource reserve are available. It must not displace the assembly/interface work or restart the completed GIS/profile programme.

Release acceptance requires both city journeys and a candidate-linked door manufacturing/QA/maintenance journey, with source-consistent geometry, BOM, costs and dossier. Tests cover invalid/stale/tampered input, resource refusal, interruption/resume, rollback, supported exports, unavailable tools and clean installation. Performance evidence includes cold/warm stage times, native rerun counts, peak memory/disk and installation footprint. An unchanged repeat should run no unnecessary native solver; a station edit should reuse unchanged vehicle and manufacturing results. Report remaining physical limits at their affected results, not as hidden assumptions or zero-valued inputs.

CI keeps fast contracts on relevant changes, integration on interface changes, and heavy numerical runs on explicit campaigns. Each Rust/provider change gets focused parity tests; repeat complete journeys only for changed dependencies or unresolved concerns. Preserve the existing sparse-test coverage guard. Tests and artifact records bind their actual source commits; never relabel old execution.

The coordinator closes tasks when outputs, review and dependency propagation are complete. If supplier data, a material model or an actual test is missing, record the specific evidence request and advance independent work. Software may publish a reproducible bounded planning capability with visible open physical findings; manufacturing release and physical qualification require their actual evidence. Use the existing maintainer software-release process, without adding a new signature bureaucracy or reopening v0.6.1.

Immediate agent dispatch

1. Coordinator retains the reviewed one-opening handoff and exact child IDs in P01/C01/M01; the 74-type, 694-piece inventory and unresolved datums are already recorded. Protect unrelated main-checkout edits. Keep the stale integrated current dossier excluded; no atlas-wide regeneration.
2. Slot A — mechanical/twin: obtain controlled header/body, rail/carriage, belt/pulley, drive and sensor datums for one opening. Replace the eleven generic header-centre zones only when mating dimensions and motion/removal clearances can be represented. Resolve the 2 m route proxy versus 2.9 m guide stock without treating the belt loop as a straight part. Start with one software-budget slice and zero native jobs; unknown inputs stay held.
3. Slot B — verifier/electrical/civil, serial bounded reviews: independently review those interfaces, distinguish fixed evidence from proposed dimensions, then identify the smallest obtainable drive/brake/thermal and rail/support/ erection inputs. Return source-bound requirements and next measurements; do not run more uniform mesh refinements before correcting the model domain.
4. Once the interface packet is reviewed, rotate slot A through SolidCraft assembly/motion/export and M01 controlled drawings. Admit FieldCraft only for the relevant material/joint/support domain; retain the optional independent FreeCAD/CalculiX checks. Every new run has its own resource/evaluation identity.
5. Rotate slot B through L01 sealed bought-source pack registration and actual native receipt, Incoming QA, serial/batch consumption, replacement and handback. The serialized purchased-child contract is now synthetically tested; pooled lots and real 694-piece module persistence are still open. Budget and inventory the disposable native environment before downloading images. M01 engineering acceptance gaps remain visible in the demonstration.
6. Coordinator integrates P02/W01 after these packets exist. The verifier tests affected city results, fourteen cost scopes, lifecycle and missing-tool paths. Run final V01 once on a frozen supported scope; keep historical CI executions bound to their original commits.

7 Optional next Rust slice: use R01's ranked phase evidence to specify one serialization/business-draft boundary, compare alternatives and land only a measured useful change. Do not default-enable the existing scheduler or recommission the rejected canonical port without the required evidence.

Primary local references: backend pilot, viaduct comparisons, GIS, release limitations, pilot delivery compiler, performance commands, agent roles, and engineering instructions.